

NOTICE INVITING e-TENDERS

The Executive Engineer-Almora, CPWD, Almora invites on behalf of President of India online **Percentage rate bids** from approved/appropriate and eligible contractors of CPWD in appropriate class of Buildings & Roads (erstwhile composite/Building/infrastructure) category for the following work :-

NIT NO.: 23/EE/ACD/CPWD/Almora/2026-27 Name of Work: Construction of Main Gate and Boundary wall with various miscellaneous repair works at Kendriya Vidyalaya Almora (UK). Estimated Cost: Rs. 49,72,713.00 Earnest money : Rs. 99,455.00 & Period of completion : 06 Months, Last date and time of submission of bid : 03:00 PM on 23.07.2026

The bid forms and other details can be obtained from the website
<https://etender.cpwd.gov.in> or <https://eauction.cpwd.go.in> or <https://eprocure.gov.in>

सहायक अभियन्ता (यो0)
के.लो.नि.वि., अल्मोड़ा

कार्यपालक अभियन्ता-अल्मोड़ा
के.लो.नि.वि., अल्मोड़ा



**GOVERNMENT OF INDIA
CENTRAL PUBLIC WORKS DEPARTMENT
OFFICE OF EXECUTIVE ENGINEER-ALMORA
CPWD, ALMORA (UTTARAKHAND)**

NIT No. 23/EE/ACD/CPWD/Almora/2026-27

NAME OF WORK	:	Construction of Main Gate and Boundary wall with various miscellaneous repair works at Kendriya Vidyalaya Almora (UK).
ESTIMATED COST	:	Rs. 49,72,713.00
EARNEST MONEY	:	Rs. 99,455.00
TIME ALLOWED	:	06 Months
DATE OF OPENING	:	23.07.2026

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Name of Work: **Construction of Main Gate and Boundary wall with various miscellaneous repair works at Kendriya Vidyalaya Almora (UK).**

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NIT amounting to Rs. 49,72,713.00 (Rs. Forty Nine Lacs Seventy-Two Thousand Seven Hundred and Thirteen) only is hereby approved.

This NIT contains pages marked as 1 to 143.

**Assistant Engineer (P)
CPWD, Almora**

**Executive Engineer-Almora
CPWD, Almora**

**INFORMATION AND INSTRUCTIONS FOR BIDDERS FOR E-TENDERING FORMING
PART OF BID DOCUMENT AND TO BE POSTED ON WEBSITE**

(Applicable for inviting open bids)

The **Executive Engineer-Almora, CPWD, Almora** on behalf of President of India invites online **Percentage rate bids** from approved/appropriate and eligible contractors of CPWD in appropriate class of Buildings & Roads (erstwhile composite/Building/infrastructure) category for the following work for:-

Sl. No.	NIT No.	Name of work & Location	Estimated cost put to bid	Earnest Money	Period of Completion	Last date & time of submission of bid, original EMD, copy of receipt for deposition of original EMD, and other Document as specified in the press notice	Time & date of opening of bid
1	23/EE/ACD/CPWD/ALMORA / 2026-27	Construction of Main Gate and Boundary wall with various miscellaneous repair works at Kendriya Vidyalaya Almora (UK).	Rs. 49,72,713.00	Rs. 99,455.00	06 Months	On 23.07.2026 upto 03:00 PM	On 23.07.2026 at 03:30 PM

- Only the enlisted contractors of Class-III, IV and V by the Enlisting authority i.e. SDG (Chandigarh), ADG (Chandigarh), ADG (Lucknow), ADG (Jammu), ADG (DR) and ADG (PRD) shall tender within their limit and tendering jurisdiction i.e. Punjab, Chandigarh, Rajasthan, Haryana, Uttarakhand, Himanchal Pradesh, Uttar Pradesh, Jammu & Kashmir, Leh & Ladakh and Delhi of North region (NR), as per the Rules for enlistment of Contractors in CPWD, 2026. Further, the enlisted contractors of Class-III, IV and V shall also have to meet the other requirements of eligibility as per NIT conditions.
- The intending bidder must read the terms and conditions of CPWD-6 carefully. He should only submit his bid if he is eligible and he is in possession of all the documents required.
- Information and Instructions for bidders posted on website shall form of bid document.

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4. The enlistment of the contractors should be valid on the last date of submission of bids. In case the last date of opening of bid is extended, the enlistment of contractor should be valid on the original date of opening of tender.
5. The bid document consisting of plans, specifications, the schedule of quantities of various types of items to be executed and the set of terms and conditions of the contract to be complied with and other necessary documents can be seen and downloaded from website <https://etender.cpwd.gov.in> or <https://eauction.cpwd.go.in> or <https://eprocure.gov.in> free of cost.
6. But the bid can only be submitted after **deposition of original EMD either in the office of Executive Engineer inviting bids or division office of any Executive Engineer, CPWD within the period of bid submission. (The EMD document shall only be issued from the place in which the office of receiving division office is situated)** and uploading the mandatory scanned documents such as Demand Draft or Pay order or Banker's Cheque or Deposit at call Receipt or Fixed Deposit Receipts and Bank Guarantee of any Scheduled Bank towards EMD in favour of Executive Engineer as mentioned in NIT, receipt for deposition of original EMD to division office of any Executive Engineer, CPWD and other documents as specified.
7. Those contractors not registered on the website mentioned above, are required to get registered beforehand. If needed they can be imparted training on online bidding process as per details available on the website.
8. Information and instruction for Contractors will form part of NIT and to be uploaded on ITI website.
9. **The contractor can deposit original EMD either in the office of Executive Engineer inviting bids or division office of any Executive Engineer, CPWD within the period of bid submission. (The EMD documents shall only be issued from the place in which the office of receiving division office is situated). The contractor shall obtain the receipt of EMD from the concerned Executive Engineer in the prescribed format uploaded by NIT issuing Executive Engineer.** The bid document as uploaded can be viewed and downloaded free of cost by anyone including intending bidder. But the bid can only be submitted after uploading the mandatory scanned documents such as Demand Draft/Pay order or Banker's Cheque/ Bank Guarantee of any Scheduled Bank towards EMD in favour of respective Executive Engineer, **copy of receipt of original EMD** and other documents specified in the press notice.
10. The intending bidder must have valid class-III digital signature to submit the bid.
11. On opening date, the contractor can login and see the bid opening process. After opening of bids he will receive the competitor bid sheets.
12. Contractor can upload documents in the form of **JPG** format and **PDF** format.
13. ~~Contractor must ensure to quote rate of each item. The column meant for quoting rate in figures appears in pink colour and the moment rate is entered, it turns sky blue.~~

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14. In addition to this, while selecting any of the cells a warning appears that if any cell is left blank the same shall be treated as "0".

Therefore, if any cell is left blank and no rate is quoted by the bidder, rate of such item shall be treated as "0" (ZERO).

15. The required documents meeting the criteria to qualify as "approved/appropriate and eligible class of contractors of CPWD" along with other documents as mentioned under para "List of Documents to be scanned and uploaded within the period of bid submission" below, as uploaded by the agency and hard copies received subsequently shall be checked first, while opening the bid. The financial bid of only those agencies shall be opened who are found to be eligible agencies, as per this NIT.
16. The department reserves the right to reject any prospective application without assigning any reason and to restrict the list of qualified contractors to any number deemed suitable by it, if too many bids are received satisfying the laid down criterion.
17. If the agency is not registered with Uttarakhand VAT/GST department earlier, he has to get register himself with Uttarakhand VAT/GST department before start of work.
18. **List of Mandatory Documents to be scanned and uploaded within the period of bid submission:**

1. **Scanned copy of Banker's Cheque of a Commercial Bank, Account Payee Demand Draft of a Commercial Bank, Fixed Deposit Receipt (FDR) of a Commercial Bank, Insurance Surety Bonds & Bank Guarantee including e-Bank Guarantee (for balance amount as prescribed in NIT) from a Commercial Bank against EMD.**
2. **Enlistment order of the contractors of CPWD in appropriate class of Buildings & Roads (erstwhile composite/Building/infrastructure) category.**
3. **GST Registration Certificate, if already obtained by the bidder.**

If the bidder has not obtained GST registration as applicable then he shall scan and upload following undertaking along with bid documents. **(Vide OM No. DG/SOP/20 Dt. 13.04.2021)**

"If work is awarded to me, I/we shall obtain GST registration Certificate as applicable, within one month from the date of receipt of award letter or before release of any payment by CPWD, whichever is earlier, failing which I/We shall be responsible for any delay in payments which will be due towards me/us on a/c of the work executed and/or for any action taken by CPWD or GST department in this regard."

4. **Copy of receipt for deposition of original EMD issued from division office of any Executive Engineer, CPWD (The EMD document shall be issued from the place in which the office of receiving division office is situated).**

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5. ERP (Enterprise Resource Planning) Training Certificate of the Contractor. The enlisted contractor will not be allowed to participate in the tendering process without uploading the Certificate of the ERP training. (OM No. DG/Enlist. Rules-2023/10 dated 22.04.2024 issued by DG, CPWD, Nirman Bhawan, New Delhi).

6. Permanent Account Number (PAN) as issued by the Income Tax Department.
7. Declaration about Site inspection in Form "A" Annexure VIII as per NIT.
8. Undertaking of association of agencies for specialized items of work:

"If work is awarded to me, I/we shall associate agencies (having desired credentials as mentioned in NIT) for specialized items of work, within one month from the date of receipt of award letter or before release of any payment by CPWD, whichever is earlier, failing which I/we shall be responsible for any delay in payments which will be due towards me/us on account of the work executed and/or for any action taken by CPWD."

The following conditions which already form part of the tender document are specially brought to the notice of all intending tenderers for compliance while filling the tender. They are requested to comply following instructions:

- After submission of the bid the contractor can re-submit revised bid any number of times but before last time and date of submission of bid as notified.
- While submitting the revised bid, contractor can revise the rate of one or more item(s) any number of times (he need not re-enter rate of all the items) but before last time and date of submission of bid as notified.
- When tenders are invited in three bid system and if it is desired to submit revised financial bid then it shall be mandatory to submit revised financial bid. If not submitted then the tender submitted earlier shall become invalid.
- The bidders should also read the **General Conditions of Contract 2023** for construction amended up to last date of submission of bid of this work, which is available as Govt. of India Publications; however provisions included in the tender document shall prevail over the provisions contained in this standard form. The set of drawings and NIT shall be available with the Executive Engineer.
- The contractor should also visit the site of work and acquaint himself with the site and soil conditions before tendering.
- The main contractor shall execute the minor component(s) also. He should be either an eligible contractor himself or associate with himself an eligible agency(s) for execution of electrical work as per CPWD-6 for e-tendering. The contractor shall indicate the names of upto three such agencies within prescribed time as "Minor component agencies". In case the details of electrical agency are not submitted by the bidder or are not proper then these shall be submitted before acceptance of the bid.

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- After acceptance of the tender by competent authority, the EE in charge of major component of the work shall issue letter of award on behalf of the President of India. After the work is awarded, the main contractor will have to enter into one agreement with EE in charge of major component and has also to sign two/or more copies of agreement depending upon number of EE's/DDH in charge of minor components. One such signed set of agreement shall be handed over to EE/DDH in charge of minor component. EE of major component will operate part A and part B of the agreement. EE/DDH in charge of minor component(s) shall operate Part C along with Part A of the agreement.
- The main contractor has to enter into agreement with the contractor(s) associated by him for execution of minor component(s). Copy of such agreement shall be submitted to EE/DDH in charge of minor component as well as to EE in charge of major component. In case of change of associate contractor, the main contractor has to enter into agreement with the new contractor associated by him.
- Tenders with any condition including that of conditional rebates in the tender document shall be rejected forthwith.
- The rate (s) must be quoted in decimal coinage. Amount must be calculated and rounded in full rupees by ignoring fifty paisa and considering more than fifty paisa as rupee one. Rate must be filled both in words and figures. Amount should be worked out for all the items. Rates in words should start with word "Rs." And finish with word "only" and there should not be undue gap in the words of the rate. For example, the rate of Rs. 2429.50 in figure should be written as "Rs. Two thousand four hundred twenty nine and paisa fifty only"
- The successful tenderer shall be required to submit a **Performance Guarantee** as per Schedule 'E' within a period of issue of letter of acceptance as specified in schedule 'F'.
- The contractor shall have to execute guarantee bonds in respect of water supply and sanitary installation works and water proofing works as per Performa at Annexure I & II.
- **GST on materials and work contract as applicable shall be paid by the contractor himself. The contractor shall quote his rates considering the increase in the GST rate of work contracts. Nothing extra shall be reimbursed on this account as per clause 34 of GCC applicable.**
- GST on materials as applicable shall be paid by the contractor himself. The contractor shall quote his rates considering GST.
- Attention is also drawn to Clauses 6 of **General Condition of Contract 2023** for construction, which prescribes the procedure for computerized recording of measurement and preparation of bill by contractor.
- Intending tenderer may submit physical mile stone on the basis of their resources and methodology at the time of tendering as indicated in the

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"Schedule F", otherwise it will be assumed that agency is agreeable to physical mile stones mentioned in the table.

- Intending tenderer may give detailed activities/programme for each mile stone fixed in the tender document while submitting the tender.
- Recovery rates for less use of materials beyond permissible limits is given in Schedule 'F' of Part-A & C & also for excess use of departmentally issued material.

19. Important Note:-

- The material for construction to be transported manually or by other means to the working locations/site, where transporting vehicle cannot approach. Nothing extra shall be payable of this account.**
- Bidder is requested to please visit the site before bidding.**

Executive Engineer-Almora
CPWD, Almora

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INFORMATION AND INSTRUCTIONS FOR EXECUTIVE ENGINEER FOR e-tendering

1. The Executive Engineer of all divisions of CPWD should receive the original EMD for tender of other division.
2. The NIT approving authority/EE at the time of issue of NIT shall also fill and upload the following prescribed format of receipt of deposition of original EMD along with NIT:-

Receipt of deposition of original EMD (Receipt No./date)	
Name of Work:- Construction of Main Gate and Boundary wall with various miscellaneous repair works at Kendriya Vidyalaya Almora (UK).	
1. NIT No.:- 23/EE/ACD/CPWD/Almora/2026-27	
2. Estimated Cost:- Rs. 49,72,713.00	
3. Amount of Earnest Money Deposit:- Rs. 99,455.00	
4. Last date of submission of bid:- 23.07.2026	
(*To be filled by NIT approving authority/EE at the time of issue of NIT and uploaded alongwith NIT)	
1. Name of contractor#	
2. Form of EMD#	
3. Amount of Earnest Money Deposit#	
4. Date of submission of EMD#	
Signature, Name and Designation of EMD receiving officer (EE/AE(P)/AAO) along with Office stamp (# To be filled by EMD receiving EE)	

3. The Executive Engineer receiving EMD in original form shall examine the EMD deposited by the bidder and shall issue a receipt of deposition of earnest money to the agency in a given format uploaded by tender inviting EE. The receipt may be issued by the AE(P)/AE/AAO.
4. The Executive Engineer receiving original EMD shall also intimate tender inviting Executive Engineer about deposition of EMD by the agency be email/fax/telephonically.
5. The original EMD receiving Executive Engineer shall release the EMD after verification from the e-tendering portal website <https://etender.cpwd.gov.in> or

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<https://eauction.cpwd.go.in> or <https://.eprocure.gov.in> that the particular contractor is not L-1 tenderer and work is awarded.

6. The tender inviting Executive Engineer will call for original EMD of the L.1 tenderer from EMD receiving Executive Engineer immediately.

Executive Engineer-Almora
CPWD, Almora

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PART-A :

**CPWD-6, CPWD-7 INCLUDING SCHEDULE 'A' TO 'F'
FOR MAJOR COMPONENT OF THE WORK, STANDARD
GENERAL CONDITIONS OF CONTRACT FOR CPWD
2023 (CONSTRUCTION WORK) AS AMENDED/
MODIFIED UP TO THE LAST DATE OF SUBMISSION OF
BID.**

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CPWD-6 FOR e-TENDERING

1. **Percentage rate bids** are invited on behalf of President of India from approved/appropriate and eligible class of contractors of Civil & composite category of CPWD only for the work of **Construction of Main Gate and Boundary wall with various miscellaneous repair works at Kendriya Vidyalaya Almora (UK).**

The enlistment of the contractors should be valid on the last date of submission of bids.

In case the last date of opening of bid is extended, the enlistment of contractor should be valid on the original date of opening of bids.

The work is estimated to cost **Rs. 49,72,713.00**

This estimate, however, is given merely as a rough guide.

The authority competent to approve NIT for the combined cost and belonging to the major discipline will consolidate NITs for calling the bids. He will also nominate Division, which will deal with all matters relating to the invitation of bids.

- 1.1 For composite tender, besides indicating the combined estimated cost put to tender, should clearly indicates the estimated cost of each component separately. The eligibility of tenderer will correspond to the combined estimated cost of different components put to tender.

1.2 ~~Conditions for Non-CPWD contractors only.~~

- 1.2.1 ~~The Contractor should have satisfactorily completed three similar works each of value not less than **Rs. 11.91 lacs** OR two similar work each of value not less than **Rs. 17.87 lacs** OR one similar work of value not less than **Rs. 23.83 lacs** in last 7 years ending last day of the previous month.~~

Similar work would means "Work of Caretaking, Housekeeping and watch & ward and Similar work executed from any Holiday home/Guest house of Govt. of India who have minimum four years experience in Hilly area only having equivalent altitude or more in last seven years (Work experience certificate issued by the competent authority)."

~~The value of executed works shall be brought to current costing level by enhancing the actual value of work at simple rate of 7% per annum, calculated from the date of completion to the last date of submission of tender.~~

1.2.2 ~~To become eligible for issue of bid, the bidders shall have to furnish an affidavit as under:-~~

~~"I/We undertake and confirm that eligible similar works(s) has/have not been got executed through another contractor on back to back basis. Further that, if such a violation comes to the notice of Department, then I/we shall be debarred for bidding in CPWD in future forever. Also, if such a violation comes to the notice of Department before date of start of work, the Engineer in Charge shall be free to~~

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forfeit the entire amount of Earnest Money Deposit/Performance Guarantee".
(Scanned copy to be uploaded at the time of submission of bid).

- 1.2.3 When tenders are invited from non CPWD contractors as per provisions of clause 1.2.1 above, it will be mandatory for non CPWD contractors to upload the work experience certificate(s) and the affidavit as per the provisions of clause 1.2.1. and 1.2.2.

Online bid documents submitted by intending bidders shall be opened only of those bidders, whose "Earnest Money Deposit, Cost of Bid Document and e-tender Processing Fee" and "other documents including eligibility documents as required as per this NIT" placed in separate sealed envelopes marked as "Earnest Money, Cost of Bid Document and Cost of Bid Processing Fee" and "Other documents", respectively are found in order. Both these envelopes shall be submitted together in another sealed envelope.

The financial bid of only those agencies shall be opened who are found to be eligible agencies, as per this NIT.

2. Agreement shall be drawn with the successful bidder on prescribed Form No. **CPWD 7** (or other Standard Form as mentioned) which is available as a Govt. of India Publication. Bidder shall quote his rates as per various terms and conditions of the said form which will form part of the agreement.
3. The time allowed for carrying out the work will be **06 Months** from the date of start as defined in schedule 'F' or from the first date of handing over of the site, whichever is later, in accordance with the phasing, if any, indicated in the bid documents.
4. (i) **The site for the work is available.**
(ii) The Architectural & Structural drawing, if any, shall be made available in phased manner. However if there is hindrance due to non available of Architectural/Structural drawing, the contractor shall communicate in writing the requirement of particular architectural/structural drawing. If requirements has not communicated in writing, no hindrance will be recorded.
5. The bid document consisting of plans, specifications, the schedule of quantities of various types of items to be executed and the set of terms and conditions of the contract to be complied with and other necessary documents except Standard General Conditions of Contract Form can be seen from website <https://etender.cpwd.gov.in> or <https://eauction.cpwd.gov.in> or <https://eprocure.gov.in> free of cost
6. After submission of the bid the contractor can re-submit revised bid any number of times but before last time and date of submission of bid as notified.
7. While submitting the revised bid, contractor can revise the rate of one or more item(s) any number of times (he need not re-enter rate of all the items)but before last time and date of submission of bid as notified.

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8. ~~When bids are invited in three stage system and if it is desired to submit revised financial bid then it shall be mandatory to submit revised financial bid. If not submitted then the bid submitted earlier shall become invalid.~~

9. Earnest Money in the form of Treasury Challan or Demand Draft or Pay order or Banker's Cheque or Deposit at Call Receipt or Fixed Deposit Receipt (drawn in favour of **Executive Engineer, Almora Central Division, CPWD, Almora**) shall be scanned and uploaded to the e-tendering website within the period of bid submission. **The original EMD should be deposited either in the office of Executive Engineer inviting bids or division office of any Executive Engineer, CPWD within the period of bid submission. (The EMD document shall only be issued from the place in which the office of receiving division office is situated.)** The EMD receiving Executive Engineer shall issue a receipt of deposition of earnest money deposit to the bidder in a prescribed format (enclosed) uploaded by tender inviting EE in the NIT.

The receipt shall also be uploaded to the e-tendering website by the intending bidder upto the specified bid submission date and time.

A part of earnest money is acceptable in the form of bank guarantee also. In such case, 50% of earnest money or Rs. 20 lac, whichever is less, shall have to be deposited in shape prescribed above, and balance may be deposited in shape of Bank Guarantee of any scheduled bank having validity for six months or more from the last date of receipt of bids which is to be scanned and uploaded by the intending bidders.

The intending bidders has to fill all the details such as Banker's name, Demand Draft/ Fixed Deposit receipt/ Pay Order/ Banker's Cheque/ Bank Guarantee number, amount and date.

The amount of EMD can be paid by multiple Demand Draft / Pay Order / Banker's Cheque / Deposit at call receipt / Fixed Deposit Receipts along with multiple Bank Guarantee of any Scheduled Bank if EMD is also acceptable in the form of Bank Guarantee.

Copy of Enlistment order and certificate of work experience and other documents as specified in the press notice shall be scanned and uploaded to the e-tendering website within the period of bid submission. **However, certified copy of all the scanned and uploaded documents as specified in press notice shall have to be submitted by the lowest bidder only within a week physically in the office of tender opening authority.**

Online bid documents submitted by intending bidders shall be opened only of those bidders, **whose original EMD deposited with any division of CPWD** and other documents scanned and uploaded are found in order.

The bid submitted shall be opened at 03:30 PM on 23.07.2026 or

10. The bid submitted shall become invalid if:

(i) The bidder is found ineligible.

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- (ii) **The bidder does not deposit original EMD with division office of any Executive Engineer, CPWD (The EMD document shall only be issued from the place in which the office of receiving division office is situated).**
 - (iii) The bidder does not upload all the documents (including GST registration) as stipulated in the bid document **including the copy of receipt for deposition of original EMD.**
 - (iv) If any discrepancy is noticed between the documents as uploaded at the time of submission of bid and hard copies as submitted physically by the lowest bidder in the office of tender opening authority.
 - (v) If bidder does not upload all the documents required as mentioned in mandatory documents in e-tender.
11. The contractor whose bid is accepted will be required to furnish **performance guarantee** as per Schedule 'E' within the period specified in Schedule F. This guarantee shall be in the form of cash (in case guarantee amount is less than Rs. 10,000/-) or Deposit at Call receipt of any scheduled bank/Banker's cheque of any scheduled bank/Demand Draft of any scheduled bank/Pay order of any Scheduled Bank of any scheduled bank (in case guarantee amount is less than Rs. 1,00,000/-) or Government Securities or Fixed Deposit Receipts or Guarantee Bonds of any Scheduled Bank or the State Bank of India in accordance with the prescribed form. In case the contractor fails to deposit the said performance guarantee within the period as indicated in Schedule 'F' including the extended period if any, the Earnest Money deposited by the contractor shall be forfeited automatically without any notice to the contractor. The earnest money deposited along with bid shall be returned after receiving the aforesaid performance guarantee. **The contractor whose bid is accepted with also be required to furnish either copy of applicable licenses/registrations or proof of applying for obtaining labour licenses, registration with EPFO, ESIC and BOCW Welfare Board and Programme Chart (Time and Progress) within the period specified in Schedule F.**
12. Intending Bidders are advised to inspect and examine the site and its surroundings and satisfy themselves before submitting their bids as to the nature of the ground and sub-soil (so far as is practicable), the form and nature of the site, the means of access to the site, the accommodation they may require and in general shall themselves obtain all necessary information as to risks, contingencies and other circumstances which may influence or affect their bid. A bidder shall be deemed to have full knowledge of the site whether he inspects it or not and no extra charge consequent on any misunderstanding or otherwise shall be allowed. The bidder shall be responsible for arranging and maintaining at his own cost all materials, tools & plants, water, electricity access, facilities for workers and all other Services required for executing the work unless otherwise specifically provided for in the contract documents. Submission of a bid by a bidder implies that he has read this notice and all other contract documents and has made himself aware of the scope and specifications of the work to be done

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and of conditions and rates at which stores, tools and plant, etc. will be issued to him by the Government and local conditions and other factors having a bearing on the execution of the work.

13. The competent authority on behalf of the President of India does not bind itself to accept the lowest or any other bid and reserves to itself the authority to reject any or all the bids received without the assignment of any reason. All bids in which any of the prescribed condition is not fulfilled or any condition including that of conditional rebate is put forth by the bidders shall be summarily rejected.
14. Canvassing whether directly or indirectly, in connection with bidders is strictly prohibited and the bids submitted by the contractors who resort to canvassing will be liable to rejection.
15. The competent authority on behalf of President of India reserves to himself the right of accepting the whole or any part of the bid and the bidder shall be bound to perform the same at the rate quoted.
16. The contractor shall not be permitted to bid for works in the CPWD Circle (Division in case of contractors of Horticulture/Nursery category) responsible for award and execution of contracts, in which his near relative is posted a Divisional Accountant or as an officer in any capacity between the grades of Superintending Engineer and Junior Engineer (both inclusive). He shall also intimate the names of persons who are working with him in any capacity or are subsequently employed by him and who are near relatives to any gazetted officer in the Central Public Works Department or in the Ministry of Urban Development. Any breach of this condition by the contractor would render him liable to be removed from the approved list of contractors of this Department.
17. No Engineer of gazetted rank or other gazetted officer employed in Engineering or Administrative duties in an Engineering Department of the Government of India is allowed to work as a contractor for a period of one year after his retirement from Government service, without the previous permission of the Government of India in writing. This contract is liable to be cancelled if either the contractor or any of his employees is found any time to be such a person who had not obtained the permission of the Government of India as aforesaid before submission of the bid or engagement in the contractor's service.
18. The bid for the works shall remain open for acceptance for a period of thirty (30) days from the date of opening of bids. If any bidder withdraws his bid before the said period or issue of letter of acceptance, whichever is earlier, or makes any modifications in the terms and conditions of the bid which are not acceptable to the department, then the Government shall, without prejudice to any other right or remedy, be at liberty to forfeit 50% of the said earnest money as aforesaid. Further the bidder shall not be allowed to participate in the rebidding process of the work.
19. This notice inviting Bid shall form a part of the contract document. The successful bidder/contractor, on acceptance of his bid by the Accepting Authority

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shall within 15 days from the stipulated date of start of the work, sign the contract consisting of:-

- a) The Notice Inviting Bid, all the documents including additional conditions, specifications and drawings, if any, forming part of the bid as uploaded at the time of invitation of bid and the rates quoted online at the time of submission of bid and acceptance thereof together with any correspondence leading thereto.
- b) Standard C.P.W.D. **Form 7** or other Standard C.P.W.D. Form as mentioned.

20. For Composite Bids

- 20.1.1 ~~The Executive Engineer in charge of the major component will call bids for the composite work. The cost of bid document and Earnest Money will be fixed with respect to the combined estimated cost put to tender for the composite bid.~~
- 20.1.2 ~~The bid document will include following three components:~~
- ~~Part A: CPWD 6, CPWD 7/8 including schedule A to F for major component of the work, Standard General Conditions of Contract for CPWD 2010 as amended/ modified up to last date of submission of bid.~~
 - ~~Part B: General/specific conditions, specifications and schedule of quantities applicable to major component of the work.~~
 - ~~Part C: Schedule A to F for minor component of the work. (SE/EE in charge of major component shall also be competent authority under clause 2 and clause 5 as mentioned in schedule A to F for major components) General/specific conditions, specifications and schedule of quantities applicable to minor component(s) of the work.~~
- 20.1.3 ~~The bidders must associate with himself, agencies of the appropriate class eligible to bid for each of the minor component individually.~~
- 20.1.4 ~~The eligible bidders shall quote rates for all items of major component as well as for all items of minor components of work.~~
- 20.1.5 ~~After acceptance of the bid by competent authority, the EE in charge of major component of the work shall issue letter of award on behalf of the President of India. After the work is awarded, the main contractor will have to enter into one agreement with EE in charge of major component and has also to sign two or more copies of agreement depending upon number of EE's/DDH in charge of minor components. One such signed set of agreement shall be handed over to EE/DDH in charge of minor component. EE of major component will operate part A and part B of the agreement. EE/DDH in charge of minor component(s) shall operate Part C along with Part A of the agreement.~~
- 20.1.6 ~~Entire work under the scope of composite bid including major and all minor components shall be executed under one agreement.~~
- 20.1.7 ~~Security Deposit will be worked out separately for each component corresponding to the estimated cost of the respective component of works. The Earnest Money will become part of the security deposit of the major components of work.~~

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- 20.1.8 ~~The main contractor has to associate agency(s) for minor component(s) conforming to eligibility criteria as defined in the tender document and has to submit detail of such agency(s) to Engineer-in-charge of minor component(s) within prescribed time. Name of the agency(s) to be associated shall be approved by Engineer in-charge of minor component(s).~~
- 20.1.9 ~~In case the main contractor intends to change any of the above agency/agencies during the operation of the contract, he shall obtain prior approval of Engineer-in-charge of minor component. The new agency/agencies shall also have to satisfy the laid down eligibility criteria. In case Engineer-in-charge is not satisfied with the performance of any agency, he can direct the contractor to change the agency executing such items of work and this shall be binding on the contractor.~~
- 20.1.10 ~~The main contractor has to enter into agreement with the contractor(s) associated by him for execution of minor component(s). Copy of such agreement shall be submitted to EE/DDH in charge of each minor component as well as to EE in charge of major component. In case of change of associate contractor, the main contractor has to enter into agreement with the new contractor associated by him.~~
- 20.1.11 ~~Running payment for the major component shall be made by EE of major discipline to the main contractor. Running payment for minor components shall be made by the Engineer in-charge of the discipline of minor component directly to the main contractor.~~
- 20.1.12 ~~Final bill of whole work shall be finalized and paid by the EE of major component. Engineer(s) in charge of minor component(s) will prepare and pass the final bill for their component of work and pass on the same to the EE of major component for including in the final bill for composite contract.~~
21. **The intending bidders are required to update their profile in CPWD e-tender portal and to upload their bids well in advance of last date of submission of tender. Any issue related to updating profile / uploading tender can be resolved through the concerned Executive Engineer (Phone no *05962-237825 & e- mail Id *eealm-cpwd-uk@gov.in) or ERP help line no. 18001803286 or e-mailldgpwd.support@techmahindra.com. The e-tendering bidders are also advised not to wait to raise any issue still the last date of submission of bid in their own interest.**

Signature of the Divisional Officer,
For and on behalf of the President of India

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CPWD-7

**GOVERNMENT OF INDIA
CENTRAL PUBLIC WORKS DEPARTMENT**

STATE	Uttarakhand	CE Office	CE-Dehradun
BRANCH	B&R	DIVISION	EE-Almora
ZONE	NZ-IV	SUB-DIVISION	Almora Central Sub Division

PERCENTAGE RATE BID AND CONTRACT FOR WORKS

Tender for the work of : Construction of Main Gate and Boundary wall with various miscellaneous repair works at Kendriya Vidyalaya Almora (UK).

- i) To be submitted up to the 03.00 PM on 23.07.2026 in the e-tendering website <https://etender.cpwd.gov.in> or <https://eauction.cpwd.go.in> or <https://eprocure.gov.in>.
- ii) The Bid shall be opened in presence of tenderers who may be present at 03.30 PM on 23.07.2026 in the office of the: **Executive Engineer-Almora, CPWD, Almora.**

TENDER

I/We have read and examined the Notice Inviting tender, schedule, A,B,C,D,E&F. Specifications applicable, Drawings & Designs, General Rules and Directions, Conditions of Contract, clauses of contract, Special conditions, Schedule of Rate & other documents and rules referred to in the conditions of contract and all other contents in the tender document for the work.

I/We hereby tender for the execution of the work specified for the President of India within the time specified in Schedule 'F', viz., schedule of quantities and in accordance in all respects with the specifications, designs, drawings and instructions in writing referred to in Rule-1 of General Rules and Directions and in Clause 11 of the Conditions of contract and with such materials as are provided for, by, and in respects in accordance with, such conditions so far as applicable.

We agree to keep the tender **open for thirty (30) days from the date of opening of bid** and not to make any modifications in its terms and conditions.

A sum of **Rs. 99,455.00** is hereby forwarded in Cash/Receipt Treasury Challan/Deposit at call Receipt of a Scheduled Bank/Fixed deposit receipt of

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scheduled bank/demand draft of a scheduled bank/bank guarantee issued by scheduled bank as earnest money. If I/we, fail to furnish the prescribed performance guarantee or fail to commence the work within prescribed period I/we agree that the said President of India or his successors in office shall without prejudice to any other right or remedy be at liberty to forfeit the said earnest money absolutely. Further, if I/we fail of commence work as specified, I/we agree that President of India or his successors in office shall without prejudice to any other right or remedy available in law, be at liberty to forfeit the said earnest money and the performance guarantee absolutely, otherwise the said earnest money shall be retained by him towards security deposit to execute all the works referred to in the tender documents upon the terms and conditions contained or referred to therein and to carry out such deviations as may be ordered, up to maximum of the percentage mentioned in Schedule 'F' and those in excess of that limit at the rates to be determined in accordance with the provision contained in Clause 12.2 and 12.3 of the tender form.

Further, I/We agree that in case of forfeiture of earnest money or both Earnest Money & Performance Guarantee as aforesaid, I/We shall be debarred for participation in the re-tendering process of the work.

I/We undertake and confirm that eligible similar work(s) has/ have not been got executed through another contractor on back to back basis. Further that, if such a violation comes to the notice of Department, then I/we shall be debarred for tendering in CPWD in future forever. Also, if such a violation comes to the notice of Department before date of start of work, the Engineer-in-Charge shall be free to forfeit the entire amount of Earnest Money Deposit/Performance Guarantee.

I/We hereby declare that I/we shall treat the tender documents drawings and other records connected with the work as secret/confidential documents and shall not communicate information derived there from to any person other than a person to whom I/we am/are authorized to communicate the same or use the information in any manner prejudicial to the safety of the State.

Dated:

Signature of
Contractor

Witness:

Postal Address

Address:

Occupation:

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ACCEPTANCE

The above tender (as modified by you as provided in the letters mentioned hereunder) is accepted by me for and on behalf of the President of India for a sum of Rs. _____ (Rupees _____

 _____)
 _____)

The letters referred to below shall form part of this contract Agreement:-

- a) _____
- b) _____
- c) _____

For & on behalf of the President of India

Dated _____

Signature _____
 Designation _____

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**PROFORMA OF SCHEDULES: A TO F
(FOR MAJOR COMPONENT OF WORK)**

SCHEDULE 'A'

Schedule of quantities (for Civil work: As per sheets attached)

SCHEDULE 'D'

Extra schedule for specific requirements/documents for the work, if any: As per NIT

SCHEDULE 'E'

Reference to General Conditions of Contract: General Conditions of Contract-2023 Construction Works as amended / modified up to the last date of submission of Tenders.

Name of Work: Construction of Main Gate and Boundary wall with various miscellaneous repair works at Kendriya Vidyalaya Almora (UK).

Estimated cost of work : Rs. 49,72,713.00

(i)	Earnest Money	₹ 99,455.00
(ii)	Performance Guarantee	5% (Five percent) of the Tendered Value or Estimated Cost put to Tender (ECPT) (Whichever is higher). Where the tendered amount is less than eighty percent (80%) of the Estimated Cost Put to Tender (ECPT), the Performance Guarantee, in addition to the requirement above, shall be increased by an amount equal to the difference between eighty percent (80%) of the ECPT and the tendered amount.
(iii)	Security Deposit	2.5% of the accepted tendered value of the work.

SCHEDULE 'F'**General Rules & Directions:**

Officer inviting Tender: Executive Engineer-Almora, CPWD, Almora

Definitions:

2(i)	Engineer-in-Charge	Executive Engineer-Almora, CPWD, Almora or successor thereof
(ii)	Accepting Authority	Executive Engineer-Almora, CPWD, Almora or successor thereof
(iii)	Percentage on cost of materials and Labour cover all to overheads and profits	15%

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(iv)	Standard Schedule of Rates	Delhi Schedule of Rates 2023 with up to date correction slips with up to date correction slips (for Civil work) + Cost Index 29%
(v)	Department	Central Public Works Department (CPWD)
(vi)	Standard CPWD Contract Form:	GCC 2023 Construction Works, CPWD Form 7 as modified & Corrected up-to the last date of submission of tenders

Clause-1:

(i)	Time allowed for submission of Performance Guarantee, programme chart (Time and Progress) and applicable labour licenses, registration with EPFO, ESIC and BOCW Welfare Board or proof of applying thereof from the date of issue of letter of acceptance.	07 (Seven) days
(ii)	Maximum allowable extension with late fee @ 0.1% per day of Performance Guarantee amount beyond the period provided in (i) above	03 (Three) days

Clause-2:

(i)	Authority for fixing compensation under clause 2	Chief Engineer-Dehradun, CPWD, Dehradun or successor thereof
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Clause-2A:

(i)	Applicable Clause	No
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Clause-5:

(i)	Number of days from the date of issue of letter of acceptance for reckoning date of start	10 (Ten) days
(ii)	Time allowed for completion of work	06 Months

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TABLE OF MILE STONES

Sl. No.	Description of mile stone	Time allowed in days (from date of start)	Amount to be withheld in case of non achievement of mile stone
1	15% of Tendered Amount	01 Month	0.5% of the accepted tendered value
2	30% of Tendered Amount	02 Month	0.5% of the accepted tendered value
3	45% of Tendered Amount	03 Month	1% of the accepted tendered value
4	60% of Tendered Amount	04 Month	1% of the accepted tendered value
5	75% of Tendered Amount	05 Month	1% of the accepted tendered value
6	100% of Tendered Amount & completed in all respect.	06 Month	1% of the accepted tendered value

Note:

1. Withheld amount shall be released if and when subsequent milestone is achieved within respective time specified. However, in case milestones are not achieved by the Bidder for the work, the amount shown against milestone shall be withheld.
2. Intending bidder may submit phasing of activities/milestones based on their resources and methodology at the time of bidding corresponding to physical milestones/stages indicated in the above table. These shall be formed part of the agreement after approval of the accepting authority, otherwise it would be assumed that agency agrees with the above mentioned physical milestones.

Authority to decide:

(i)	Authority to convey the decision of shifting of milestone and Extension of time	Executive Engineer-Almora, CPWD, Almora or his successor thereof
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(ii)	Authority to decide Rescheduling of mile stones and extension of time.	Executive Engineer-Almora, CPWD, Almora or his successor thereof
(iii)	Shifting of date of start in case of delay in handing over of site	Chief Engineer-Dehradun, CPWD, Dehradun or his successor thereof

PROFORMA OF SCHEDULES Clause 5 Schedule of handing over of Site:

Part	Portion of site	Description	Time period for handing over reckoned from date of issue of letter of intent
Part A	Portion without any hindrance	Full Site	Immediately on issue of Lol
Part B	Portions with encumbrances	NA	NA
Part C	Portion dependent on work of other agencies	NA	NA

Schedule of issue of designs:

Part	Portion of site	Description	Time period for handing over reckoned from date of receipt of tenders
Part A	Portion already included in NIT	All designs	Available
Part B1	Portions of Architectural Designs to be issued	All design	Available
Part B2	Portion of civil designs to be issued	All design	Available
Part B2	Portion of E&M Designs to be issued	All design	To be submitted by the agency

Clause-5.1:

Schedule of rate of recovery for delay in submission of the Time and Progress Chart and Progress Report:

Sl. No.	Contract Value	Recovery (Rs.)
I	For works costing upto Rs. 20 Crore	Rs. 2500 per week or part week

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II	For works costing more than Rs. 20 Crore	Rs. 5000 per week or part week
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Clause-5.2:

Nature of Hindrance Register (either Physical or Electronic): Electronic /Online system through CPWD website under “Contractors Login” tab.

Clause-5.4:

Schedule of rate of recovery for delay in submission of the modified programme in terms of delay days:

Sl. No.	Contract Value	Recovery (Rs.)
I	Less than or equal to Rs. 1 Crore	Rs. 500 per day
II	More than Rs. 1 Crore but less than or equal to Rs. 5 Crore	Rs. 1000 per day
III	More than Rs. 5 Crore but less than or equal to Rs. 20 Crore	Rs. 2500 per day
IV	More than Rs. 20 Crore	Rs. 5000 per day

Clause-6: Applicable

Clause-7:

Gross work to be done together with net payment /adjustment of advances for material collected, if any, since the last such payment for being eligible to interim payment	8.00 Lacs
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Clause-7A:

No Running Account Bill shall be paid for the work till the applicable labour licenses, registration with EPFO, ESIC and BOCW Welfare Board, whatever applicable as submitted by the Bidder to the Engineer-in Charge.	Yes
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Clause-8A:

Authority to decide compensation on account if contractor fails to submit completion plans	Executive Engineer, Almora CPWD, Almora or his successor thereof
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Clause-10A:

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List of testing equipment's to be provided by the contractor at site lab	As given in General Condition of Contract 2023 Construction Work under Quality Assurance & as per CPWD Specification-2019 Volume-I & II
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Clause-10B (ii): Not Applicable

Clause-10C: Applicable

Component of labour expressed as percent of value of work	25%
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Clause-10CA: Not Applicable

Clause-10CC: Not Applicable

Clause-11:

Specifications to be followed for execution of work:	
Work	CPWD Specifications 2019 Vol. 1 and Vol. 2 with up-to-date correction slips (Hereinafter called CPWD specifications).

Clause-12.2(C)

12.2 (C)	Deviation Limit beyond which clauses 12.2(C) shall apply for building work	100%.
(i)	Deviation Limit beyond which clauses 12.2 (C) shall apply for foundation work (except items mentioned in earth work subhead in DSR and related items)	100%.
(ii)	Deviation Limit for items mentioned in earth work subhead of DSR and related items	100%.

Clause-16:

Competent Authority for deciding reduced rates:	Chief Engineer-Dehradun, CPWD, Dehradun or his successor thereof
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Clause-18:

Suggestive List of Machinery, Tools & Plants to be deployed by the Contractor at site as applicable:

S. No.	Name of Equipment	Numbers
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1	Excavators (various sizes)	As per Requirement	
	Equipment for hoisting & lifting		
2	Tower Crane or Builder's hoist (Desirable)	Nil	
	Equipment for Concrete work		
3	Automatic Concrete Batching Plant sufficient Capacity: as per direction of Engineer-in-Charge	1 No.	
4	Concrete pump (Desirable)	1 No.	
5	Concrete transit mixer	As per Requirement	
6	Concrete mixer (diesel)	1 No.	
7	Concrete mixer (electrical)	As per Requirement	
8	Needle vibrator (electrical)	2 No.	
9	Needle vibrator (petrol)	2 No.	
10	Surface vibrator	As per Requirement	
	Equipment for Building work		
11	Bar bending Machine	2 No.	
12	Bar cutting machine	2 No.	
13	Drilling machine	2 No.	
14	Welding machine i/c transformer	1 No.	
15	Cube testing machines	1 No.	
16	M.S. pipes	As per direction of Engineer in charge	As per Requirement
17	Steel shuttering		
18	Steel scaffolding		
19	Grinding/polishing machines	As per Requirement	
	Equipment for transportation		
20	Tippers	As per Requirement	
21	Trucks	As per Requirement	

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	Pneumatic equipment	
22	Air compressors (diesel)	Nil
	Dewatering equipment	
23	Pump (diesel)	As per Requirement
24	Pump (electric) (Desirable)	As per Requirement
	Power equipment	
25	Diesel generator	1No.
26	Tool kit for field test of water	1No.

Note:

1. Workshop facilities for fabrication/addition and alterations, and other allied works shall be arranged by the contractor at his own cost.
2. The list of equipment/T&P/machinery as per above is for general guidance. In addition to these, machinery / equipment as required shall be arranged by the contractor in case the requirement at any stage exceeds as per the Programme finalized at his own cost and nothing extra whatsoever on this account shall be paid. This includes equipment for arrangement of concrete from RMC producing plants also.
3. All the equipment, T&P and machinery shall be kept in good working conditions.
4. Equipment like batching plant, concrete pump excavators/Transit mixer etc. shall be allowed to be moved away from the site when, the same are no longer required at site of work in the opinion of Engineer-in-charge.
5. In addition to above list, contractor is bound to brought at site any test equipment for any item of work, at his own cost, which Engineer-in-Charge may direct him. Nothing extra shall be paid to contractor in this regard. Direction of Engineer-in-Charge in this regard shall be final & binding.
6. If contractor fails to comply such directions within time specified by Engineer-in-Charge, the same shall be brought to site by department by any means at cost of contractor itself and nothing shall be paid in this regard.

Clause	Penalty for each default as mentioned in relevant clause of GCC
Clause-19 C	Rs. 400.00 Each default

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Clause-19 D	Rs. 2500.00 Each default
Clause-19 G	Rs. 400.00 Each default
	Rs. 500.00 Each default (For enhanced penalty in case of continuous default)
Clause-19 K	Rs. 200.00 Each default

Clause-25:

i	Conciliator:	ADG Chandigarh Region.
li	Arbitrator appointing authority:	Chief Engineer-Dehradun
iii	Place of Arbitration:	Dehradun.

Clause-30, 30A & 31: Applicable

Clause-32:

Requirement of Technical Representative(s) and Recovery Rates:

Cost of work (Rs. In Crore)	Requirement of Technical Staff		Minimum experience (Years)	Designation of Technical Staff	Rate at which recovery shall be made from the contractor in the event of not fulfilling provision of clause 32(i)
	Qualification	Number (of Major + Minor Component)			
More than 15 lac to 1.5 crore	Graduate Engineer or Diploma Engineer	1 of Major Component	2 or 5 respectively	Project Manager cum Planning/ Quality/site/ billing Engineer	Rs.15,000/- per month per person

Note:

1. Assistant Engineers retired from Government services that are holding Diploma will be treated at par with Graduate Engineers.
2. Diploma holder with minimum 10-years relevant experience with a reputed construction company can be treated at par with Graduate Engineers for the purpose of such deployment subject to that such diploma holder should not exceed 50% of requirement of degree engineers.

Clause-38:

Applicable as given below:

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i) (a)	Schedule/statement for determining theoretical quantity of cement & bitumen based on Delhi Schedule of Rates	Delhi Schedule of Rates 2023 with amendments up to the date of submission of bid.
ii)	Variations permissible on theoretical quantities:	
(a)	Cement	
	For works with estimated cost put to tender upto Rs. 25 Lakh.	3% Plus/Minus
	For works with estimated cost put to tender more than Rs. 25 Lakh.	2% Plus/Minus
(b)	Bitumen for all works.	2.5% Plus only and nil on minus side.
I	Steel reinforcement and structural steel sections for each diameter, section and category.	2% Plus/minus
(d)	All other materials	Nil

Note:- In the schedule 'F' those clauses which are not applicable shall means that they will not be part of GCC 2023 for construction work as amended / modified up to the last date of submission of bid.

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SALIENT / MANDATORY REQUIREMENTS FOR THE TENDERER

Name of Work:

Construction of Main Gate and Boundary wall with various miscellaneous repair works at Kendriya Vidyalaya Almora (UK).

- 1 The tenderer is advised to read and examine the tender documents for the work and the set of drawings available with Engineer-in-charge. He should inspect and examine the site and its surroundings by himself before submitting his tender.
- 2 Separate schedule of quantity is included in this tender for civil and electrical items of work. If the tenderer wants to offer any unconditional rebates on their rates, the same should also be offered in the respective components of civil and electrical schedule separately. The contractor shall quote the **percentage rates** in figures and words accurately so that there is no discrepancy in rates written in figures and words.
- 3 Time allowed for the execution of work is **06 Months.**
- 4 The contractor(s) shall submit a detailed program of execution in accordance with the master programme/milestone within ten days from the date of issue of award letter.
- 5 Quality of the project is of utmost importance. This shall be adhered to in accordance with the provisions of CPWD specifications and guidelines given in the relevant paras.
- 6 The contractor (s) shall make his own arrangements for electricity and water required for the execution of work.
- 7 **Cement shall be arranged by the contractor himself.**
- 8 **Steel Reinforcement shall be arranged by the contractor himself.**
- 9 Contractor has to deploy required Plant and machinery on the project. Minimum number of plant and machinery to be deployed by him is indicated in this NIT. However, if any additional type and number of plants and machines are required at site, the same shall be deployed by the agency without any extra cost which shall be in order to adhere to the time schedule, as specified in this NIT. In case the contractor fails to deploy the plant and machinery whenever required and as per the direction of the Engineer-in-charge, he (Engineer-in-charge) shall be at a liberty to get the same deployed at the risk and cost of the contractor.
- 10 The contractor shall submit the running bills in the shape of the eveling ed MB in pages of A-4 size as per the standard format of department and shall act as per modified clause 6 of GCC 2023 Construction work.
- 11 Contractor has to provide reinforcement cover blocks made of approved proprietary pre packed free flowing mortars (Conbextra as manufactured by

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M/s Fosroc Chemical India Ltd. or approved equivalent) of high early strength.

~~The contractor shall submit list of works which are in hand (progress) in the following form (For non CPWD Contractors for work upto 15 crores & for CPWD as well as non CPWD contractors for work beyond 15 crores.):~~

Name of work	Name and particulars of Divn. Where work is being executed	Value of work	Position of works in progress		Remarks
			Stipulated	Actual	
4	2	3	4	5	6

- 12 The officer inviting tenders shall have the right of rejecting all or any of the tenders and will not be bound to accept the lowest or any other tender.
- 13 In the case of any tender where unit rate of any item / items appear unrealistic, such tender will be considered as unbalanced and in case the tenderer is unable to provide satisfactory explanation such a tender is liable to be disqualified and rejected.
- 14 The Contractor, whose tender is accepted, will be required to furnish performance guarantee of 5% (Five Percent) of the tendered amount within the period specified in Schedule "F".
- 15 The Contractor, whose tender is accepted, will also be required to furnish by way of Security deposit for the fulfillment of his contract, an amount equal to 2.5% of the tendered value of the work. The Security deposit will be collected by deductions from the running bills of the contractor at the rates mentioned above and the earnest money deposited at the time of tender, will be treated as a part of the security deposit.
- 16 On acceptance of the tender, the name of the accredited representative(s) of the contractor who would be responsible for taking instructions from the Engineer-in-Charge shall be communicated in writing to the Engineer-in-charge.
- 17 Sales Tax/VAT, purchase tax, turnover tax or any other tax on material in respect of this contract shall be payable by the Contractor and Government will not entertain any claim whatsoever in respect of the same.
- 18 The contractor shall give a list of both Gazetted and Non-Gazetted CPWD employees related to him.

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- 19 The tender for composite work includes in addition to building work all other works such as sanitary and water supply installations drainage installation, electrical work, horticulture work, roads and paths etc. The tenderer apart from being a registered contractor (B&R) of appropriate class, must associate himself with agencies of appropriate class which are eligible to tender for sanitary and water supply drainage, electrical and horticulture works in the composite tender.
- 20 The contractor shall comply with the provisions of the Apprentices Act 1961, and the rules and orders issued there under from time to time. If he fails to do so, his failure will be a breach of the contract and the Superintending Engineer/Executive Engineer may in his discretion, without prejudice to any other right or remedy available in law, cancel the contract. The contractor shall also be liable for any pecuniary liability arising on account of any violation by him of the provisions of the said Act.

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PART-B :

**ADDITIONAL CONDITIONS, GENERAL CONDITIONS,
ADDITIONAL SPECIFICATION, QUALITY ASSUARANCE
OF THE WORK, SPECIAL CONDITIONS, PARTICULAR
SPECIFICATION, LIST OF ACCEPTABLE/PREFERRED
MAKES OF MATERIALS, ANNEXURE-I TO VIII AND
SCHEDULE OF QUANTITIES APPLICABLE TO THE
WORK**

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ADDITIONAL CONDITIONS

1. Unless otherwise provided in the Schedule of Quantities/Specifications, the rates tendered by the contractor shall be all inclusive and shall apply to all heights, lifts, leads and depths of the work and nothing extra shall be payable to him on account of the same. Extra payment for centering/shuttering, if required to be done for heights greater than 3.5 m shall however be admissible at the rates arrived at in accordance with clause 12 of the agreement, if not already specified.
2. The contractor shall make his own arrangement for obtaining electric connection(s) if required and make necessary payments directly to the department concerned.
3. Other agencies doing works related with this project may also simultaneously execute their works and the contractor shall afford necessary facilities for the same. The contractor shall leave such necessary holes, openings etc. for laying/burying in the work, pipes cables, conduits, clamps, boxes and hooks for fan clamps etc. as may be required for the other agencies. Nothing extra over the Agreement rates shall be paid for doing these.
4. Some restrictions may be imposed by the security staff etc. on the working and for movement of labour, materials etc. The contractor shall be bound to follow all such restrictions/instructions and nothing extra shall be payable on account of the same.
5. The contractor shall fully comply with all legal orders and directions of the Public or local authorities or municipality and abide by their rules and regulations and pay all fees and charges for which he may be liable in this regard. Nothing extra shall be paid/reimbursed for the same.
6. The building work shall be carried out in the manner complying in all respects with the requirements of the relevant bylaws and regulations of the local body under the jurisdiction of which the work is to be executed or as directed by the Engineer-in-charge and nothing extra shall be paid on this account.
7. The work of water supply, internal sanitary installations and drainage etc. shall be carried out as per the bylaws of the Municipal Corporation or any other local body and the contractor shall produce necessary completion certificates from such authority after completion of work.
8. All water tanks, taps, sanitary, water supply and drainage pipes fittings and accessories etc. shall conform to the bylaws and specifications of the Municipal Body/Corporation where CPWD specifications are not available.
9. The contractor shall engage licensed plumbers for the work and the materials (fixtures/fittings) tested by the local Municipal Body/Corporation wherever required at his own cost. Nothing extra shall be paid/reimbursed for the same.

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10. The contractor shall give a performance test of the entire installation(s) as per standing specifications before the work is finally accepted by making his own arrangements for water supply, electricity etc. and nothing extra whatsoever shall be payable for the same.
11. If as per local Municipal regulations, huts for labour are not to be erected at the site of work, the contractor shall be required to provide such accommodation at a place as is acceptable to the local body and nothing extra shall be paid on this account.
12. The structural and architectural drawings shall at all times be properly co-related before executing any work. However, in case of any discrepancy in the item given in the schedule of quantities appended with the tender and Architectural drawings relating to the relevant item, the former shall prevail unless otherwise given in writing by the Engineer-in-charge.
13. The contractor shall bear all incidental charges for cartage, storage and safe custody of materials issued by department.
14. Samples of various materials required for testing shall be provided free of charges by the contractor. Testing charges shall be borne by the contractor. All other expenditure required to be incurred for taking the samples; conveyance, packing etc. shall be borne by the contractor himself.
15. Any cement slurry added over base surface (or) for continuation of concreting for better bond is deemed to have been in-built in the items and nothing extra shall be payable (or) extra cement considered in consumption on this account.
16. For the purpose of recording measurements and preparing running account bills, the abbreviated nomenclature indicated in the publications Abbreviated Nomenclature of Items of DSR 2023 shall be accepted. The abbreviated nomenclature shall be taken to cover all the materials and operations as per the complete nomenclature of the relevant items in the agreement and relevant specifications.

In case of items for which abbreviated nomenclature is not available in the aforesaid publication and also in case of extra and substituted items for which abbreviated nomenclature are not provided for in the agreement, full nomenclature of item shall be reproduced in the measurement books and bill forms for running account bills.

For the final bill, however, full nomenclature of all the items shall be adopted in preparing abstract in the measurement books and in the bill forms.
17. The contractor shall have to make approaches to the site, if so required and keep them in good condition for transportation of labour and materials as well as inspection of works by the Engineer in charge. Nothing extra shall be paid on this account.
18. No payment will be made to the contractor for damage caused by rains, or other natural calamities during the execution of the works and no such claim on this account will be entertained.

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19. The contractor shall take instructions from the Engineer-in-charge for stacking of materials. No excavated earth or building materials etc. shall be stacked/collected in areas where other buildings, roads, services, compound walls etc. are to be constructed.
20. Any trenching and digging for laying sewer lines/water lines/cables etc. shall be commenced by the contractor only when all men, machinery's and materials have been arranged and closing of the trench(s) thereafter shall be ensured within the least possible time.
21. It shall be ensured by the contractor that no electric live wire is left exposed or unattended to avoid any accidents in this regard.
22. In case the supply of timber/steel frames/shutters for doors, windows etc. is made by some other agency, the contractor shall make necessary arrangements for their safe custody on the direction of the Engineer-in-charge till the same are fixed in position by him & nothing extra shall be paid on this account.
23. The contractor shall maintain in perfect condition, all portions executed till completion of the entire work allotted to him. Where however phased delivery of work is contemplated these provisions shall apply separately to each phase.
24. The entire royalty at the prevalent rates shall have to be paid by the contractor on all the boulders, metals, shingle sand etc. collected by him for execution of the work, directly to the Revenue authority or authorized agents of the State Government concerned or the Central Government, as the case may be.
25. Various factory made materials shall be procured from reputed and approved manufacturers or their authorized dealers. Decision of Engineer in charge in this regard shall be final and binding.
26. It must be ensure that all materials to be used in work bear BIS certification mark. In cases where BIS certification system is available for a particular material/product but not even a single producer has so far approached BIS for certification the material can be used subject to the condition that it should confirm to CPWD specification and relevant BIS codes. In such case written approval of the Technical sanctioning Authority may be obtained before use of such material in the work.
27. **Contractor shall have to execute a Guarantee Bond in respect of Water Proofing works as per Performa attached in this N.I.T. He shall also have to execute guarantee bonds for water supply and sanitary installations work on the Performa attached in this NIT.**
28. The terms machine batched, machine mixed and machine vibrated concrete used elsewhere in agreement shall mean the concrete produced in concrete batching and mixing plant and if necessary transported by transit concrete mixers, placed in position by the concrete pumps, tower crane and vibrated by surface vibrator /needle vibrator / plate vibrator, as the case may be to achieve required strength and durability.

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29. Wherever work is specified to be done or material procured through specialized agencies, their names shall be got approved well in advance from Engineer in charge. Failure to do so shall not justify delay in execution of work. It is suggested that immediately after award of work, contractor should negotiate with concerned specialist agencies and send their names for approval to Engineer in charge. Any material procured without prior approval of Engineer in charge in writing is liable to be rejected. Engineer in charge reserves right to get the materials tested in laboratories of his choice before final acceptance. Non standard materials shall not be accepted.
30. Doors and frames shall be procured from specialist firms and name of such agencies shall be got approved from the Engineer in charge well in advance.
31. The construction joints shall be provided in predetermined locations only as decided by Engineer in charge. The cost of shuttering for these construction joints shall be included in item of Concrete work / RCC work and nothing extra shall be payable on this account to the contractor.
32. The gradation of fine sand to be used in plaster work shall be strictly as per CPWD Specifications 2019, conforming to IS 1542-1977. The plastered surface shall be fairly smooth without any undulation of any kind for applying paint/white wash.
33. No chase cutting/dismantling of plaster/RCC/CC shall be allowed, so contractor has to execute the electrical work accordingly.
34. The contractor shall invariably prepare the samples of finishing items i.e. flooring of different types, external & internal finishing i/c colour scheme of paint, tiles in dado, flooring in platforms & staircase, water supply & sanitary fittings and any other item as per direction of Engineer-in-charge. The contractor shall proceed with further finishing items only after getting the samples of these items approved in writing from Engineer-in-charge. In case of construction of residential quarters, one sample quarter complete in all shape for each category, shall be prepared by the contractor and got approved from Engineer-in-charge in writing. The contractor shall be allowed to proceed with further quarters only after getting the sample quarters approved in writing from Engineer-in-charge. No extra claim whatsoever beyond the payments due at agreement rates will be entertained from the contractor on this account.
35. The contractor or his authorized representative shall associate in collection, preparation, forwarding and testing of such samples. In case, he or his authorized representative is not present or does not associate him, the results or such tests and consequences thereon shall be binding on the contractor.
36. **The contractor shall get the water tested with regard to its suitability of use in the works and get written approval from the Engineer in charge before he proceeds with the use of same of execution of works. If the tube-well water is not suitable, the contractor shall arrange Municipal water or from any other sources at his own cost and nothing extra shall be paid to the**

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contractor on this account. The water shall be got tested at frequency specified in latest CPWD specifications/BIS code.

37. Departmental Issue of Materials:

- 37.1 The materials shall be issued to the contractor at the place of delivery as mentioned in the Schedule-B during the working hours as per the rules of CPWD stores in force from time to time. If these are delivered at any other place, adjustments on accounts of the difference in cost of cartage shall be affected as per the terms of clause 12 of the contract agreement. The contractor shall however have to cart the materials to the site of work at his cost as soon as these are issued.
- 37.2 Materials like reinforcing bars, flats, tees, angles, sheets, CI and SCI pipes etc., if contemplated to be issued shall be issued in available sizes and lengths and the contractors shall bear the cost of cutting and shaping them according to the requirements of work. No claim for the wastage on this account shall be entertained.
- 37.3 The contractor shall bear all incidental charges for cartage, storage and safe custody of materials issued by the departments and shall construct suitable godowns, yards at the site of work for storing all materials as to be safe against damage by sun, rain, dampness, fire, theft etc. at his own cost and also employ necessary watch and ward establishment for the purpose, at his own cost. Materials to be charged directly to work and stipulated for issue free of cost shall also be issued to the contractor as soon as those are received at site or at the stipulated place of issue. The provision of this para shall apply equally and fully to those as well.
- 37.4 All materials obtained from the Govt. stores or otherwise on receipt shall be got checked by the Engineer-in-charge of the work or his representations before use.
- 37.5 Registers for the materials to be issued by the department shall be maintained as required by the Engineer-in-charge and these shall be signed by the contractor or his authorized agent and representative of Engineer-in-charge on each day of transactions.
- 37.6 Cement bags shall be stored in separate godowns to be constructed by the contractor at his own cost as per sketch (which is only indicative and actual size will depend on the site requirements) given in CPWD specifications with weather proof roofs and walls. Each godown shall be provided with a single shutter door with two locks. The key of one lock shall remain with Engineer-in-charge or his authorized representative and that of the other lock with the authorized agent of the contractor at the site of work so that the cement is issued from the godown according to the daily requirements with the knowledge of both parties and proper account for the same is maintained in the standard proforma.

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PROFORMA FOR THE CEMENT REGISTER
PARTICULARS OF RECEIPT

Date of receipt	Quantity received	Progressive total	Date of issue	Quantity issued	Items of work for which issued
1	2	3	4	5	6

PARTICULARS OF ISSUE

Qty. returned at the end of the day	Total issued	Daily balance in hand	Contractor's initial	JE's initial	Remarks (AE/EE's periodical check)
7	8	9	10	11	12

Separate cement registers showing the receipt of the OPC and PPC shall be maintained at site. The contractor shall construct separate godowns for storage of OPC & PPC at site and nothing extra on this account shall be payable.

37.7 Cement issued shall be for consumption at site only. No cement for factory made items and those not manufactured at site shall be issued.

~~38. The contractor shall make his own arrangement for conducting geological survey for determining the possibility of underground water adequately at his own cost. In case the failure of boring (Not available underground water) department will not pay anything extra on this account. The Department is liable to pay one time only for successful boring as per BOQ items and rate quoted by the contractor. The contractors are advised to quote his rates including geological survey charges.~~

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GENERAL CONDITIONS

- 1 The contractor (s) shall inspect the site of work before tendering and acquaint himself with the site conditions and no claim on this account shall be entertained by the department.
The contractor (s) shall get himself acquainted with nature and extent of the work and satisfy himself about the availability of materials from kiln or approved quarries for collection and conveyance of materials required for construction.
2. **General Conditions of Contract (GCC 2023-Construction work) for CPWD Works which is available at M/s Jain Book Agency, 4574/15, 2nd Floor, Padam Chand Marg, Darya Ganj, New Delhi-110002as CPWD publication shall be deemed to be part of the Tender document.**
3. **The Standard GCC is amended from time to time through issue of Oms under series DG/CON which are available on CPWD official website at [http://www.cpwd.gov.in/Documents/Official Circulars/](http://www.cpwd.gov.in/Documents/Official_Circulars/) DG CON. The said Circulars bearing no. DG/CON/309and onwards issued upto last date of tender submission shall also be deemed to be part of tender document.**
4. The contractor (s) shall study the soil investigation report for the site, available in the office of the Engineer-in-Charge and satisfy himself about complete characteristics of soil and other parameters of site. However, no claim on the alleged inadequacy or incorrectness of the soil data supplied by the department shall be entertained.
5. The tenderer shall see the approaches to the site. In case any approach from main road is required by the contractor, the same shall be provided, improved and maintained by the contractor at his own cost. No payment shall be made on this account.
6. The contractor (s) shall give to the Municipality, Police and other authorities all necessary notices etc. that may be required by law and obtain all requisite Licenses for temporary obstructions, enclosures etc. and pay all fee, taxes and charges which may be leviable on account of these operations in executing the contract. He shall make good any damage to the adjoining property whether public or private and shall supply and maintain light and other illumination on for cautioning the public at night.
7. The contractor shall take all precautions to avoid accidents by exhibiting necessary caution boards day and night speed limit boards red flags, red lights and providing barriers. He shall be responsible for all dangers and incidents caused to existing / new work due to negligence on his part. No hinderances shall be caused to traffic during the execution of the work.
8. The contractor shall provide at his own cost suitable weighing surveying and leveling and measuring arrangements as may be necessary at site for checking. All such equipments shall be got calibrated in advance from laboratory, approved by the Engineer-in-Charge. Nothing extra shall be payable on this account.

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9. Contractor shall provide permanent bench marks and other reference points for the proper execution of work and these shall be preserved till the end of work. All such reference points shall be in relation to the levels and locations, given in the Architectural and plumbing drawings
10. On completion of work, the contractor shall submit at his own cost four prints of "as built" drawings to the Engineer-in-Charge within 30 days of completion of work. These drawings shall have the following information:
 - a) Run of all piping and their diameters including soil waste pipes & vertical stacks.
 - b) Ground and invert levels of all drainage pipes together with locations of all manholes and connections upto outfall.
 - c) Run of all water supply lines with diameters, location of control valves, access panels etc.

If above said drawings are not submitted by the contractor within the above specified time then final bill will not be paid and Security Deposit shall not be released.

11. The contractor shall submit for the approval of Engineer-in-Charge names of leveling agencies of repute along with their technical capacity proposed to be engaged by him for executing specialized works, who must have executed satisfactorily works of value as specified in mandatory conditions.
12. The works shall be carried out in accordance with the Architectural drawings and structural drawings, to be issued from time to time by the Engineer-in-Charge. Before commencement of any item of work, the contractor shall correlate all the relevant architectural and structural drawings issued for the work and satisfy himself that the information available there of is complete and unambiguous. The discrepancy, if any shall be brought to the notice of the Engineer-in-Charge before execution of the work. The contractor alone shall be responsible for any loss or damage executing by the commencement of work on the basis of any erroneous and or incomplete information.
13. The contractor shall take all precautions to avoid accidents by, exhibiting caution boards day and night, speed limit boards, red flags, red light and providing necessary barriers and other measures required from time to time. The contractor shall be responsible for all damages and accidents due to negligence on his part.
14. Other agencies will also simultaneously execute and install the works of electrification, air conditioning, lifts, fire-fighting etc. for this work and the contractor shall provide necessary facilities for the same. The contractor shall leave such recesses, holes openings etc. as may be required for the electric, air-conditioning and other related works (for which inserts, sleeves, brackets, conduits base pinion, clamps etc. shall be supplied free of cost by the department unless otherwise specifically mentioned) and the contractor shall fix

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the same at time of casting of concrete, stone work & brick work, if required and nothing extra shall be payable on this account.

15. All materials obtained from Govt. stores or otherwise shall be get checked by the Engineer-in-Charge or his any authorized supervisor staff on receipt of the same at site before use.
16. The contractor shall conduct work so as not to interfere with or hinder the progress or completion of the work being performed by other contractor(s) or by the Engineer-in-Charge and shall as far as possible arrange his work and shall place and dispose off the materials being used or removed so as not to interfere with the operations of other contractor or he shall arrange his work with that of the others in an acceptable and coordinated manner and shall perform it in proper sequence to the complete satisfaction of others.
17. All Architectural drawings given in the tender other than those indicated in nomenclature of items are only indicative of the nature of the work and materials/fixings involved unless and otherwise specifically mentioned. However, the work shall be executed in accordance with the drawings duly approved by the Engineer-in-Charge.
18. **PROGRAMME CHART**
 - i) The contractor shall prepare an integrated programme chart for the execution of work, showing clearly all activities from the start of work to completion, with details of manpower, equipment and machinery required for the leveling of the programme within the stipulated period or earlier as indicated in the mile stones under clause 5 of the contract and submit the same for approval to the Engineer-in-Charge within ten days of the award of the contract.
 - ii) The programmes chart should include the following:-
 - a) Descriptive note explaining sequence of various activities.
 - b) Net work (PERT/CPM/BAR CHART)
 - c) Programme for procurement of materials by the contractor
 - d) Programme of procurement of machinery/equipment's having adequate capacity commensurate with the quantum of work to be done within the stipulated period by the contractor.
19. If appears to the Engineer-in-Charge that the actual progress of work does not conform to the approved programme referred above the contractor shall produce a revised programme showing the modifications to the approved programme to ensure completion of the work within the stipulated time for completion.
20. The submission for approval by the Engineer-in-Charge of such programme or the furnishing of such particulars shall not relieve the contractor of any of his duties or responsibilities under the contract. This is without prejudice to the right of Engineer-in-Charge to take action against the contractor as per terms and conditions of the agreement.

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21. If the work is carried out in more than one shift or during night no claim on this accounts shall be entertained.
22. Existing drains, pipes, cables, over-head wires, sewer lines, water lines and similar services encountered in the course of the execution of work shall be protected against the damage by the contractor at his own expense. The contractor shall not store materials or otherwise occupy any part of the site in a manner likely to hinder the operation of such services.
23. The contractor shall be responsible for the watch and ward/guard of the buildings, safety of all fittings and fixtures including sanitary and water supply fittings and fixtures provided by him against pilferage and breakage during the period of installations and thereafter till the building is physically handed over to the department. No extra payment shall be made on this account.
24. The day to day receipt and issue accounts of different grade/brand of cement shall be maintained separately in the standard proforma by the Jr. Engineer-in-Charge of work and which shall be duly signed by the contractor or his duly authorized representative.
25. The contractor shall render all help and assistance in documenting the total sequence of this project by way of photography, slides, audio-video recording etc. Nothing extra shall be payable to the contractor on this account. However cost of photographs, slides, audio/video photography etc shall be born by the department.
26. The contractor shall be fully responsible for the safe custody of materials brought by him issued to him even though the materials are under double lock key system.
27. The rate of items of flooring is inclusive of providing sunk flooring at bath rooms kitchen etc. and nothing extra on this accounts is admissible.
28. No payment shall be made to the contractor for any damage caused by rain, snowfall, floods, earthquake or any other natural causes whatsoever during execution of work. The damages of the work will be made good by the contractor at his own cost and no claim on this account shall be entertained.
29. For construction works which are likely to generate malba/rubbish to the tune of more than a tempo/truck load, contractor shall dispose of malba, rubbish & other unserviceable materials and wastes at their own cost to the notified/specified dumping ground and under no circumstances these shall be stacked/dumped, even temporarily outside the construction premises.
30. The Plinth Level of Building is to be kept as per Architectural drawings. All the items of works such as PCC, RCC, Brickwork and shuttering etc. in foundation upto this plinth level will be measured and paid as the work done upto plinth level. Nothing extra due to higher plinth will be paid and contractors rates quoted for all these items shall, therefore, be deemed to cater for extra height of plinth.

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31. Till the work is almost completed to the satisfaction of Engineer-in-charge.
- a) Contractor shall not divert any advance payments or part thereof for any other purpose other than needed for completion of the contracted work. All advance payments received as per terms of the contract (i.e. mobilization, secured against materials brought at site, secured against plant & machinery and/ or for work done during interim stages, etc) are required to be re-invested in the contracted work to ensure advance availability resources in terms of materials, labour, plant & machinery needed for required pace of progress for timely completion of work.
 - b) All running account bills preferred by the contractor for advance payments shall be processed only if Engineer-in-charge is satisfied that upto date investments (excluding security deposit & performance guarantee, which are not considered as investments) made by the contractor against contracted work are more than the payments received. Accordingly, all running account bills shall be supported with an account of up to date payments received vis-à-vis up to date investments made on the work to enable engineer-in-charge to check to his satisfaction that the payments made by engineer-in-charge are properly utilized only on the work and nowhere else.

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ADDITIONAL SPECIFICATIONS

1. The work in general shall be executed as per the description of the item, specifications and conditions attached and CPWD specifications 2019, Vol I & II with up-to-date correction slips and instructions of Engineer-in-Charge.
2. The order of preference in case of any discrepancy as indicated in condition No. 8.1 under "Conditions of Contract" give in standard CPWD contract form may be read as the following:
 - i) Nomenclature of items as per schedule of quantities.
 - ii) Additional conditions, General Conditions, Additional Specification and Particular Specifications attached with the tender document.
 - iii) Architectural/Structural Drawings.
 - iv) CPWD Specifications 2019, Vol I & II with up-to-date correction slips.
 - v) Indian standard specifications of B.I.S.
 - vi) National Building Code-2016
 - vii) Manufacturer specifications.
 - viii) Sound Engineering Practices.
 - ix) Decision of Engineer-in-charge.

A reference made to any Indian Standard specification in these documents, shall imply to the latest version of that standard. Including such revision/amendments as issued by the bureau of Indian standard upto last date of receipt of tenders. The contractor shall keep at his own cost all such publications of relevant Indian standard applicable to the work at site.

3. The proposed building is a prestigious project and quality of work is paramount importance. Contractor shall have to engage well experienced skilled labour and deploy modern T&P and other equipment to execute the work. Many items like stone masonry & stone cladding works, stone flooring, structural glazing, PVDF coating aluminium composite panel and other specialized flooring work, Wood work will specially require engagement of skilled workers having experience particularly in execution of such items.
4. Samples of all materials and fittings to be used in the work in respect of brand manufacturer and quality shall be got approved from the Engineer-in-Charge, well in advance of actual execution and shall be preserved till the completion of the work. Articles bearing BIS certifications mark shall only be used unless no manufacturer has got BIS mark for the particular material. Any material/fitting whose sample has not been approved in advance and any other unapproved material brought by the contractor shall be immediately removed as soon as directed.

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5. The rates for all items of work shall unless clearly specialized otherwise include cost of all labour, material tools and plants and other inputs involved in the execution of the item.
6. The contractor (s) shall quote all inclusive rates against the items in the schedule of quantities and nothing extra shall be payable for any of the conditions and specifications mentioned. In the tender documents unless specifically specified otherwise.
7. Unless otherwise specified in the schedule of quantities the rates for all items shall be considered as inclusive of pumping/baling out water, if necessary, for which no extra payment shall be made. Those conditions shall be considered to include water from any source such as inflow of flood, surface and sub-soil water etc. and shall apply to the execution in any season.
8. The rate for all items in which the use of cement is involved is inclusive of charges for curing.
9. The foundation trenches shall be kept free from water while works below ground level are in progress.
10. The work shall be executed and measured as per metric dimensions given in the schedule of quantities, drawings etc. (FPS units wherever indicated are for guidelines only).
11. Payment for items of "RCC work", brick work and concrete work above different floor shall be made at the rates provided for those items. For operation of these rates, the floor level shall be considered as top of the main structural slab in that floor viz. top of RCC slab in main room and not top of any sunk or depressed floor for lavatory slabs.
12. The rate of items of flooring is inclusive of providing sunk flooring in Bath-rooms, kitchen, etc. and nothing extra on this account shall be payable.
13. The SCI/CI pipes and GI pipes wherever necessary shall be fixed to RCC columns, beams etc. with rawl plugs and nothing extra shall be paid for this.
14. The contractor shall be responsible of the protection of the sanitary and water supply fittings and other fittings and fixtures against pilferage and breakage during the period of installation and thereafter until the building is handed over.

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QUALITY ASSURANCE OF THE WORK

1. The contractor shall ensure quality construction in a planned and time bound manner. Any sub-standard material/work beyond set out tolerance limit shall be summarily rejected by the Engineer-in-Charge.
2. The contractor shall ensure quality control measures on different aspects of construction including materials, workmanship and correct construction methodologies to be adopted. He shall have to submit quality assurance programme within two weeks of the award of work. The quality assurance programme should include method statement for various items of work to be executed along with check lists to enforce quality control.
3. Wherever work is specified to be done through specialized agencies, their names shall be got approved well in advance from Engineer in charge. Failure to do so shall not justify delay in execution of work. It is suggested that immediately after award of work, contractor should negotiate with concerned specialist agencies and send their names for approval to Engineer in charge within 10 days of issue of letter of award of work.
4. All materials obtained from Govt. Stores or otherwise shall be got checked by the Engineer-in-Charge or his any authorized supervisory staff on receipt of the same at site before use.
5. The contractor shall provide at his own cost suitable weighing and measuring arrangements at site for checking the weight / dimensions as may be necessary for execution of work. The sealed samples are to be handed over to the testing lab by contractor in the presence of Junior Engineer/Assistant Engineer-in-Charge of work.
6. **Samples of various materials required for testing shall be provided free of charges by the contractor. Testing charges, if any, unless otherwise specified shall be born by the contractor. All other expenditure required to be incurred for taking the samples; conveyance, packing etc. shall also be born by the contractor himself.**
7. The contractor or his authorized representative shall associate in collection, preparation, forwarding and testing of such samples. In case, he or his authorized representative is not present or does not associate him, the results or such tests and consequences thereon shall be binding on the contractor.

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8. For certain items, if frequency of tests not mentioned in the CPWD Specifications and then relevant IS code shall be followed and tests shall be carried out as per the frequency specified therein.
9. If any load testing or special testing is to be done for any sample whose strength is doubtful, the cost of the same shall also be borne by the contractor.
10. Samples of all materials and fittings to be used in the work in respect of brand manufacturer and quality shall be got approved from the Engineer-in-Charge, well in advance of actual execution and shall be preserved till the completion of the work. If a particular brand of material is specified in the item of work in Schedule of Quantity, the same shall be used after getting the same approved from Engineer-in-Charge. Wherever brand / quality of material is not specified in the item of work, the contractor shall submit the samples as per approved list of brand names given in the tender document / additional specifications for approval of Engineer-in-charge. For all other items, materials and fittings of ISI Marked shall be used with the approval of Engineer-in-Charge. Wherever ISI Marked material / fittings are not available, the contractor shall submit samples of materials / fittings manufactured by firms of repute conforming to relevant specifications or IS codes and use the same only after getting the approval.
11. To avoid delay, contractor should submit samples as stated above well in advance so as to give timely orders for procurement. If any material, even though approved by Engineer-in-Charge is found defective or not conforming to specifications shall be replaced / removed by the contractor at his own risk & cost.
12. The contractor shall get the source of all other materials, not specified elsewhere in the document, approved from the Engineer-in-Charge. The contractor shall stick to the approved source unless it is absolutely unavoidable. Any change shall be done with the prior approval of the Engineer-in-Charge for which tests etc. shall be done by the contractor at his own cost. Similarly, the contractor shall submit brand/ make of various materials not specified in the agreement, to be used for the approval of the Engineer-in-Charge along with samples and once approved, he shall stick to it.
13. The contractor shall get the water tested with regard to its suitability and conforming to the relevant IS Code. The contractor shall obtain written approval from the Engineer-in-Charge before he proceeds by using the same for execution of work. The water testing charges shall be borne by the contractor. If

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tube well water is not suitable, the contractor shall arrange Municipal water or from any other source at his own cost and nothing extra shall be paid on this account. The water shall be got tested at frequency specified in latest CPWD Specifications / BIS Code.

14. The Stone aggregate/stone, sand shall be brought from any quarries subjected to the said materials conforming to CPWD specifications.
15. The gradation of fine sand to be used in plaster work shall be strictly as per CPWD Specifications 2019, conforming to IS:1542-1977. The plastered surface shall be fairly smooth without any undulation of any kind for applying paint/white wash.
16. No concreting shall be done until the mix-design is approved by Engineer-in-charge. In case of change of source or characteristic properties of the ingredients used in the concrete mix-design during the work, a revised concrete mix-design conducted by laboratory established at site shall be submitted by the contractor as per the direction of the Engineer-in-charge. Nothing extra shall be paid on this account. No claim for extra time or delay, whatsoever, on this account shall be entertained.
17. The contractor shall submit shop drawings of staging and shuttering arrangement, aluminum work and other works as desired by Engineer-in-charge for his approval before execution. The contractor shall also submit bar bending schedule for approval of Engineer-in-charge before execution.
18. The contractor shall invariably prepare the samples of finishing items i.e. flooring of different types, external & internal finishing i/c colour scheme of paint, tiles in dado, flooring in platforms & staircase, water supply & sanitary fittings and any other item as per direction of Engineer-in-charge. The contractor shall proceed with further finishing items only after getting the samples of these items approved in writing from Engineer-in-charge. Toilets and in case of construction of residential quarters, one sample quarter complete in all shape for each category, shall be prepared by the contractor and got approved from Engineer-in-charge in writing. The contractor shall be allowed to proceed with further quarters only after getting the sample quarters approved in writing from Engineer-in-charge No extra claim whatsoever beyond the payments due at agreement rates will be entertained from the contractor on this account.
19. The construction joints shall the provided in predetermined locations only as decided by Engineer in charge. The cost of shuttering for these construction

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joints shall be included in item of Concrete work / RCC work and nothing extra shall be payable on this account to the contractor.

20. Water tanks, taps, sanitary, water supply and drainages pipes, fittings and accessories should conform to the specifications provided in bidding documents. The contractor(s) should engage approved, licensed plumbers for the work and get the materials (fixtures/fittings) tested, by the municipal Body/Corporation authorities, wherever required, at his own cost. The Contractor(s) shall submit for the approval to the Engineer-in-Charge the name of the plumbing agency proposed to be engaged by him.
21. All the hidden items such as water supply lines, drainage pipes, conduits, sewers etc. are to be properly tested as per the design conditions submitted before covering.
22. The contractor shall give a performance test of the entire installation(s) as per standing specifications before the work is finally accepted by making his own arrangements for water supply, electricity etc. and nothing extra whatsoever shall be payable for the same.
23. Door/window frames/shutters and other factory-made materials shall be procured from reputed and approved manufacturers or their authorized dealers. Decision of Engineer-in-charge in this regard shall be final and binding.
24. The contractor shall render all help and assistance in documenting the total sequence of this project by way of photography, slides, audio-video recording etc. nothing extra shall be payable to the contractor on this account. However, cost of photographs, slides, audio-video graphy etc. shall be borne by the department.
25. **Testing of Materials:**
 - (i) **Laboratory at site:** The contractor shall establish a testing lab at site and provide testing equipment and materials for the field tests mentioned in the list of mandatory tests given in CPWD Specifications 2019 Vol. 1 & 2. Nothing extra shall be payable to him on this account. The representatives of the department shall be at liberty to inspect the testing facilities at site and conduct testing at random in consultation with Engineer in charge. The contractor shall provide all necessary facilities for the purpose. The laboratory shall be equipped, inter alia, with the following equipment:

a) **Balances:**

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- i) 7 kg to 10 kg capacity, indicating semi-self-type – Accuracy 10 gm.
- ii) 500 gm capacity, semi-self-indicating type Accuracy 1 gm.
- iii) Pan Balance- 5 kg Capacity- Accuracy 10 gm.
- b) **Ovens-** Electrically operated, thermostatically controlled up to 1100C- Sensitivity 10C.
- c) **Sieves:** as per IS: 460
 - i) IS Sieves – 450 mm internal dia of sizes 100 mm, 80 mm, 63 mm, 50 mm, 40 mm, 25 mm, 20 mm, 12.5 mm, 10 mm, 6.3 mm, 4.75 mm, complete with lid and pan.
 - ii) IS Sieves – 200 mm internal dia (brass frame) consisting of 2.36 mm, 1.18 mm, 500 microns, 425 microns, 300 microns, 212 microns, 150 microns, 90 microns, 75 microns with lid and pan.
- d) Sieve shaker capable of 200 mm and 300 mm dia sieves, manually operated with timing switch assembly.
- e) Equipment for slump test- slump cone, steel plate, taping rod, steel scale, scoop.
- f) Equipment for concrete testing
 - i) Concrete cube moulds 15x15x15cm : 18Nos.
 - ii) Pruning Rods 2Kg weight length 40cm and ramming face 25mm : 1 No.
 - iii) Extra Bottom plates for 15cm cube mould : 6 Nos.
 - iv) Standard Vibration table for cubes : 1 No
 - v) Dial gauges 25 mm travel- 0.01 mm/division Least count- : 1 No.
 - vi) Compression testing machine of 100 tonne capacity. : 1 No.
- g) *Digital Vernier*
- h) *Digital micrometer screw gauge.*
- i) *Elcho meter for measuring all type of paint/powder coating thickness on all materials.*
- j) *Ultrasonic pulse velocity apparatus for testing of concrete.*
- k) *Digital distance measuring meter with range of 80 meter.*

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- l) *Digital moisture meter.*
- m) *Digital infrared temperature meter.*
- n) *Multi line rotating laser instrument of Hilti or equivalent for plumbing, aligning, leveling and squaring functions for indoor and outdoor.*
- o) *Pressures pump apparatus for testing of concealed water supply lines for leakage.*

Testing apparatus with all required chemicals for testing of water for construction including quick ready to use compact kits

Not less than 90% tests for material be performed at site lab with above stated equipment's, however at least 10% testing of materials shall be got done from external laboratories. However, for the tests to be carried out by the external laboratories, the contractor shall supply free of charge all the materials required for testing, including transportation. If the tests which were to be conducted in the site laboratory are conducted in other laboratories for any the reasons the cost of such tests shall be born by the contractor.

- (ii) **Other Laboratories:** The contractor shall arrange carrying out all tests required under the agreement through the laboratory as approved by the Engineer-in-Charge and shall bear all charges in connection therewith including charges for testing for all materials except cement for which separate condition is provided in tender document.

26. Maintenance of Site Registers:

- (i) All the Site Registers including Tests registers for tests to be carried out at construction site or in outside laboratories shall be maintained by the contractor which shall be issued to the contractor by Engineer-in-Charge in the same manner as being issued to CPWD field staff.
- (ii) The various registers to be issued to the contractor are:
 - Materials at site account register.
 - Cement register.
 - Master test registers.
 - Cube test register.
 - Paint register.
 - Inspection register.

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- Drawing register.
- (iii) All the entries in the registers will be made by the designated engineering staff of the contractor and same should be regularly reviewed by JE/AE/EE.
- (iv) All the registers of tests carried out at construction site or in outside laboratories shall be maintained by the contractor which shall be issued to the contractor by Engineer-in-Charge in the same manner as being issued to CPWD field staff.
- (v) All samples of materials including cement concrete cubes shall be taken jointly with contractor by JE and out of this at least 50% samples shall be taken in presence of AE in charge. If there is no JE, all samples of materials including cement concrete cubes shall be taken by AE jointly with contractor. All the necessary assistance shall be provided by the contractor. Cost of sampling & testing are to be borne by the contractor and he shall be responsible for safe custody of samples to be tested at site/ outside laboratory.
- (vi) All the tests in field lab at construction site shall be carried out by the Engineering staff deployed by the contractor and shall be 100% witnessed by JE and 50% of tests shall be witnessed by AE –in-Charge. At least 10% of the tests are to be witnessed by the Executive Engineer.
- (vii) Submission of copy of all test registers, material at site register along with each alternate running account bill and final bill shall be mandatory. These registers should be duly checked by division office.

27. Ultrasonic Pulse Velocity Method of Test for RCC

- (i) The under lying principle of assessing the quality of concrete is that comparatively higher velocities are obtained when the quality of concrete in terms of density, homogeneity and uniformly is good. The consistency of the concrete as regards its general quality gets established. In case of poorer quality lower velocities are obtained. If there are cracks, voids or flaws inside the concrete which come in the way of transmission of pulse, lower velocities are obtained.
- (ii) **Velocity criterion for Concrete Quality Grading:** The quality of concrete in terms of uniformity, incidence or absence of internal flaws, cracks and segregation etc. indicative of the level of workmanship

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employed, can thus be assessed using the guidance given in table below, which have been evolved for characterizing the quality concrete in structure in term of the ultrasonic pulse velocity.

S. No.	Pulse Velocity by Cross	Concrete Quality
	Probing (Km/Sec.)	Grading
1	Above 4.5	Excellent
2	4.5 to 3.5	Good
3	3.5 to 3.0	Medium
4	Below 3.0	Doubtful

- Note:** In Case of “doubtful” quality it may be necessary to carry further tests
- (iii) Pulse velocity method of test of concrete is to be conducted for CPWD works as a routine test. The acceptance criteria as per the above table will be applicable which is as per IS 13311 (part-1): 1992. From the above “Good” and “Excellent” grading are acceptable and below these grading the concrete will not be acceptable
- (iv) 5% of the total number of RCC members in each category i.e. beam, column, slab and footing may be tested by UPV test method for establishing quality of concrete. It is suggested that test be conducted on RCC beam near joint with column, on RCC column near joint with beam, on RCC footings and rafts. On RCC rafts a suitable grid can be worked out for determining number of tests. In addition, doubtful areas such as honeycombed locations, locations, where continuous seepage is observed, construction joints and visible loose pockets will also be tested.
- (v) The test results are to be examined in view of the above acceptance criteria “Good” and “Excellent” and wherever concrete is found with less than required quality as per acceptance criteria, repairs to concrete will be made. Honeycombed areas and loose pockets will be repaired by grouting using Portland Cement Mortar/Polymer Modifies Cement Mortar /Epoxy Mortaretc. After chipping loose concrete in appropriate manner. In areas where concrete is found below acceptance criteria and defects are not apparently visible on surface, injecting approved grout in appropriate proportion using epoxy grout /acrylic Polymer modified cements slurry made with shrinkage compensating cement / plain cement slurry etc. will be resorted to for repairs. (refer relevant chapters from CPWD Hand Book on Repairs and Rehabilitation of RCC Buildings). Repair to concrete will be done till satisfactory results are obtained as per the acceptance criteria by retesting of the repaired area. If satisfactory results are not obtained dismantling and relaying of concrete will be done.

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28. Removal of Rejected / Sub-standard Materials:

The following procedure should be followed in case of removal of rejected/sub-standard materials from the site of work.

- Whenever any material brought by the contractor to the site of work is rejected, entry thereof should invariably be made in the site order book under the signature of the AE/AEE giving approximate quantity of such materials.
- As soon as the material is removed, a certificate to that effect may be recorded by the AE/AEE against the original entry, giving the date of removal a mode of removal i.e. whether by truck, carts or by manual labour. If removal is by truck, the registration number of the truck should be recorded.
- When it is not possible for the AE/AEE to be present at the site of work at the time of actual removal of the rejected/sub-standard materials from the site the required certificate should be recorded by the Junior Engineer and the AE/AEE should countersign the certificate recorded by the Junior Engineer.

29. QUALITY & QUARRIES OF STONE AGGREGATE & SAND

The Stone aggregate/stone shall be brought from quarries as approved by the Engineer-in-charge, as shown below against each or any other quarry. The materials shall however confirm to CPWD specifications.

Stone/Sand	From
Stone aggregate 40mm nominal size and above	Material to be obtained from the Bonafide Quarries and matching to the CPWD Specifications and terms and conditions of this agreement as per direction of Engineer-in-charge.
Stone aggregate 20mm nominal size and below	 do
Stone for soling	 do
Coarse sand of fineness modulus 2.5 to 3.5 and confirming to CPWD specifications 2019.	 do

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Fine sand of fineness modulus 1.2 to 1.6 and confirming to CPWD specifications 2019.	 do
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30. The contractor shall provide approved type of supports for maintaining the bars in position and ensuring required spacing and correct cover of concrete to reinforcement as called for in the drawings. Spacer blocks of required shape and size, MS chairs and spacer bars shall be used in order to ensure accurate positioning of reinforcement. Spacer blocks shall be cast well in advance with approved proprietary pre-packed free flowing mortars (conbextra as manufactured by M/s Fosroc Chemicals India Ltd. or approved equivalent) of high early strength. Blocks of polymer shall not be used as spacer blocks unless specially approved by the Engineer-in-Charge. Rate of item of steel reinforcement is inclusive of cost of such cover blocks.

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SPECIAL CONDITIONS**1. SPECIAL CONDITIONS FOR CEMENT:**

1.1 The contractor shall procure Portland Pozzolana Cement (PPC) conforming to IS: 1489 (Part-1) / 43 grade ordinary Portland Cement (OPC) conforming to IS: 269 as required in the work from reputed manufacturers of cement as mentioned in "List showing preferred brands / Manufacturers / Makes" or from any other reputed cement manufacture having a production capacity not less than 1 million Ton per annum as approved by CE, CPWD, Dehradun or his successor. The cement of approved make as aforesaid in 50 kg. bags bearing manufacturer's name and ISI marking, along with manufacturers test certificate for each lot shall be procured by the contractor. However, if the contractor uses higher grade of cement or uses OPC only nothing extra shall be paid. The use of PPC shall be regulated as per the following conditions.

- a) IS:456-2000 Code of Practice for Plain and Reinforced Concrete (as amended upto date) shall be followed in regard to Concrete Mix Portion and its production as under:
 - (i) The concrete mix design shall be done as "Design Mix Concrete" as prescribed in clause-9 of IS 456 mentioned above.
 - (ii) Concrete shall be manufactured in accordance with clause 10 of above mentioned IS:456 covering quality assurance measures both technical and organizational, which shall also necessarily require a qualified Concrete Technologist to be available during manufacture of concrete for certification of quality of concrete.
- b) The mechanical properties such as modulus of elasticity, tensile strength, creep and shrinkage of flyash mixed concrete or concrete using flyash blended cements (PPCs) are not likely to be significantly different and their values are to be taken same as those used for concrete made with OPC.
- c) To control higher rate of carbonation in early ages of concrete both in flyash admixed as well as PPC based concrete,

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water/binder ratio shall be kept as low as possible, which shall be closely monitored during concrete production.

If necessitated due to low water/binder ratio, required workability shall be achieved by use of chloride free chemical admixtures conforming to IS:9103. The compatibility of chemical admixtures and super plasticizers with each set OPC, fly ash and /or PPC received from different sources shall be ensured by trails.

- d) In environment subjected to aggressive chloride or sulphate attack in particular, use of flyash admixed or PPC based concrete is recommended. In case, where structural concrete is exposed to excessive magnesium sulphate, flyash substitution/content shall be limited to 18% by weight. Special type of cement with low C3A content may also be alternatively used. Durability criteria like minimum binder content and maximum water/binder ratio also need to be given due consideration in such environment.
- e) Wet curing period shall be enhanced to a minimum of 10 days or its equivalent. In hot & arid regions, the minimum curing period shall be 14 days or its equivalent.
- f) Subject to General Guidelines detailed out as above, PPC manufactured conforming to IS:1489 (Part-I) shall be treated at par with OPC for manufacture of Design Mix Concrete for structural use in RCC.
- g) Till the time, BIS makes it mandatory to print the %age of flyash on each bag of cement, the certificate from the PPC manufacturer indicating the same shall be insisted upon before allowing use of such cements in works.
- h) While using PPC for structural concrete work, no further admixing of flyash shall be permitted

- 1.2 Supply of cement shall be made in 50 kg. bags bearing manufacturer's name and ISI marking. Samples of cement arranged by the contractor shall be taken by the Engineer-in-Charge and got tested in accordance with provisions of the relevant BIS codes. In case the test results indicate that the cement arranged by the contractor does not confirm to the relevant BIS code the same shall stand rejected and shall be removed

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from the site by the contractor at his own cost within a week's time of written order from the Engineer-in-charge to do so.

- 1.3 The cement shall be brought at site in bulk supply of approximately **50 tonnes** or as decided by the Engineer-in-Charge.
- 1.4 The cement godown of the capacity to store a minimum of **2000 bags** of cement shall be constructed by the contractor at site of work for which no extra payment shall be made.
- 1.5 Double lock provision shall be made to the door of the cement godown. The keys of one lock shall remain with the Engineer-in-charge or his authorized representative and the key of the other lock shall remain with the contractor. The contractor shall be responsible for the watch and ward and safety of the cement godown. The contractor shall facilitate the inspection of the cement godown by the Engineer-in-Charge at any time.
- 1.6 The cement shall be got tested by the Engineer-in-Charge and shall be used on the work only after satisfactory test results have been received. The contractor shall supply free of charge the cement required for testing including its transportation cost to test laboratories. *The cost of tests shall be borne by the contractor.*
- 1.7 The actual issue and consumption of cement on work shall be regulated and proper accounts maintained as provided in clause 10A of the contract. The theoretical consumption of cement shall be worked out as per procedure prescribed in clause 38 of the contract and shall be governed by conditions laid therein. In case the cement consumption is less than theoretical consumption including permissible variation, recovery at the rate show prescribed shall be made after ensuring structural soundness and stability on the basis of testing. In case of excess consumption, no adjustment need to made.
- 1.8 The cement brought to site and the cement remaining unused after completion of the work shall not be removed from site without the written permission of the Engineer-in-Charge.
- 1.9 The damaged cement shall be removed from the site immediately by the contractor on receipt of a notice in writing from the Engineer-in-Charge. If he does not do show within three days of receipt of such notice, the Engineer-in-Charge shall get it removed at the cost of the contractor.

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- 1.10 Cement bags shall be stored in separate godowns to be constructed by the contractor at his own cost as per sketch (which is only indicative and actual size will depend on the site requirements) given in CPWD specifications with weather proof roofs and walls. Each godown shall be provided with a single shutter door with two locks. The key of one lock shall remain with Engineer-in-charge or his authorized representative and that of the other lock with the authorized agent of the contractor at the site of work so that the cement is issued from the godown according to the daily requirements with the knowledge of both parties and proper account for the same is maintained in the standard proforma.
- 1.11 Separate cement registers showing the receipt of the OPC and PPC shall be maintained at site. The contractor shall construct separate godowns for storage of OPC & PPC at site and nothing extra on this account shall be payable.
- 1.12 In order to have an effective control over the issue of cement, the following drill should be observed:
- (i) The cement godown(s) should be properly and effectively double locked, keys of one of the locks remaining with the department and that of the other with the contractor.
 - (ii) The pages of the Cement Register should be as per Annexure-30 of CPWD Works Manual for SOP-2019, machine numbered and each page initialed by the Executive Engineer.
 - (iii) Periodical checking of cement godown: The cement godown and the Cement Register should be checked by the Assistant Engineer/Executive Engineer in-charge of the work as per following schedule:
 - (a) At least weekly or fortnightly, respectively, in case of works at the headquarters of the Assistant Engineer/Executive Engineer.
 - (b) Whenever they visit the site of work in case of works that are located outside the Sub-Divisional/Divisional headquarters.

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2 SPECIAL CONDITIONS FOR STEEL REINFORCEMENT BARS:

- 2.1 For reinforced cement concrete or pre-stressed concrete works, the reinforcement bars shall consist of the following grades conforming to IS1786 : 2008 (Indian Standard specification for high strength deformed steel bars and wires for concrete reinforcement): Fe 415, Fe 415D, DE 415S, Fe500D, De 500S and Fe 550D.
- 2.2 The contractor shall obtain manufacture's certificate stating the process of manufacture, chemical composition and test sheet giving result of each mechanical test applicable to the material purchased and submit it to the Engineer-in-charge. Each test certificate shall indicate the number of the cast to which it applies, corresponding to the number or identification mark to be found on the material.
- 2.3 The Engineer-in-Charge shall get each consigning tested for both chemical composition and physical properties (including bend and re-bend test) as specified in IS:1786 from NABL accredited laboratory or any Government laboratory.
- 2.4 Reinforcement bar is one of the most important building materials. Its quality directly affects the life span, resistance to earthquake and durability of the structure. In order to ensure consistent quality of rebars, the NIT approving authority may prescribe reputed brands like SAIL, RINL, Tata, JSW, JSPL in the list of preferred makes. Only corrosion resistant steel rebars shall be used. The contractor shall have to obtain and furnish test certificates to the Engineer-in-charge in respect of all supplies of steel brought by him to the site of work.
- 2.5 Samples shall also be taken and got tested by the Engineer-in-Charge as per the provisions in this regard in relevant BIS codes. In case the test results indicate that the steel arranged by the contractor does not conform to the specifications, the same shall stand rejected, and it shall be removed from the site of work by the contractor at his cost within a week time or written orders from the Engineer-in-Charge to do so.
- 2.6 The steel reinforcement bars shall be brought to the site in bulk supply of **50 tonnes** or more as decided by the Engineer-in-charge.
- 2.7 The steel reinforcement shall be stored by the contractor at site of work in such a way as to prevent distortion and corrosion and nothing extra shall

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be paid on this account. Bars of different sizes and lengths shall be stored separately to facilitate easy counting and checking.

- 2.8 For checking nominal mass, tensile strength, bend test, re-bend test etc. specimen of sufficient length shall be cut from each size of the bar at random at frequency not less than that specified below:

Size of bar	For consignment below 100 tonnes	For consignment above 100 tonnes
Under 10mm dia. bars	One sample for each 25 tonnes or part thereof	One sample for each 40 tonnes or part thereof
10mm to 16mm dia bars	One sample for each 35 tonnes or part thereof	One sample for each 45 tonnes or part thereof
Over 16mm dia bars	One sample for each 45 tonnes or part thereof	One sample for each 50 tonnes or part thereof

- 2.9 The contractor shall supply free of charge the steel required for testing including transportation to testing laboratories. The cost of tests shall be borne by the contractor.
- 2.10 The actual issue and consumption of steel on work shall be regulated and proper accounts maintained as provided in clause 10A of the contract. The theoretical consumption of steel shall be worked out as procedure prescribed in clause 38 of the contract and shall be governed by conditions laid therein. In case the consumption is less than theoretical consumption including permissible variations recovery at the rate so prescribed shall be made. In case of excess consumption, no adjustment need to be made.
- 2.11 Steel brought to site and steel remaining unused shall not be removed from site without the written permission of the Engineer-in-charge.
- 2.12 Steel bars brought by the contractor for use in the work shall be got checked from the Engineer-in-Charge or his authorized representative of the work on receipt of the same at site before use.
- 2.13 In case the contractor brings surplus quantity of steel the same after completion of the work will be removed from the site by the contractor at his own cost after approval of the Engineer-in-Charge.

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- 2.14 Reinforcement including authorized spacer bars and overlaps shall be measured in length of different diameters, as actually (not more than as specified in the drawing) used in the work, nearest to a centimeter. Wastage and unauthorized overlaps shall not be measured.
- 2.15 The standard sectional weights referred to as in Table 5.4 under para 5.3.4 in CPWD Specifications for works 2019 Vol. 1 will be considered for conversion of length of various sizes of MS bars, Tor steel bars and TMT bars into standard weight.
- 2.16 Records of actual sectional weight shall also be kept dia-wise & lot-wise. The average sectional weight for each diameter shall be arrived at from samples from each lot of steel received at site. The decision of the Engineer-in-Charge shall be final for the procedure to be followed for determining the average sectional weight of each lot. Quantity of each diameter of steel received at site of work each day will constitute one single lot for the purpose. The weight of steel by conversion of length of various sizes of bars based on the actual weighted average sectional weight shall be termed as derived actual weight.
- 2.17 If the derived weight as in para 2.16 above is lesser than the standard weight as in para 2.15 above, the derived actual weight shall be taken for payment. If the derived actual weight is found more than the standard weight then the standard weight as para 2.15 above shall be taken for payment. In such case nothing extra shall be paid for the difference between the derived actual weight and the standard weight.
- 2.18 Mixing of different type of steel/different grades of steel shall not be allowed in the same structural members as main reinforcement to satisfy clause 26.1 of IS:456.
- 2.19 Tolerances on Nominal Mass (individual sample) shall be as per table 5.4A of CPWD Specification-2019, Vol-I.

3 SPECIAL CONDITIONS FOR PREVENTION OF AIR POLLUTION AS PER DIRECTIVES OF NATIONAL GREEN TRIBUNAL (NGT):

- 3.1 The contractor shall not store/dump construction material or debris on metalled road.
- 3.2 The contractor shall get prior approval from Engineer-in-charge for the area where the construction material or debris can be stored beyond the

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metalled road. This area shall not cause any obstruction to the free flow of traffic/inconvenience to the pedestrians. It should be ensured by the contractor that no accidents occur on account of such permissible storage.

- 3.3 The contractor shall take appropriate protection measures like raising wind breakers of appropriate height on all sides of the plot/area using CGI sheets or plastic and/or other similar material to ensure that no construction material dust fly outside the plot area.
- 3.4 The contractor shall ensure that all the trucks or vehicles of any kind which are used for construction purpose or are carrying construction material like cement, sand and other allied materials are fully covered. The contractor shall take all necessary precautions that the vehicles are properly cleaned and dust free to ensure that enroute their destination, the dust, sand or any other particles are not released in air / contaminate air.
- 3.5 The contractor shall provide mask to every worker working on the construction site and involved in loading, unloading and carriage of construction material and construction debris to prevent inhalation of dust particles.
- 3.6 The contractor shall provide all medical help, investigation and treatment to the workers involved in the construction of building and carry of construction material and debris relatable to dust emission.
- 3.7 The contractor shall ensure that C&D waste is transported to the C&D waste site only and due record shall be maintained by the contractor.
- 3.8 The contractor shall ensure compulsory use of wet jet in grinding and stone cutting.
- 3.9 The contractor shall comply all the preventive and protective environmental steps as stated in the MoEF guidelines, 2010.
- 3.10 The contractor shall carry out on road inspection for black smoke generating machinery. The contractor shall use cleaner fuel.
- 3.11 The contractor shall ensure that all DG sets comply emission norms notified by MoEF.
- 3.12 The contractor shall use vehicles having pollution under control certificate. The emissions can be reduced by a large extent by reducing

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the speed of a vehicle to 20 KMPH. Speed bumps shall be used to ensure speed reduction. In cases where speed reduction cannot effectively reduce fugitive dust, the contractor shall divert traffic to nearby paved areas.

3.13 The contractor shall ensure that the construction material is covered by tarpaulin. The contractor shall take all other precaution to ensure that no dust particles are permitted to pollute air quality as a result of such storage.

3.14 The paving of the path for plying of vehicles carrying construction material is more permanent solution to dust control and suitable for longer duration projects.

4 CONDITIONS OF CONTRACT SPECIFIC TO GREEN BUILDING PRACTICE

The contractor shall strictly adhere to the following conditions as part of his contractual obligations:-

4.1 SITE

4.1.1 The contractor shall ensure that all the top soil excavated during construction works is neatly stacked and is not mixed with other excavated earth. The contractors shall take the clearance of the Engineer in Charge before any excavation. Top soil should be stripped to a depth of 20 cm (centimeters) from the areas to be disturbed, for example proposed area for buildings, roads, paved areas, external services and area required for construction activities etc. It shall be stacked designated areas, covered or stabilized with temporary seeding for erosion prevention and shall be reapplied to site during plantation, landscaping etc. of the proposed vegetation. Top soil shall be separated from subsoil, debris and stones larger than 50 mm diameter. The stored top soil may be used as finished grade for planting areas.

4.1.2 The Contractor should follow the construction plan as proposed by the Architect / Engineer in Charge to minimize the site disturbance such as soil pollution due to spilling. Use staging and spill prevention and control plan to restrict the spilling of the contaminating material on site. Protect top soil from erosion by collection storage and reapplication of top soil, constructing sediment basin, contour trenching, mulching etc.

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- 4.1.3 No excavated earth shall be removed from the campus unless suggested by Engineer in Charge. All subsoil shall be reused in backfilling/landscape, etc as per the instructions of the Engineer in Charge.
- 4.1.4 The contractor shall not change the natural gradient of the ground unless specifically instructed by the Engineer in Charge. This shall cover all natural features like water bodies, drainage gullies, slopes, mounds, depressions, etc. Existing drainage patterns through or into any preservation area shall not be modified unless specifically directed by the Engineer-in-charge.
- 4.1.5 The contractor shall not carry out any work which results in the blockage of natural drainage.
- 4.1.6 The contractor shall ensure that existing grades of soil shall be maintained around existing vegetation and lowering or raising the levels around the vegetation is not allowed unless specifically directed by the Engineer-in-charge
- 4.1.7 Contractor shall reduce pollution and land development impacts from automobiles use during construction.
- 4.1.8 Overloading of trucks is unlawful and creates the erosion and sedimentation problems, especially when loose materials like stone dust, excavated earth, sand etc. are moved. Proper covering must take place. No overloading shall be permitted.
- 4.1.9 The dismantle material/building rubbish received from dismantling/ demolishing shall be dumped to the dumping ground in properly covered truck with precaution. Agency shall submit the hard copy of photograph showing the properly covered truck disposing the dismantles material/building rubbish. Failure of which shall be sternly dealt and a penalty @ Rs.500/- per trip of truck shall be levied and the decision of Engineer-in-Charge shall be final & binding.
- 4.1.10 Agency/contractor shall not dump the construction material on the metaled road and shall keep the construction material on the physically demarcated space by the Engineer-in- Charge.

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4.1.11 All the building material responsible for pollution shall be brought at site from sources covered by tarpaulin and shall take all precautionary measure to ensure that no dust particles are permitted to pollute the air quality, failure of which Agency shall be liable to pay damages as decided by Engineer-in-Charge. The decision of Engineer-in-Charge shall be final & binding.

4.1.12 There shall be no burning of leaves, plastic etc. at construction site.

4.2 CONSTRUCTION PHASE AND WORKER FACILITIES

4.2.1 The contractor shall specify and limit construction activity in preplanned/ designated areas and shall start construction work after securing the approval for the same from the Engineer in Charge. This shall include areas of construction, storage of materials, and material and personnel movement.

4.3 PRESERVE AND PROTECT LANDSCAPE DURING CONSTRUCTION

4.3.1 The contractor shall ensure that no trees, existing or otherwise, shall be harmed and damage to roots should be prevented during trenching, placing backfill, driving or parking heavy equipment, dumping of trash, oil, paint, and other materials detrimental to plant health. These activities should be restricted to the areas outside of the canopy of the tree, or, from a safe distance from the tree/plant by means of barricading. Trees will not be used for support; their trunks shall not be damaged by cutting and carving or by nailing posters, advertisements or other material. Lighting of fires or carrying out heat or gas emitting construction activity within the ground, covered by canopy of the tree is not to be permitted.

4.3.2 The contractor shall take steps to protect trees or saplings identified for preservation within the construction site using tree guards of approved specification.

4.3.3 Contractor should limit all construction activity within the specified area as per the Construction Management Plan (CMP) approved by Engineer in Charge.

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- 4.3.4 The contractor shall avoid cut and fill in the root zones, through delineating and fencing the drip line (the spread limit of a canopy projected on the ground) of all the trees or group of trees. Separate the zones of movement of heavy equipment, parking, or excessive foot traffic from the fenced plant protection zones.
- 4.3.5 The contractor shall ensure that maintenance activities during construction period shall be performed as needed to ensure that the vegetation remains healthy.
- 4.3.6 Contractor shall be required to develop and implement a waste management plan, quantifying material diversion goals. He shall establish goals for diversion from disposal in landfills and incinerators and adopt a construction waste management plan to achieve these goals. A project-wide policy of nothing leaves the Site, should be followed, in such a case when strictly followed, care would automatically be taken in ordering and timing of materials such that excess does not become waste. The Contractor ingenuity is especially called towards meeting this prerequisite/ credit (as per 3 star rating of GRIHA/ GHAR CPWD). Consider recycling cardboard, metal, brick, acoustical tile, concrete, plastic, clean wood, glass, gypsum wallboard, carpet and insulation. Designate a specific area(s) on the construction site for segregated or commingled collection of recyclable material, and track recycling efforts throughout the construction process. Identify construction haulers and recyclers to handle the designated materials. The diversion may include donation of materials to charitable organizations and salvage of materials on-site.
- 4.3.7 Contractor shall collect all construction waste generated on site. Segregate these wastes based on their utility and examine means of sending such waste to manufacturing units which use them as raw material or other site which require it for specific purpose. Typical construction debris could be broken bricks, steel bars, broken tiles, spilled concrete and mortar etc.
- 4.3.8 The contractor shall provide potable water for all workers.

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4.3.9 The contractor shall provide the minimum level of sanitation and safety facilities for the workers at their camp/labour site. The contractor shall ensure cleanliness of workplace with regard to the disposal of waste and effluent; provide clean drinking water and latrines and urinals as per applicable standard. Adequate toilet facilities shall be provided for the workman within easy access of their place of work. The total no. to be provided shall not be less than 1 per 30 employees in any one shift. Toilet facilities shall be provided from the start of building operations, connection to a sewer shall be made as soon as practicable. Every toilet shall be so constructed that the occupant is sheltered from view and protected from the weather and falling objects. Toilet facilities shall be maintained in a sanitary condition. A sufficient quantity of disinfectant shall be provided. Natural or artificial illumination shall be provided. **Agency has to ensure all preventive measures to prevent the spread of COVID-19 and nothing extra shall be payable in this account.**

4.3.10 The contractor shall ensure that air pollution due to dust/generators is kept to a minimum, preventing any adverse effects on the workers and other people in and around the site. The contractor shall ensure proper screening, covering stockpiles, covering brick and loads of dusty materials, wheel-washing facility, gravel pit, and water spraying. Contractor shall ensure the following activities to prevent air pollution during construction:

- (i) Clear vegetation only from areas where work will start right away
- (ii) Vegetate / mulch areas where vehicles do not ply.
- (iii) Apply gravel / landscaping rock to the areas where mulching / paving is impractical
- (iv) Identify roads on-site that would be used for vehicular traffic. Upgrade vehicular roads (if these are unpaved) by increasing the surface strength by improving particle size, shape and mineral types that make up the surface & base. Add surface gravel to reduce source of dust emission.

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Limit amount of fine particles (smaller than 0.075mm) to 10 - 20%

- (v) Water spray, through a simple hose for small projects, to keep dust under control. Fine mists should be used to control fine particulate. However, this should be done with care so as not to waste water. Heavy watering can also create mud, which when tracked onto paved public roadways, must be promptly removed. Also, there must be an adequate supply of clean water nearby to ensure that spray nozzles don't get plugged.
- (vi) Water spraying shall be done on:
 - (a) Any dusty materials before transferring, loading and unloading
 - (b) Area where demolition work is being carried out
 - (c) Any un-paved main haul road
 - (d) Areas where excavation or earth moving activities are to be carried out
- (vii) The contractor shall ensure that the speed of vehicles within the site is limited to 10 km/hr.
- (viii) All material storages should be adequately covered and contained so that they are not exposed to situations where winds on site could lead to dust / particulate emissions.
- (ix) Spills of dirt or dusty materials will be cleaned up promptly so the spilled material does not become a source of fugitive dust and also to prevent of seepage of pollutant laden water into the ground aquifers. When cleaning up the spill, ensure that the clean-up process does not generate additional dust. Similarly, spilled concrete slurries or liquid wastes should be contained / leaned up immediately before they can infiltrate into the soil / ground or runoff in nearby areas
- (x) Provide barricading of not less than 6.00 metre or higher as per direction of Engineer-in- charge, along the site

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boundary, next to a road, around batching plant or other public area. Nothing extra shall be paid on this account.

- (xi) Provide dust screens, sheeting or netting to scaffold along the perimeter of the building
- (xii) Cover stockpiles of dusty material with impervious sheeting
- (xiii) Cover dusty load on vehicles by impervious sheeting before they leave the site.

4.3.11 Contractor shall be required to provide an easily accessible area that serves the entire building and is dedicated to the separation, collection and storage of materials for recycling including (at a minimum) paper, corrugated cardboard, glass, plastics, and metals.

4.3.12 The contractor shall ensure that no construction leachate (e.g. cement slurry etc.), is allowed to percolate into the ground. Adequate precautions are to be taken to safeguard against this including, reduction of wasteful curing processes, collection, basic filtering and reuse. The contractor shall follow requisite measures for collecting drainage water run-off from construction areas and material storage sites and diverting water flow away from such polluted areas. Temporary drainage channels, perimeter dike/swale, etc. shall be constructed to carry the pollutant-laden water directly to the treatment device or facility (municipal sewer line).

4.3.13 Staging (dividing a construction area into two or more areas to minimize the area of soil that will be exposed at any given time) should be done to separate undisturbed land from land disturbed by construction activity and material storage.

4.3.14 The contractor shall comply with the safety procedures, norms and guidelines (as applicable) as outlined in the document Part 7 Constructional practices and safety, National Building code -2016 Bureau of Indian Standards. A copy of all pertinent regulations and notices concerning accidents, injury and first-aid shall be prominently exhibited at the work site. Depending upon the scope & nature of work, a person qualified

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in first-aid shall be available at work site to render and direct first-aid to casualties. A telephone may be provided to first-aid assistant with telephone numbers of the hospitals displayed. Complete reports of all accidents and action taken thereon shall be forwarded to the competent authorities.

4.3.15 The contractor shall ensure the following activities for construction workers safety, among other measures:

- (i) Guarding all parts of dangerous machinery.
- (ii) Precautionary signs for working on machinery
- (iii) Maintaining hoists and lifts, lifting machines, chains, ropes, and other lifting tackles in good condition.
- (iv) Durable and reusable formwork systems to replace timber formwork and ensure that formwork where used is properly maintained.
- (v) Ensuring that walking surfaces or boards at height are of sound construction and are provided with safety rails or belts.
- (vi) Provide protective equipment; helmets etc.
- (vii) Provide measures to prevent fires. Fire extinguishers and buckets of sand to be provided in the fire-prone area and elsewhere.
- (viii) Provide sufficient and suitable light for working during night time.

4.3.16 The storage of material shall be as per standard good practices as specified in Part 7, Section 3. Storage, stacking and Handling practices, NBC 2016 and shall be to the satisfaction of the Engineer in Charge to ensure minimum wastage and to prevent any misuse, damage, inconvenience or accident. Watch and ward of the Contractor's materials shall be his own responsibility. There should be a proper planning of the layout for stacking and storage of different materials, components and equipments with proper access and proper maneuverability of the vehicles carrying the materials. While planning the layout, the requirements of various materials,

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components and equipments at different stages of construction shall be considered.

- 4.3.17 The contractor shall provide for adequate number of garbage bins around the construction site and the workers facilities and will be responsible for the proper utilisation of these bins for any solid waste generated during the construction. The contractor shall ensure that the site and the workers facilities are kept litter free. Separate bins should be provided for plastic, glass, metal, biological and paper waste and labelled in both Hindi and English with suitable symbols.
- 4.3.18 The contractor shall prepare and submit spill prevention and control plans before the start of construction, clearly stating measures to stop the source of the spill, to contain the spill, to dispose the contaminated material and hazardous wastes, and stating designation of personnel trained to prevent and control spills. Hazardous wastes include pesticides, paints, cleaners, and petroleum products.
- 4.3.19 Contractor shall collect & submit the relevant material certificates for materials with high recycled (both post-industrial and post-consumer) content, including materials like RMC mix with fly-ash, glass with recycled content, calcium silicate boards etc.
- 4.3.20 Contractor shall collect the relevant material certificates for rapidly renewable materials such as bamboo, wool, cotton insulation, agrifiber, linoleum, wheat board, strawboard and cork etc.
- 4.3.21 Where possible, the contractor shall select materials/vendors, harvested and manufactured regionally.
- 4.3.22 Contractor shall adopt an IAQ (Indoor Air Quality) management plan to protect the HVAC system during construction, control pollutant sources, and interrupt pathways for contamination. He shall sequence installation of materials to avoid contamination of absorptive materials such as insulation, carpeting, ceiling tile, and gypsum wallboard. He shall also protect stored on-site or installed absorptive materials from moisture damage.
- 4.3.23 The contractor shall ensure that neat and deep cleaning of all internal spaces is conducted prior to handover. This shall

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comprise an opening of all doors and windows for 14 days to vent out any toxic fumes due to paints, varnishes, polishes, etc.

4.3.24 Contractor shall make efforts to reduce the quantity of indoor air contaminants that are odorous or potentially irritating harmful to the comfort and well-being of installer and building occupants. Contractor shall ensure that the VOC (Volatile Organic Compounds) content of paints, coatings and primers used must not exceed the VOC content limits mentioned below:

Paints

Non-flat - 150 g/L Flat (Mat) - 50 g/L

Anti corrosive/ anti rust - 250 g/L

Coatings / Clear wood finishes

Varnish - 350 g/L Lacquer - 550 g/L

Floor coatings - 100 g/L Stains - 250 g/L

Sealers

Waterproofing sealer - 250 g/L Sanding sealer - 275 g/L

Other sealers - 200 g/L

The VOC (Volatile Organic Compounds) content of adhesives and sealants used must be less than VOC content limits mentioned:

Architectural Applications VOC Limit (g/l less water)

Indoor Carpet adhesives - 50 g/L Carpet Pad Adhesives - 50 g/L

Wood Flooring Adhesive - 100 g/L Rubber Floor Adhesives - 60 g/L Sub Floor Adhesives . 50 g/L Ceramic Tile Adhesives - 65 g/L

VCT and Asphalt Tile adhesives - 50 g/L DryWall and Panel Adhesives - 50 g/L Structural Glazing Adhesives - 100 g/L Multipurpose Construction Adhesives . 70 g/L

Substrate Specific Application VOC Limit (g/l less water) Metal to Metal - 30 g/L

Plastic Foams - 50 g/L

Porous material (except wood) - 50 g/L Wood - 30 g/L

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Fiber Glass . 80 g/L

4.3.25 Wherever required, Contractor shall meet and carry out documentation of all activities on site, supplementation of information, and submittals in accordance with IGBC LEED India New Construction v1.0 & GRIHA / or equivalent of GHAR, CPWD program standards and guidelines. Towards meeting the aforementioned building environmental rating standard(s) expert assistance shall be provided to him up on request.

4.3.26 Water Use during Construction

Contractor should spray curing water on concrete structure and shall not allow free flow of water. Concrete structures should be kept covered with thick cloth/gunny bags and water should be sprayed on them. Contractor shall do water ponding on all sunken slabs using cement and sand mortar bunds.

4.3.27 The Contractor shall remove from site all rubbish and debris generated by the Works and keep Works clean and tidy throughout the Contract Period. All the serviceable and non-serviceable (malba) material shall be segregated and stored separately. The malba obtained during construction shall be collected in well formed heaps at properly selected places, keeping in a view safe condition for workmen in the area. Materials which are likely to cause dust nuisance or undue environmental pollution in any other way, shall be removed from the site at the earliest and till then they shall be suitably covered. Glass & steel should be dumped or buried separately to prevent injury. The work of removal of debris should be carried out during day. In case of poor visibility artificial light may be provided.

4.3.28 The contractor shall provide O & M Manuals wherever applicable.

4.4 MATERIALS & FIXTURES FOR THE PROJECT

4.4.1 Contractor will produce wherever feasible certificate regarding distance of the source of the relevant material .

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- (a) Unless otherwise stated cement used at site for reinforced concrete, precast members, mortar, plaster, building blocks, etc shall be PPC.
- (b) The contractor shall ensure that all paints, polishes, adhesives and sealants used both internally and externally, on any surface, shall be Low VOC products. The contractor shall get prior approval from the Engineer in Charge before the application of any such material.
- (c) All plumbing and sanitary fixtures installed shall be as per the direction of the Engineer in Charge and shall adhere to the minimum LPM (litres per minute) and LPF (litres per flush) mentioned.

4.5 RESOURCES CONSUMED DURING CONSTRUCTION

- (a) The contractor shall ensure that the water and electricity is not wasted during construction. The Engineer in Charge can bring to the attention any such wastage and the contractor will have to ensure that such bad practices are corrected.
- (b) The contractor shall ensure that all run-off water from the site, during construction is collected and reused to the maximum.
- (c) The contractor shall use treated recycled water of appropriate quality standards for construction, if available.

4.6 CONSTRUCTION WASTE

- (a) All construction debris generated during construction shall be carefully segregated and stored in a demarcated waste yard. Clear, identifiable areas shall be provided for each waste type. Employ measures to segregate the waste on site into inert, chemical, or hazardous wastes.
- (b) No construction debris shall be taken away from the site, without the prior approval of the Engineer in Charge.
- (c) The contractor shall recycle the unused chemical/hazardous wastes such as oil, paint, batteries, and asbestos.
- (d) If and when construction debris is taken out of the site, after prior permissions from the Engineer in Charge, then the contractor

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shall ensure the safe disposal of all wastes and will only dispose of any such construction waste in approved dumping sites.

4.7 Documentation:

- a) Digital photo documentation to demonstrate compliance of safety guidelines.
- b) The contractor shall, during the entire tenure of the construction phase, submit the following records to the Engineer in Charge on daily basis:
 - i) Quantities of material brought into the site, including the material issued to the contractor by the Engineer in charge.
 - ii) Inventories of materials used in the work i/c. flyash, flyash bricks etc.
 - iii) Digital photographs of the works at site, the workers facilities, the waste and other material storage yards, pre-fabrication and block making works, etc as guided by the Engineer in Charge.
- c) The contractor shall submit a document after construction of the buildings, a brief description along with photographic records to show that other areas have not been disturbed during construction.
- d) The contractor shall submit to the Engineer in Charge, before the start of construction, a site plan along with a narrative to demarcate areas on site from which top soil has to be gathered, designate area where it will be stored, measures adopted for top soil preservation and indicate areas where it will be reapplied after construction is complete.
- e) Provide supporting document from the manufacturer of the cement specifying the flyash content in PPC used in reinforced concrete.

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4.8 EQUIPMENT

- a) To ensure energy efficiency during and post construction all pumps, motors and engines used during construction or installed, shall be subject to approval of the Engineer-in-Charge.
- b) All lighting installed by the contractor around the site and at the labour quarters during construction shall be appropriate illumination levels. This condition is a must, unless specifically prescribed.

4.9 Submission of Pollution Control Plan

- (a) The contractor shall submit the detailed action plan for control of pollution and for adherence to all the environmental guidelines/Laws/statutes/Court Orders/NGT orders/orders of pollution control authorities through the entire period of construction at site. The detailed action plan shall be submitted to the Engineer-in- Charge within 15 days of the stipulated date of start of work and shall be got approved from the Engineer-in-Charge.
- (b) The contractor shall arrange for control measures of all dust/noise/emission from the construction activities at site of work and shall install screens/curtains/ covers/dust trappers etc. as per guidelines/orders of the NGT/Court of law/ statutory authorities etc. No hindrance shall be allowed, arising out of any stay/stopping of work from any court/statutory authority/NGT/Govt. Authorities as a consequence of the contractor not adhering to any pollution control guideline/law/order of the state bodies during the construction period. Nothing shall be paid to the contractor on account of expenses for any dust/pollution/emission control measures at the site of work or any delay in work due to any orders passed by any court/ statutory authority/Govt. Authorities during the period of construction.
- (c) *A compensation of Rs. 2,000/- per day will be levied and recovered from the dues of the contractor for each day of delay beyond 15 days for non submission of pollution control plan.*

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- (d) The contractor is strongly advised to study all dust/Noise/emission/pollution control norms/laws/Court Orders before bidding for the work and quote his rates accordingly for any liability which may arise on this account during the period of construction.

Note 1: The special conditions detailed above are in addition to general /additional conditions as specified elsewhere in tender document.

Note 2: Nothing extra shall be paid on the account of these special conditions.

5.0 Special Condition for paint

1. The contractors are advised to get acquainted with the proposed work and its site and also study the specification and special conditions carefully before tendering. The claim of any part shall not be entertained on account of any site conditions and ignorance of specifications and special conditions.
2. The site for the work shall be made available in parts. The contractor should programme his work accordingly.
3. In the event of any discrepancy between the additional and special conditions attached and the CPWD specifications 2019 Vol.-I & II with correction slips up to the day previous to last date of submission of tender the additional and special condition attached shall prevail.
4. Unless otherwise provided in the schedule of quantities, the rates tendered by the contractor shall be inclusive of all leads and lifts and shall apply to all heights and depths of the buildings and nothing extra shall be payable to him on this account.
5. The rates for all items of work shall, unless clearly specified otherwise, includes the cost of all labour, material and inputs involved in the execution of the item of work.
6. The rates quoted by the contractor shall be inclusive of all taxes and nothing extra shall be paid on any account i.e. Royalty, Cartage, Sales Tax and stacking of required places etc.
7. Full quantity of materials such as Acrylic distemper, Paint, Plastic Emulsion paint and Acrylic Emulsion etc. shall be deposited at site in sealed container in advance and got checked by the Engineer-in-charge before use of following

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manufacturers:-

Sl. No.	Name of Manufacturer	Synthetic Enamel (in 4 litre packing)	Plastic Emulsion (in 4 litre packing)	Acrylic Emulsion Paint	Oil bound (in 20 kg packing)	Cement Primer (in 20 litre)	Premium/ Acrylic Smooth Exterior Paint
1.	Berger	Luxol Hi-gloss	Easy clean	Rangoli total care	Bison Acrylic Distemper	BP White Primer	
2.	Goddies Nerolac Ltd.	Synthetic Hi-gloss	Nerolac Impression	--	Beauty Acrylic Distemper	--	
3.	Asian Paints	Apcolite premium gloss		Apcolite Premium	Professional Acrylic Distemper	Decoprime WT	Apex Ultima
4.	ICI Dulux	Gloss synthetic enamel		Super Cover Ultra			Weather shield max

- i) Approved paint, Distemper, Water Proofing Cement Paint etc. shall be brought to the site of work by the contractor from the Companies offices. If a company does not have retail outlet in Lucknow then the material may be brought from authorised dealer in their original containers in sealed conditions, the material shall be brought on time in adequate quantities for the whole work. The contractor shall produce necessary documentary proof / original invoice to the effect that the material has been brought from the companies retail outlet / authorised dealer. The material shall be kept in the custody of the department. The material shall be got issued to the contractor by the site in-charge.

6.0 Condition for Structural Drawings:

Structural drawings will be provide by the agency of the work as per requirement from the CPWD approved structural designer or as approved by the Engineer-in-charge. All the expenditure shall be borne by the agency and nothing extra shall be paid on this account by the department to the agency.

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PARTICULAR SPECIFICATIONS

1.0 EARTH WORK

- 1.1** The work shall be done in accordance with CPWD Specifications – 2019 – Vol.I& Vol. II with upto date correction slips.
- 1.2** All the major excavation shall be carried out by mechanical excavator. No extra payment shall be made for that.
- 1.3** Any trenching and digging for laying sewer lines/water lines/cables etc. shall be commenced by the contractor only when all men, machinery's and materials have been arranged and closing of the trench(s) thereafter shall be ensured within the least possible time.
- 1.4** The Jungle clearance and clearance of grass are not payable separately for any type of Earth Work in excavation and Preparation of sub-grade.

2.0 CONCRETE WORK

- 2.1** The work shall be done in accordance with CPWD Specifications – 2019 – Vol.I& Vol. II with upto date correction slips.

3.0 RCC/CC WORK (DESIGN MIX CONCRETE)

- 3.1** The work shall be done in accordance with CPWD Specifications – 2019 – Vol.I& Vol. II with upto date correction slips.
- 3.2** The contractor shall engage one of the IIT/NIT/Reputed Govt. Engineering Institutions/Approved Laboratories as directed by the Engineer-in-charge at his own expenses for designing the concrete mix in accordance with relevant IS Codes and to conduct laboratory test to ensure the target strength and workability criteria for a given grade of concrete.
- 3.3** In case of change of source or characteristic properties of the ingredients used in the concrete mix during the work, a revised laboratory mix design report conducted at laboratory established at site shall be submitted by the contractor as per the direction of the Engineer-in-Charge.
- 3.4** All cost of mix designing and testing connected therewith including charges payable to the laboratory shall be borne by the Contractor including redesigning of the concrete mix wherever required and directed by Engineer-in-Charge.

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3.5 APPROVAL OF DESIGN MIX:

The mix design for a specified grade of concrete shall be done for a target mean compressive strength

$$T_{ck} = F_{ck} + 1.65s$$

Where, F_{ck} = Characteristic compressive strength at 28 days.

S = Standard deviation which depends on degree of quality control.

The degree of quality control for this work is "good" for which the standard deviation (s) obtained for different grades of concrete shall be as follows:

GRADE OF CONCRETE	STANDARD DEVIATION(S)
M-25	4.0
M-30	5.0
M-35	5.0
M-40	5.0

- The average strength of each test sets is not less than the specified target mean compressive strength (T_{ck}).
- The strength of any specimen cube is not less than $0.85 T_{ck}$.
- The concrete mix is required degree of workability and acceptance concrete finish.

3.6 All cost of mix designing and testing connected therewith including charges payable to the laboratory shall be borne by the Contractor.

3.7 WORK STRENGTH TEST:

TEST SPECIMEN:

Work strength test shall be conducted in accordance with IS:516 on random sampling. Each test shall be conducted on six specimens, three of which shall be tested at 7 days and remaining three at 28 days.

TEST RESULTS OF SAMPLES:

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The test results of the sample shall be the average of the strength of three specimen. The individual variation shall not be more than + 15% percent of the average. If variation is more, the test results shall be treated as invalid. 90% of the total tests shall be done at the laboratory established at site by the contractor and remaining 10% in the laboratory of IIT/NIT or in any other laboratory as directed by the Engineer-in-Charge.

LOT SIZE:

The minimum frequency of sampling of concrete of each grade shall be in accordance with the following:

QUANTITY OF CONCRETE IN THE WORK (CUBIC METRE PER DAY).	NUMBER OF SAMPLES
1-5	1
6-15	2
16-30	3
31-50	4
51 & above	4 Plus one additional sample for each additional 50 cubic meters of part thereof

NOTE: - At least one sample shall be taken from each shift.

3.8 STANDARD OF ACCEPTANCE:

- (i) The concrete shall be accepted if both the following condition are met.
 - a. Any individual test result is not less than $(F_{ck} - 3) \text{ N/mm}^2$.
 - b. The mean of test results from any group of four consecutive samples is more than $(F_{ck} + 3) \text{ N/mm}^2$.
 - c. **Note:-** For concrete of quantity up to 30 cum (where the number of samples to be taken is less than four as per

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frequency of sampling given in clause 5.11.5(d) CPWD Specification-2019 Vol-I, the mean of test results of all such samples shall be $f_{ck} + 4 \text{ N/mm}^2$, minimum.

- (ii) Concrete of each grade shall be assessed separately.
- (iii) Concrete is liable to be rejected, if it is porous or honeycombed, its placing has been interrupted without providing a proper construction joint, the reinforcement has been displaced beyond the tolerances specified, or construction tolerances have not been met.

3.9 Nothing extra shall be paid for the centering and shuttering, circular in shape whenever the form work is having a mean radius exceeding 6m in plan.

3.10 In order to keep the floor finish as per architectural drawings and to provide required thickness of the flooring as per specifications, the level of top surface of RCC shall be accordingly adjusted at the time of its centering, shuttering and casting for which nothing extra shall be paid to the Contractor.

3.11 As per general engineering practice, level of floors in toilet / bath, balconies, shall be kept 25 mm as required lower than general floors shuttering should be adjusted accordingly and slabs should be laid with slope towards the drainage point. Nothing extra is payable on this account.

3.12 LAND FOR TEMPORARY USE

The land for labour camps & batching Plant shall be arranged by the contractor. The lease/rent charges shall be borne by the contractor. The Engineer-in-Charge shall extend necessary help and issue necessary recommendations etc. to the concerned clients / department for temporary allotment of land during construction period if land is available with them. In such cases, the contractor shall vacate the land after completion of work in same condition as was at the time of allotment.

4.0 SHUTTERING / FORM WORK

4.1 The work shall be done in accordance with CPWD Specifications – 2019 Vol. I & Vol. II with upto date correction slips.

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- 4.2** Steel shuttering as approved by the Engineer-in-Charge shall be used by the contractor. Minimum size of shuttering plates shall be 600mm x 900mm except for the case when closing pieces required to complete the shuttering panels. Dented, broken, cracked, twisted or rusted shuttering plates shall not be allowed to be used on the work.
- 4.3** The shuttering plates shall be cleaned properly with electrically driven sanders to remove any cement slurry or cement mortar or rust. Proper shuttering oil or debonding compound shall be applied on the surface of the shutter plates in the requisite quantity before assembly of steel reinforcement.
- 4.4** The joint filler shall be resilient closed cell expanded polyethene and non-tainting as per manufacturer specification and direction of Engineer-in-Charge.
- 4.5** Providing joint filler of required thickness in position to substrate using either double sided foam adhesive tape or neoprene synthetic rubber adhesive. When forming expansion joint with the Board in in-situ concrete, joint sealing slots can be readily formed in the following matter-
- Before installing, simply cut off a strip of the required depth. Then install the filler flush with the finished surface.
 - Prior to sealing, the top strip can then be pulled easily from the joint to provide an uncontaminated sealing slot ready for preparation and sealing.

5.0 REINFORCEMENT

- 5.1** The reinforcement shall be done as per CPWD Specifications – 2019 – Vol.I& Vol. II with upto date correction slips.
- 5.2** The item of reinforcement of RCC work includes all operations including straightening, cutting, bending, welding, binding with annealed steel or welding and placing in position at all the floors with all leads and lift complete as per CPWD Specification – 2019 – Vol.I& Vol. II with upto date correction slips.
- 5.3** To avoid displacement of bars in any direction and to ensure proper cover, only factorymade round type/rectangular cover blocks shall be used by the contractor. Nothing extra shall be payable on this account.

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6.0 BRICK WORK

- 6.1** Unless otherwise specified FPS Bricks shall be used in all items of brick work. The classification of bricks brought by the contractor shall strictly conform to CPWD Specifications – 2019 Vol-1 & II with upto date correction slips or as specified. The work shall also include for leaving chases / notches for dowels / cramps for all kinds of cladding to come over brick work.

7.0 RANDOM RUBBLE STONE MASONRY

7.1 Material

7.1.1 Stone

The stone shall be of the type specified such as granite, trap, limestone, sand stone, quartzite, etc. and shall be obtained from the quarries, approved by the Engineer-in-Charge. Stone shall be hard, sound, durable and free from weathering decay and defects like cavities, cracks, flaws, sand holes, injurious veins, patches of loose or soft materials and other similar defects that may adversely affect its strength and appearance. As far as possible stones shall be of uniform colour, quality or texture. Generally stone shall not contain cryptocrystalline silica or chert, mica and other deleterious materials like iron-oxide organic impurities etc.

Stones with round surface shall not be used.

The compressive strength of common types of stones shall be as per Table 7.1 and the percentage of water absorption shall generally not exceed 5% for stones other than specified in Table 7.1. For laterite this percentage is 12%.

7.1.2 Size of Stones

Normally stones used should be small enough to be lifted and placed by hand. Unless otherwise indicated, the length of stones for stone masonry shall not exceed three times the height and the breadth on base shall not be greater than three-fourth of the thickness of wall, or not less than 150 mm. The height of stone for rubble masonry may be upto 300 mm.

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The selection and grading of stones for rubble masonry is largely done at site and the smaller stones are used in the hearting of wall.

7.1.3 Random Rubble Masonry shall be uncoursed or brought to courses as specified (Fig. 7.8 and 7.9). Uncoursed random rubble masonry shall be constructed with stones of sizes as referred to in para 7.0 and shapes picked up random from the stones brought from the approved quarry. Stones having sharp corners or round surfaces shall, however, not be used.

7.1.4 Random rubble masonry brought to the course is similar to uncoursed random rubble masonry except that the courses are roughly levelled at intervals varying from 300 mm to 900 mm in height according to the size of stones used.

7.1.5 Dressing

Each stone shall be hammer dressed on the face, the sides and the beds. Hammer dressing shall enable the stones to be laid close to neighbouring stones such that the bushing in the face shall not project more than 40 mm on the exposed face.

- (i) **Face stone:** At least 25% stones shall be headers tailing into the work at least 2/3rd the thickness of wall in super structure masonry. Such stones shall not be less than 200 sq. cm in cross sections.
- (ii) **Hearting Stones:** The hearting or interior filling of a wall face shall consist of rubble stones not less than 150 mm in any direction, carefully laid, hammered down with a wooden mallet into position and solidly bedded in mortar. The hearting should be laid nearly level with facing and backing.
- (iii) **Quoin Stone:** Quoin stone shall be less than 0.03 cum in volume.
- (iv) **Jamb stones:** The jambs shall not be made with stones specified for quoins except that the stones which were required to be provided at 1 metre centre to centre on both the exposed faces shall here be provided only on the jamb

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and the length shall be equal to the thickness of the wall for wall upto 60 cm and a line of headers shall be provided for walls thicker than 60 cm as specified for bond.

7.1.5 (A) Courses

The masonry shall be carried out in regular courses of height not exceeding 50 cm and masonry on any day will not be raised more than 60 cm in height when using mortars having compressive strength less than 20 kg./sq. cm at 28 days and 100 cm when using mortars exceeding this strength.

7.1.5 (B) Thickness of Joints

The joint thickness shall not exceed 30 mm at any point on the face. Chips of the stone and spalls shall be wedged into seating bed of face stones to avoid excessive bed thickness. No pinning shall be allowed to avoid excessive joint thickness.

7.1.6 Mortar

The mortar used for joining shall be as specified.

7.1.7 Laying

Stone shall be laid on their natural bed and shall be solidly bedded full in mortar with close joints, chips of stone spalls be wedged into the work wherever necessary. No dry work or hollow spaces shall be allowed and every stone whether large or small shall be carefully selected to fit snugly the interstices between the large stones. Masonry shall be built breaking joints in all the three directions.

Bond stone and headers shall be properly laid into the work and shall be marked by the contractor with white lead paint. The bond stones shall be provided as specified in para 7.1.8. The masonry work in wall shall be carried up true to plumb or to specified batter.

Random rubble masonry shall be brought to the level courses at plinth, window sills, lintel and roof levels. Levelling shall be done with concrete comprising of one part of the mortar as used for masonry and two parts of graded stone aggregate of 20 mm nominal size.

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The masonry in structure shall be carried uniformly. Where the masonry of one part is to be delayed, the work shall be raked back at an angle not steeper than 45°.

7.1.7 (A) Raking out joints

All the joints on the faces to be pointed or plastered shall be raked out with racking tool to a depth of 20mm while the mortar is still green.

7.1.8 Bond Stones

Though bond stones shall be provided in walls upto 600 mm thickness, a set of two or more bond stones overlapping each other by at least 150 mm shall be provided in a line from face to back.

In case of highly absorbent types of stones (porous lime stone and sand stone etc.) the bond stone shall extend about two-third into the wall, as through stones in such walls a set of two or more bond stones overlapping each other by at least 150 mm shall be provided. Each bond stone or a set of bond stones shall be provided for every 0.5 sqm of the wall surface and shall be provided at 1.5 m to 1.8 m apart clear in every course.

In case of highly absorbent types of stones (porous lime stone and sand stone etc.) single piece bond stones may give rise to dampness. For all thicknesses of such walls a set of two or more bond stones overlapping each other by at least 15 cm shall be provided. Length of each such bond stone shall not be less than two-third of the thickness of the wall.

Where bond stones of suitable lengths are not available pre-cast cement concrete block of 1:3:6 mix (1 cement : 3 coarse sand: 6 graded stone aggregate 20 mm nominal size) of cross section not less than 225 square centimeters and length equal to the thickness.

At least one bond stone or a set of bond stones shall be provided at 1.5 m to 1.8 m apart clear in every course. (Bond stones shall be marked suitably with paint as directed by the Engineer-in-Charge).

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7.2 PLUM CONCRETE

The work shall be done in accordance with CPWD Specifications – 2019 Vol. I & Vol. II with upto date correction slips.

7.3 GUARANTEE:

Work of RR Stone Masonry/Plum concrete shall carry five years guarantee to be reckoned from the date after the expiry of maintenance period prescribed in the contract of the work against faulty workmanship, finishing, unsound materials, structural instability and other related problems.

Five Years Guarantee bond in prescribed Performa attached herewith as **Annexure-I** shall be submitted by the contractor to meet their liability / liabilities under the guarantee bond.

Five percent of the cost of RR Stone Masonry/Plum concrete work shall also be retained as security deposit and the amount so withheld would be released after Five Years from the date of completion of the entire work under the agreement, if the performance of the work done is found satisfactory. If any defect is noticed during the guarantee period, it shall be rectified by the contractor within seven days of receipt of intimation of defects in the work. If the defects pointed out are not attended to within the specified period, the same will be got done from another agency at the risk and cost of contractor.

However, the security deposit deducted may be released in full against bank guarantee of equivalent amount in favour of Engineer in charge, if so decided by the Engineer in charge.

The Security deposit against this item of work shall be in addition to the security deposit mentioned elsewhere in contract form.

8.0 STONE / MARBLE /GRANITE WORK (OTHER THAN MASONARY)

8.1 The execution of stones work shall be in general as per CPWD Specifications – 2019 – Vol.I& Vol. II with upto date correction slips.

8.2 All holes, rebates, recesses etc. for providing fixing and inserts shall be predrilled and precut and worked using precision machine tools. Nothing extra on this account shall be payable.

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8.3 SAMPLES FOR STONE WORK: Samples of each item of stone work either individually or in combination shall be prepared for approval of Engineer-in-Charge before commencement of work.

8.4 Sequence of execution for cladding work shall be suggested by the contractor for approval of Engineer-in-Charge.

9.0 SCAFFOLDING

9.1 Double steel scaffolding having two sets of vertical supports shall be provided. The supports shall be sound and strong, tied together with horizontal pieces over which scaffolding planks shall be fixed.

10.0 WOOD WORK

10.1 The wood work in general shall be carried out as per CPWD Specifications – 2019 Vol.I & II with upto date correction slips.

10.2 The samples of species of timber to be used shall be got approved and deposited by the contractor with the EE before commencement of the work. The contractor shall produce cash vouchers and certificates from kiln seasoning or/and chemical treatment plants about the timber section to be used on the work having been kiln seasoned or/and chemically treated by them.

10.3 Factory made shutter as specified shall be obtained from factories approved by the Engineer in charge. The contractor shall inform well in advance to the Engineer-in-charge the names and address of the factory from where the contractor intends to get the shutters manufactured. The contractor will place order for manufacture of shutters only after written approval of the Engineer-in-charge in this regard is given. The contractor is bound to abide by the decision of the Engineer-in-charge and recommend a name of another factory from the approved list in case the factory already proposed by the contractor is not found competent to manufacture quality shutters. Shutters will however be accepted only if this meet the specified tests. The contractor will also arrange stage wise inspection of the shutters at factory to the Engineer-in-charge or his authorized representative. Contractor will have no claim if the shutters brought at site are rejected by Engineer-in-charge in part or in full lot due to bad workmanship / quality even after inspection of factory. Such shutters will not be measured and paid and the contractor shall remove the same from the site of work within 7 days after the written instruction in

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this regard are issued by Engineer in Charge or his authorized representative.

10.4 All fittings and fixtures shall be got approved from the Engineer-in Charge before procurement well in advance and the approved samples shall be kept at site till completion of the work.

10.5 Glazing for toilets shall be of translucent type.

10.6 The shape and size of beading shall be as per drawings. The joints of beading shall be mitred.

11.0 STEEL WORK

11.1 Work shall be carried out as per CPWD Specifications – 2019 – Vol.I& Vol. II with upto date correction slips.

11.2 The rate of T- angle iron frame shall include the following.

- (a) M.S. sill/tie of 10mm dia bar welded to T-iron frames to keep the frames vertical in correct position. The sill / tie shall be embedded in floor concrete. No tie is necessary for window frames.
- (b) Each T – iron frame for doors shall have 4 Nos. M.S. lugs 15x3mm, 10 cm long welded to each vertical member of the frame.
- (c) M.S. flat 6 x 25mm, 100mm long having threaded holes (No. of flats shall correspond to the no. of butt hinges to be fixed to door / window shutters) shall be welded at appropriate places at the back of the T-iron frames for fixing the required butt hinges to the frame with machine screws.

11.3 All welded structural steel work shall be tested for quality of weld as laid down in IS:822-1970 before actual erection if required.

12.0 FLOORING

12.1 All work in general shall be carried out as per CPWD Specifications- 2019 – Vol.I& Vol. II with upto date correction slips.

12.2 Whenever flooring is to be done in patterns tiles/ stone, the contractor shall get samples of each pattern laid and approved by the Engineer-in-Charge before final laying of such flooring for which nothing extra shall be paid.

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12.3 Different stones/ tiles used in pattern flooring as per the approved architectural drawings and nothing extra for laying pattern flooring shall be paid. No additional wastage if any shall be accounted for any extra payment.

12.4 The proper gradient shall be given to flooring for toilets, varandah, kitchen, court yard, etc. as per the directions of Engineer-in-Charge. For this there may be extra thickness of dry mortar below the tiles/stone slabs. These gradients should be insured in the shuttering itself. Nothing extra shall be paid for this as this is included in the tendered cost.

12.5 The rate of items of flooring is inclusive of providing sunken flooring in bathrooms, kitchen etc. and nothing extra on this account is admissible. The samples of flooring, dado & skirting as per approved pattern shall be prepared & got approved from the Engineer-in-charge before execution of work.

12.6 Ceramic Tiles/Vitrified Tiles Work/ Granite stone flooring

12.6.1 Work shall be carried out as per CPWD Specifications- 2019 Vol I & II with up to date correction slips and as per manufactures specifications.

12.6.2 Rates shall be inclusive of all operations including labour, material, T&P, scaffolding etc. complete. Nothing extra shall be payable on any account.

12.6.3 One-piece Granite stone for treads / risers in staircase shall be used and nothing extra shall be paid on this account.

13.0 STAINLESS STEEL RAILING/HANDRAILS:

13.1 GENERAL: The contractor shall apply all materials, labour, tools, ladders, scaffolding and other equipment necessary for the completion and protection of all stainless-steel work.

13.2 MATERIAL: All stainless-steel pipes and plates shall conform to AISI 316 grade and the relevant clauses associated with this grade of steel to be followed.

13.3 SURFACE FINISH: Surface finish of all the stainless-steel materials will be in 240 grit satin finish / matt finish.

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- 13.4 ACCESSORIES:** Fixing will be done by stainless steel expansion bolts of approved size and make as per Engineer-in-charge and welding to be done by using organ welding rods and the surface being duly finished and cleaned by K2 passivation, which is nitric acid plus fluoric acid solution treatment by which the chances of corrosion will be eliminated and any burn out makes on the metal will also be eliminated.
- 13.5 COATING MASS:** All stainless-steel material will have to be coated by a solution of Inox to avoid finger in prints and avoidance of settlement of environment / atmospheric dust.

14.0 WATER PROOFING TREATMENT

- 14.1** Work shall be executed as per CPWD Specifications, 2019 Vol I & II with upto date correction slips.
- 14.2** The contractor shall associate himself with the specialized firm, to be approved by the Engineer-in-charge in writing, for water proofing treatment for basement/lower ground floor, underground tank and on roofs.
- 14.3** The finished surface after water proofing treatment for roof slab shall have smooth slope with minimum gradient of 1 in 80.
- 14.4** Before commencement of treatment on roof surface, it shall be ensured that the outlet drain pipes/ spouts have been fixed and the spout opening have been eased and rounded off properly for easy flow of water.
- 14.5** The surface where the water proofing is to be done shall be thoroughly cleaned with wire brushes. All loose scales mortar splashes etc. shall be removed and dusted off. The surface shall be treated with neat cement slurry admixed with proprietary water proof compound to penetrate into crevices and fill up all the pores in the surface.
- 14.6** Curing of water proofing treatment shall be done for a minimum period of two weeks by flooding the water by making compartments etc.
- 14.7 GUARANTEE:**
- The work shall carry **Ten Years guarantee** to be reckoned from the date of completion of the entire work under the contract against faulty workmanship, finishing, unsound materials, efficiency of water proofing treatment and other related problems.

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Ten Years Guarantee bond in prescribed Performa attached herewith as **Annexure-III** shall be submitted by the contractor which shall also be signed by both the specialized agency and the contractor to meet their liability / liabilities under the guarantee bond. However, the sole responsibility about efficiency of water proofing treatment shall rest with the main contractor.

10% (Ten percent) of the tender amount of work done under the head of water proofing shall be retained as security deposit and the amount so withheld would be released after ten years from the date of completion of the entire work under the contract, if the performance of the work done is found satisfactory. If any defect is noticed during the guarantee period, it shall be rectified by the contractor within seven days of receipt of intimation of defects in the work. If the defects pointed out are not attended within the specified period, the same will be got done from another agency at the risk and cost of contractor.

However, the security deposit deducted may be released in full against bank guarantee of equivalent amount in favour of Engineer in charge, if so decided by the Engineer in charge.

The Security deposit against this item of work shall be in addition to the security deposit mentioned elsewhere in contract form.

15.0 FINISHING:

15.1 The work shall be carried out as per CPWD Specifications- 2019 Vol.-I & Vol. II with upto date correction slips.

15.2 All painting material shall be brought to the site of work in the original sealed containers. The material brought to the site of work shall be sufficient for at least 30 days of work. The material shall be kept under the joint custody of contractor and representative of the Engineer-in-Charge. The empty contains shall not be removed from the site till the completion of the work without permission of the Engineer-in-Charge.

16.0 SANITARY INSTALLATIONS, WATER SUPPLY AND DRAINAGE

16.1 The scope of work comprises supply, laying, installation, commissioning and testing of water supply, sewerage and drainage works including sanitary fixtures and fittings. These works shall be executed as per the

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specifications of items attached and CPWD specifications- 2019 Vol. I & II with up-to-date correction slips up to the date of tender submission.

- 16.2** The work of water supply and sanitary installations shall be got executed by the agency as approved by Engineer-in-Charge.
- (i) The entire plumbing drawing and sanitary installation drawing/details shall be submitted by the contractor and got approved by the Engineer-in-Charge before the execution.
- (ii) The entire responsibility for the quality of work will however rest with the building contractor only.
- 16.3** The work of water supply, internal sanitary installations and drainage etc. shall be carried out as per the bylaws of the Municipal Corporation or any other local body and the contractor shall produce necessary completion certificates from such authority after completion of work.
- 16.4** All water tanks, taps, sanitary, water supply and drainage pipes fittings and accessories etc. shall conform to the bylaws and specifications of the Municipal Body/Corporation where CPWD specifications are not available.
- 16.5** The contractor shall engage licensed plumbers for the work and the materials (fixtures/fittings) tested by the local Municipal Body/Corporation wherever required at his own cost. Nothing extra shall be paid/reimbursed for the same.
- 16.6** All sanitary wares and fittings shall conform to IS standards and to be procured from approved makes. The contractor shall submit samples of all fittings and fixtures proposed to be used to the Engineer-in-charge for his approval. The approved samples shall remain with the Engineer-in-charge till the completion of the work.
- 16.7** All fixtures and accessories shall be fixed in accordance with a set pattern matching the tiles or interior finish as per architectural requirements. Wherever necessary the fittings centered to dimensions and pattern desired.
- 16.8** The rates shall include the cost of cutting chases, holes in walls, floors, RCC slabs etc. Wherever required and making good the same for which nothing extra shall be paid. The work in general shall be carried out as per CPWD specifications.

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- 16.9** Rate includes all materials, labour and all the operations mentioned in the respective items unless and otherwise specifically mentioned.
- 16.10** The CI pipe wherever necessary shall be fixed to RCC columns, beams etc. with rawl plugs of approved quality and nothing extra shall paid for on this account.
- 16.11** All the works shall be completely concealed either within shafts or chases or in fills and dropped ceilings, unless specifically shown in drawings or required otherwise.
- 16.12** All the works shall be adequate protected against corrosion, so that the whole work is free from damage throughout.
- 16.13** The contractor shall give a performance test of the entire installation(s) as per standing specifications before the work is finally accepted by making his own arrangements for water supply, electricity etc. and nothing extra whatsoever shall be payable for the same.
- 16.14** The contractor shall give a satisfactory performance test of the entire installation (s) before the work is finally accepted and nothing extra shall be payable to the contractor on this account.
- 16.15** The contractor shall be responsible for all the protection of sanitary, water supply fittings and fixtures against pilferage and breakage during the period of installation until the completion / handing over of the work.
- 16.16** Before the work is handed over, the contractor shall clean all fixtures removing all plaster, stickers, rust stains and other foreign matter, leaving every part in acceptable condition and ready for use to the satisfaction of the Engineer-in-charge.
- 16.17** The contractor shall submit completion plans for water supply internal sanitary installations and building drainage work within thirty days of the date of completion. These plans are to be submitted on drawings prepared preferably through computers (1 original copy + 3 photocopies) on suitable scales to show the general arrangement and desired details.
- 16.18 INSPECTION AND TESTING**
- Inspection and testing of water supply sewerage and drainage installations shall be carried out as per CPWD Specification & National Building Code 2016 with up to date amendments.

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16.19 GUARANTEE:

The work shall carry **Ten Years guarantee** to be reckoned from the date of completion of the entire work under the contract against faulty workmanship, finishing, unsound materials, efficiency of water proofing treatment and other related problems.

Ten Years Guarantee bond in prescribed Performa attached herewith as **Annexure-II** shall be submitted by the contractor which shall also be signed by both the specialized agency and the contractor to meet their liability / liabilities under the guarantee bond. However, the sole responsibility about efficiency of water proofing treatment shall rest with the main contractor.

10% (Ten percent) of the tender amount of work done under the head of water supply, sanitary and drainage installations shall be retained as security deposit and the amount so withheld would be released after ten years from the date of completion of the entire work under the contract, if the performance of the work done is found satisfactory. If any defect is noticed during the guarantee period, it shall be rectified by the contractor within seven days of receipt of intimation of defects in the work. If the defects pointed out are not attended to within the specified period, the same will be got done from another agency at the risk and cost of contractor.

17.0 LIST OF SPECIALISED ITEMS

List of Specialized Items has been provided in **Annexure-6** of Standard Operating Procedures (SOP) for CPWD Work Manual -2024.

17.1 PROCEDURE FOR EXECUTION OF THE SPECIALIZED ITEMS:

Specialized items shall be got executed only through associated agencies specialized in these fields. The contractor shall indicate the name(s) of his associated specialized agencies those fulfilling the eligibility conditions as early as possible and within 10 days of award of work to Engineer-in-Charge for approval.

The contractors shall quote the rates after careful study of contract conditions, specifications, drawings & schedule of quantities.

It shall be the responsibility of main contractor to sort out any dispute / litigation with the Specialized Agencies without any time & cost overrun to

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the Department. The main contractor shall be solely responsible for settling any dispute / litigation arising out of his agreement with the Specialized Agencies. The contractor shall ensure that the work shall not suffer on account of litigation/ dispute between him and the specialized agencies / sub-contractor(s). No claim of hindrance in the work shall be entertained from the Contractor on this account. No extension of time shall be granted and no claim what so ever, of any kind, shall be entertained from the Contractor on account of delay attributable to the selection/rejection of the Specialized Agencies.

For specialized items, the main contractor cannot work as a specialized agency unless his name is already included in the list of approved specialized agencies for these items. The contractor shall get these items executed through the specialized agencies as approved by Engineer in Charge.

17.2 ELIGIBILITY CONDITIONS FOR APPROVAL OF SPECIALIZED AGENCIES:

The Contractor(s) shall submit his proposal for the approval of the Engineer-in-Charge, the names of specialized agencies of repute along with their technical capability /experience proposed to be engaged by him. The agency must be currently actively engaged in execution of the said specialized item and must have who have satisfactorily executed similar work during last 7 (Seven) years as below:

- (i) Three works each of value not less than 40% of corresponding cost of the specialized items.

Or

- (ii) Two works each of value not less than 60% of corresponding cost of the specialized items.

Or

- (iii) One work of value not less than 80% of corresponding cost of the specialized items.

For calculation purpose only, cost of the specialized item as per accepted tendered value of the all the items corresponding to the specialized item under consideration.

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For ascertaining the above criteria, contractor shall provide certified copy of Income tax TDS certificates for payments made, work order copy, completion certificate, performance certificate against each completed similar work for verification by Engineer in charge. In case required and directed by Engineer in charge, contractor shall get such similar completed work inspected by Engineer in charge and contractor shall extend all necessary help for same at no extra cost to the department. Decision as to approve or reject the proposed specialized agency based on the inspection of the completed work shall be final and binding on the contractor and no claim, whatsoever, shall be admissible on this account and the contractor shall be required to propose another specialized agency as per the required eligibility criteria.

18.0 VARIATION IN CONSUMPTION OF MATERIALS

- 18.1 The pig lead for caulking of joints of SCI pipes shall be used as per the theoretical consumption for SCI pipes of sizes 100mm, 75mm, 50mm at 0.98Kg, 0.88 Kg and 0.77Kg. per joint respectively. Over and above the theoretical quantities of lead as worked out a variation of 5% shall be allowed for wastage etc. Any difference between the actual consumption of pig lead and the theoretical consumption worked out on the above basis shall be recovered at double the issue rate. Where the pig lead is arranged by the contractor, also a variation of 5% shall be allowed. In case the variation is on the lower side, the quantity of pig lead used less shall be recovered from the contractor at market rate to be determined by the Engineer-in-charge whose decision in this matter shall be final.
- 18.2 The theoretical quantity of cement to be utilized in item of concrete involving use of single aggregate and mixed by volume batching shall be computed on the basis of the coefficient for cement to be used in different item of the work provided in DSR reducing each of the co-efficient by 5%. However, where the concrete is mixed by weight batching no such reduction shall be made from theoretical co-efficient given in **DSR 2023** for concrete with crushed stone aggregate.

19.0 GUARANTEE BOND FOR STONE WORK AND PLUM CONCRETE:

All types of Stone work shall carry **five years guarantee** to be reckoned from the date after the expiry of maintenance period prescribed in the contract of the work against faulty workmanship, finishing, unsound materials, structural instability

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and other related problems as per guarantee bond Performa attached in this tender document on **Annexure-I**.

Five years guarantee in prescribed proforma attached must be given by the contractor in token of his overall responsibility. **2.5% (Two Point Five Percent) of the cost of stone work and plum concrete** would be retained as guarantee to the performance of the work done. The guarantee against this item of work shall be in addition to the security deposit mentioned elsewhere in the contract form. If any defects or deficiencies are noticed during the guarantee period, the same shall be rectified by the contractor within seven days of issue of the written notice by the Engineer-in-charge, failing which the defects/deficiencies would be got removed by the Engineer-in-charge from another agency at the risk and cost of the contractor. However this amount of the guarantee can be released in full, if bank guarantee of equivalent amount for the required period is produced and deposited with the department.

20.0 G.I. CHAIN LINK FABRIC FENCING

20.1 Material

G.I. Chain link fabric fencing of required width in mesh size 50 x 50 or 25 x 25 mm or specified otherwise of approved brand and made of specified dia GI wire PVC coated of specified thickness / or not as specified in item of required colour or shade to be used and it should be conform to IS 2721.

20.1.1 Fixing : GI chain link shall be stretched and fixed in specified width, strengthening with 2mm dia wire or nuts bolts & washers as required to be done complete as per the direction of Engineer-in-Charge.

20.1.2 Measurements : The length and width shall be measured correct to a cm. The area shall be calculated in square metre, correct to two places of decimal.

20.2 The rate shall include the cost of material and labour involved in all the operations described as above.

21.0 CORRUGATED GALVANISED STEEL SHEET ROOFING

21.1 C.G.S. Sheets

These shall be of the thickness specified in the description of the item and shall conform to IS 277.

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The sheets shall be of 275 grade of coating (See Appendix-A) unless otherwise specified in the description of item.CPWD SPECIFICATIONS 2019

The sheets shall be free from cracks, split edges, twists, surface flaws etc. They shall be clean,bright and smooth. The galvanizing shall be non-injured and in perfect condition. The sheets shall not show signs of rust or white powdery deposits on the surface. The corrugations shall be uniform in depth and pitch and parallel with the side.

21.2 Laying and Fixing

21.2.1 The sheets shall be laid and fixed in the manner described below, unless otherwise shown in the working drawings or directed by the Engineer-in-Charge.

21.2.2 The sheets shall be laid on the purlins to a true plane, with the lines of corrugations parallel or normal to the sides of the area to be covered unless otherwise required as in special shaped roofs.

21.2.3 The sheets shall be laid with a minimum lap of 15 cm at the ends and 2 ridges of corrugations at each side. The above minimum end lap of 15 cm shall apply to slopes of 1 vertical to 2 horizontal and steeper slopes. For flatter slopes the minimum permissible end lap shall be 20 cm. The minimum lap of sheets with ridge, hip and valley shall be 20 cm measured at right angles to the line of the ridge, hip and valley respectively. These sheets shall be cut to suit the dimensions or shapes of the roof, either along their length or their width or in a slant across their lines of corrugations at hips and valleys.

They shall be cut carefully with a straight edge chisel to give a smooth and straight finish.

21.2.4 Lapping in C.G.S. sheets shall be painted with a coat of approved steel primer and two coats of painting with approved paint suitable for G.S. sheet, before the sheets are fixed in place.

21.2.5 Sheets shall not generally be fixed into gables and parapets. They shall be bent up along their side edges close to the wall and the junction shall be protected by suitable flashing or by a projecting drip course, the later to cover the junction by at least 7.5 cm.

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21.2.6 The laying operation shall include all scaffolding work involved.

21.2.7 Sheets shall be fixed to the purlins or other roof members such as hip or valley rafters etc. with galvanised J or L hook bolts and nuts, 8 mm diameter, with bitumen and G.I. limpet washers or with a limpet washer filled with white lead as directed by the Engineer-in-Charge. While J hooks are used for fixing sheets on angle iron purlins, and L hooks are used for fixing the sheet to R.S. joists, timber or precast concrete purlins. The length of the hook bolt shall be varied to suit the particular requirements.

The bolts shall be sufficiently long so that after fixing they project above the top of the nuts by not less than 10 mm. The grip of J or L hook bolt on the side of the purlin shall not be less than 25 mm. There shall be a minimum of three hook bolts placed at the ridges of corrugations in each sheet on every purlin and their spacing shall not exceed 30 cm. Coach screws shall not be used for fixing sheets to purlins.

21.2.8 The galvanised coating on J or L hooks, and bolts shall be continuous and free from defects such as blisters, flux stains, drops, excessive projections or other imperfections which would impair serviceability.

The galvanised coating should conform to IS 1367 (Pt. XIII) The mass of coating per square meter of the surface shall be as per specification.

21.2.9 Where slopes of roofs are less than 21.5 degrees (1 vertical to 2.5 horizontal) sheets shall be joined together at the side laps by galvanised iron bolts and nuts 25 × 6 mm size, each bolt provided with a bitumen and a G.I. limpet washer or a G.I. limpet washer filled with white lead. As the overlap at the sides extends to two corrugations, these bolts shall be placed zig-zag over the two overlapping corrugations, so that the ends of the overlapping sheets shall be drawn tightly to each other. The spacing of these seam bolts shall not exceed 60 cm along each of the staggered rows. Holes for all bolts shall be drilled and not punched in the ridges of the corrugations from the underside, while the sheets are on the ground.

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21.3 Finish

The roof when completed shall be true to lines, and slopes and shall be leak proof.

21.4 Measurements

21.4.1 The length and breadth shall be measured correct to a cm. Area shall be worked out in sqm correct to two places of decimal.

21.4.2 The superficial area of roof covering shall be measured on the flat without allowance for laps and corrugations. Portion of roof covering overlapping the ridge hip etc. shall be included in the measurements of the roof.

21.4.3 Roof with curved sheets shall be measured and paid for separately. Measurements shall be taken on the flat and not girthed.

21.4.4 No deduction in measurement shall be made for opening upto 0.4 sqm and nothing extra shall be allowed for forming such openings. For any opening exceeding 0.4 sqm in area, deduction in measurements for the full opening shall be made and in such cases the labour involved in making these openings shall be paid for separately. Cutting across corrugation shall be measured on the flat and not girthed. No additions shall be made for laps cut through.

21.5 Rate

The rate shall include the cost of all the materials and labour involved in all the operations described above including a coat of approved steel primer and two coats of approved steel paint on overlapping of C.G.S. sheets. This includes the cost of roof sheets, galvanised iron J or L hooks, bolts and nuts, galvanised iron seam bolts and nuts, bituminous and galvanised iron limpet washers etc.

21.6 RIDGES AND HIPS OF PLAIN GALVANISED STEEL SHEETS**21.6.1 Ridges and Hips**

Ridges and hips of C.G.S. roof shall be covered with ridge and hip sections of plain G.S. sheet with a minimum lap of 20 cm on either side over the C.G.S. sheets. The end laps of the ridges and

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hips and between ridges and hips shall also be not less than 20 cm. The ridges and hips shall be of 60 cm overall width plain G.S. sheet, 0.6 mm or 0.8 mm thick as given in the description of the item and shall be properly bent in shape.

21.6.2 Fixing

21.6.2.1 Ridges shall be fixed to the purlins below with the same 8 mm dia G.I. hook bolts and nuts and bitumen and G.I. limpet washers which fix the sheets to the purlins.

21.6.2.2 Similarly, hips shall be fixed to the roof members below such as purlins, hip and valley rafters with the same 8 mm dia G.I. hook bolts and nuts and bitumen and G.I. limpet washers which fix the sheets to those roof members. At least one of the fixing bolts shall pass through the end laps of ridges and hips, on either side. If this is not possible extra hook bolts shall be provided.

21.6.2.3 The end laps of ridges and hips shall be joined together with C.G.S sheet by galvanised iron seam bolts 25 x 6 mm size each with a bitumen and G.I. washer or white lead as directed by the Engineer-in-Charge. There shall be at least two such bolts in each end lap.

21.6.2.4 Surface of C.G.I. sheets of ridge and hip sections and the roofing sheets which overlap each other shall be painted with a coat of approved primer and two coats of approved paint suitable for painting G.S. Sheets before they are fixed in place.

21.6.3 Finish

The edges of the ridges and hips shall be straight from end to end and their surfaces should be plane and parallel to the general plane of the roof. The ridges and hips shall fit in squarely on the sheets.

21.6.4 Measurement

The measurements shall be taken for the finished work in length along the centre line of ridge or hip, as the case may be, correct

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to a cm. The laps in ridges and hips and between ridges and hips shall not be measured.

21.6.5 Rate

The rate shall include the cost of all labour and materials specified above, including painting, cost of seam bolts and any extra G.I. hook bolts, nuts and washers, required.

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LIST OF ACCEPTABLE/PREFERRED MAKES OF MATERIALS

Acceptable makes of materials to be used in the work are as given in the table below. In case of non-availability of these makes, the Engineer-in-charge may allow use of alternative makes. Only BIS marked materials shall be used in the work. Non-BIS marked materials may be permitted by the Engineer-in-charge only when BIS marked materials are not manufactured.

LIST OF ACCEPTABLE MAKES OF MATERIALS FOR CIVIL WORKS

S. No.	Materials	Name of Manufacturers
1.	(i) Ordinary Portland Cement/ Portland Pozzolana Cement.	A.C.C., Ultratech, Ambuja Cement, J.K. Cement, Century Cement, Shree Cement, Jaypee Cement, Vikram Cement, Wonder Cement, Cement Corporation of India Ltd (CCI)
	(ii) White Cement	Birla White, J.K. White.
2.	Reinforcement Steel	SAIL, Tata Steel, Rashtriya Ispat Nigam Ltd (RINL), JSW Steel Ltd., Jindal Steel Power Ltd. (JSPL).
3.	Water Proofing compounds, Admixtures, Plasticizer, Super Plasticizer, Curing Compounds	Fosroc, ROFF/Dr. Fixit (Pidilite Industries), CICO, Sika BASF, ArdexEndura (Bal Endura), Home Pride
4.	Integral water proofing compound with cement (For Plaster & Mortar)	Fosroc: Conplast 421 Dr. Fixit : LW+ Sika: Sikacin Asian Paints: Smart care Vitalia& equivalent product of BASF, CICO, ArdexEndura.
5. a)	a) Water proofing for bathroom/ toilet/ balcony & other wet areas	Fosroc: Brush Bond Dr. Fixit :Pidifine 2 K Sika: Nito Bond Asian Paints: Damp Block 2 K & equivalent product of BASF, CICO: (Tapecrete), ArdexEndura.
b)	TWO-PART POLYSULPHIDE SEALANT, Epoxy Repair Mortar, Epoxy Bonding Adhesive, Micro concrete, Primer for Polysulphide sealant.	Fosroc, Dr. Fixit, Sika, BASF, MAPEI.

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6.	Crystalline water proofing compound	Fosroc: Fosroco Crystalline Dr. Fixit: Dr. Fixit Crystalline Sika: Sika Crystalline Asian Paints: Crystalline Quart& equivalent product of BASF, CICO, ArdexEndura, Pentron.
7.	Grouts, Tile Adhesive	Latecrete, Kerokoal, BASF, ArdexEndura, Ferrous Crete, Home Pride
8.	Structural steel & M.S. Hollow tubes	SAIL, Tata Steel, RashtiryalspatNigan Ltd (RINL), JSW Steel Ltd., Jindal Steel & Power Ltd., APOLLO
9.	Polycarbonate Sheet	GE Plastic, LEXAN, MG Polyplast.
10.	Profile Steel Sheet (Precoated)	Ezydeck of TATA, Lloyd Superdesk, JSW/ Jindal
11.	Particle Board	Merino, Action TESA, Greenlam,
12.	Laminated Particle Board/ Laminates	Action TESA, Greenlam, Century Ply, Merino, Sunmica.
13.	Flush door Shutters	Duro Century, Durian, Green Ply, Archidply, Jayna (Jain Wood Industries), Jain Doors Pvt. Ltd., Orion doors. Note: Only ISI marked flush door to be used.
14.	Fire Rated Doors	Signum fire protection, Shakti Metdoor, NAVAIR, Sukri, Promat International, Jain Doors Pvt. Ltd.
15.	False Ceiling system	Armstrong, Hunter Douglas, USG Boral, Saint Gobain, Aerolite, Durlum, Interarch, GYPTECH
16.	Plywood/Veneer	Green Ply, Century, Merino, Duro, Durian.
17.	Melamine Polish	Asian Paints Melamine Gold, Wudfin of Pidilite, Timbertone of ICI Dulux.
18.	Floor Spring & Door Closure	Godrej, Dorma, Doorset, Kich.
19.	(a) Aluminium section	Hindalco, Jindal, Indian Aluminium Co.
	(b) Anodised Aluminium Hardware (Heavy Duty)	Kilong, Alualpha, Ebco, Classic.
	Clear/ Float/ Frosted/ Refractive/ Toughened/Coated Glass	Saint Gobain, AIS, Modiguard, Ashai Float, Gold Plus.
20.	Stainless Steel Railing, Accessories etc.	JINDAL, Dorma, Kich, GEZE, Hardwyn, Godrej.
21.	S.S. Door & window Fittings	JINDAL, Dorma, Kich, Doorset, Godrej,

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		Ozone.
22.	Silicon Based water repellent/ weather sealant	G.E. Plastics, Dow Corning, Wacker, BASF, Pidilite (Dr. Fixit/Roff)
23.	Poly-Sulphide Sealant	Fosroc, Pidilite (Dr. Fixit/roff), Sika, BASF.
24.	Mosaic Tiles/ Chequered Tiles	Ultra Tiles, NITCO, Hyper (Mayur), Pavcon.
25.	Ceramic Tiles	Somany, Kajaria, NITCO, Johnson, Marbitto, AGL (Asian), Orient bell, Somany. Note: The tiles manufactured at mother plant to be used, Sunheart Ceramix
26.	Vitrified Tiles (Antiskid/ Matt/ Glazed)	Somany, Kajaria, RAK, NITCO, Johnson, Marbitto, AGL (Asian), Orient bell. Note: The tiles manufactured at mother plant to be used, Sunheart Ceramix
27.	Paver Block & Kerb Stone	Pavcon, Hyper tile/Dynamic Industries/Mayur, KK, Power, Sarda, Navya, Krishna tiles (Awesome), Tulip (Local make to be modified by each zone), Bharat Tiles (B.T)
28.	Dash/Anchoring Fasteners	HILTI, Fischer, Bosch, Wurth.
29.	Cement Based wall putty	Birla Wall care, JK white, Berger, Asian Paints, Home Pride.
30.	Oil Bound Washable Acrylic Distemper	Asian Paints: (Professional Acrylic Distemper) Nerolac: Beauty Acrylic Distemper Berger: Bison Acrylic Distemper Dulux ICI: Maxilite
31.	1 st Quality Acrylic Distemper (washable/Ready mix/Low VOC)	Asian Paints (Tractor Aqua Lock Paint) Berger: Commando Or Equivalent paints of Nerolac or ICI-Dulux.
32.	Acrylic Emulsion Paints	Asian Paints (Professional Premium Interior Emulsion Paint) Nerolac: Beauty Gold Berger: Rangoli Total care ICI-Dulux: Super Cover
33.	Plastic Emulsion paint	Asian Paints (Apolite Heavy Duty Premium Emulsion Paint) Nerolac: Impression Berger: Easy Clean ICI-Dulux: 3 in 1

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34.	Premium Acrylic Emulsion Paints (Interior)	Asian Paints (Royale Luxury Emulsion) Nerolac: Impression Berger: Silk ICI-Dulux: Velvet touch.
35.	Textured Exterior Paint	Asian Paints, Nerolac, Berger Paints, Ultratech Paints, Luxture.
36.	Acrylic Smooth Exterior Paint	Asian Paints (Apex/Professional Exterior Emulsion) Nerolac: XL Berger: Weather Coat ICI-Dulux: Weather Shield
37.	Premium Acrylic Smooth Exterior Paints with Silicon Additive.	Asian Paints (Apex Ultima) Nerolac: XL Total Berger: Weather Coat all guard ICI-Dulux: Weather Shield Max
38.	Synthetic Enamel paint	Asian (ApolitePremium gloss enamel) Nerolac: Synthetic Hi gloss Berger: Luxol Hi gloss ICI-Dulux: Gloss Synthetic enamel.
39.	Cement Primer	Nerolac, Berger, BP White (Beger), Decoprime WT (Asian), White primer (ICI).
40.	Steel Primer (Red Oxide Zinc Chromate Primer)	Asian Paints, Nerolac, Berger, ICI
41.	Wood Primer	Asian paint (wood primer-white/pink), Berger, ICI, Nerolac.
42.	Epoxy Paint	Asian, Nerolac, Berger, ICI, Kansai Akzo Nobel.
43.	Fire Paint	Asian Paints, Akzo Nobel Coating India Ltd, PROMAT, Jotun
44.	G.I./ M.S Pipe	Tata, Jindal (Hisar), PrakashSurya.
45.	G.I. Fittings	Unik, AVR, Zoloto, PLATO
46.	HDPE Pipes	Reliance, Jain Pipes, ORIPLAST, Supreme
47.	DI PIPES	Elctrosteel, Jindal, TATA DUCTURA, Kapilansh, Kesoram.
48.	DI Fittings	Elctrosteel, Jindal, TATA DUCTURA, Kapilansh, Kesoram
49.	UPVC pipe and fittings	Astral, Supreme, Ashirwad, Prince, McCoy 8M
50.	Centrifugally cast (spun) Iron Pipes	NECO, Kapilansh, Electrosteel, SKF, HIF,

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	& fittings	RPMF.
51.	C.I Manhole Covers, Frames & GI Gratings	NECO, Raj Iron Foundary Agra, BIC, SKF, Kapilansh, RPMF.
52.	SFRC Manhole Cover & gratings	KK, JAIN, PARGATI, (Local make to be modified by each zone)
53.	WPC DOORS & FRAMES	ALSTONE, RAJSHRI
54.	WPC BOARDS	ALSTONE, RAJSHRI, AMULYA
55.	WPC Doors, Door Frames, Board, WPC Jaali	ALSTONE WPC
56.	Silicon Sealants	ALSTONE WPC
57.	CP Brass Fittings	Hindware, Jaquar, Parryware, CERA, ESS, Plumber, PLATO, PRIMA
58.	Sanitary ware, Fittings & accessories .	Hindware, CERA, Parryware, Jaquar, Somany. Kohler, Roca,
59.	Mirror Glass	Atul, Modi Guard, Golden Fish.
60.	CPVC Pipe & Fitting	Astral, Supreme, Ashirwad, Prince.
61.	Stainless Steel sink	Neelkanth, Nirali, Jayna, Silver Shine, PLATO
62.	RCC Pipes (NP-2)	Lakshmi, Sood&Sood, Jain Pipe Co. (Newai), Mahaveer Enterprises (Newai), work well spun pipes (Pali)
63.	UPVC Doors and windows (Profile makers and their authorized fabricators only)	Fenesta, VEKA, KOEMMERLING, RHEAU, Aluplast, Wintech, Duroplast, Marcolini (Mathura poly pack pvt. Ltd.),
64.	Extruded Polystyrene Insulation Board	DowCorning, Supreme, Texas, Analco.
65.	Heat Resistant Tiles	Swastik, Thermatek
66.	Gypsum Plaster	Ferrous Crete, Gyproc, Saint Gobain, Ultratech
67.	Floor Hardener	Ironite, Ferrok, Hardonate
68.	Modular Expansion Joint	Herculus, Sanfield India Ltd. Vexcolt,
69.	Glass Wool	Dow Corning, U.P. Twiga, Isover
70.	uPVC doors and window Hardware	Rotto, Dorset, Kinlong
71.	AAC Block Adhesive	Ultratech, ArdexEndura, Ferrous Crete.
72.	AAC Block	Biltech, Eco Green, Finecrete, J.K. Lakshmi Cement Ltd, Indo Bhutan
73.	Acoustical Wall Panelling, False	GYPTech

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	Ceiling system, Acoustical Doors & Windows.	
74.	Aluminium Composite Panel (ACP)	TimexBond, REYNOARCH
75.	PVC Water Storage Tank	SPL, Sintex, Vectus, Supreme, Ashirvad
76.	GI Chain Link fencing	Tata Wiron, Zonate Wire Industries, Armstrong Wires and Engineers Pvt. Ltd. or equivalent conform to IS 2721

- Note:-**
- 1) In case approved make for any material/item is not specified in the NIT, the decision of finalizing a particular brand shall rest with EE-Almora, CPWD, Almora.
 - 2) In case of non-availability of a particular material/Item from specified manufactures/make, the decision of EE-Almora, CPWD, Almora in selection of alternate manufacturer/make is final.
 - 3) Contractor shall provide minimum three samples of three different makes for each item of work as per the list of approved makes for the approval of Engineer-in-charge. It shall be the discretion of Engineer-in-charge to approve any of the three samples to be submitted by the contractor and decision of Engineer-in-charge in this regard shall be final and binding and nothing extra, whatsoever, shall be admissible on this ground.

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Annexure-I

**TO BE EXECUTED BY THE CONTRACTOR FOR REMOVAL OF
DEFECTS AFTER COMPLETION IN RESPECT OF STONE
WORKS/PLUM CONCRETE**

The agreement made this _____ day of _____ Two Thousand and _____ between _____ son of _____ (hereinafter called the GUARANTOR of the one part) and the PRESIDENT OF INDIA (hereinafter called the Government of the other part.)

WHEREAS THIS agreement is supplementary to a contract (Hereinafter called the Contract) dated _____ and made between the GUARANTOR OF THE ONE PART AND the Government of the other part, whereby the contractor inter alia, undertook to render the work in the said contract recited structurally stable workmanship, finishing and use of sound materials.

AND WHEREAS THE GUARANTOR agreed to give a guarantee to the effect that the said work will remain structurally stable and guaranteed against faulty workmanship, finishing and unsound materials and other related problems.

NOW THE GUARANTOR hereby guarantee that work executed by him will remain structurally stable after the expiry of maintenance period prescribed in the contract for the **minimum life of five years** to be reckoned from the date after expiry of maintenance period prescribed in the contract.

The decision of the Engineer-in-charge with regard to nature and cause of defect shall be final.

During this period of guarantee, the guarantor shall make good all defects to the satisfaction of the Engineer-in-charge calling upon him to rectify the defects failing which the work shall be got done by the Department by some other contractor at the Guarantor's risk and cost. The decision of the Engineer-in-Charge as to the cost payable by the Guarantor shall be final and binding.

That if the guarantor fails to make good all the defects, commits breach thereunder, then the guarantor will indemnify the principal and his successor against all loss, damage, cost expense or otherwise which may be incurred by him by reason of any default on the part of the GUARANTOR in performance and observance of this supplementary agreement. As to the amount of loss and/or damage and/or cost incurred by the Government, the decision of the Engineer-in-charge will be final and binding on both the parties.

IIN WITNESS WHEREOF these presents, have been executed by the obligator _____ and _____ by _____ for and on behalf of the PRESIDENT OF INDIA on the day, month and year first above written.

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SIGNED, sealed and delivered by OBLIGATOR in the presence of:

1. _____
2. _____

SIGNED FOR AND ON BEHALF OF THE PRESIDENT OF INDIA BY

_____ in the presence of :

1. _____
2. _____

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Annexure-II

GUARANTEE TO BE EXECUTED BY THE CONTRACTOR FOR REMOVAL OF DEFECTS AFTER COMPLETION IN RESPECT OF WATER SUPPLY, SANITARY AND DRAINAGE INSTALLATIONS

The agreement made this _____ day of _____ Two Thousand and _____ between _____ son of _____ (hereinafter called the GUARANTOR of the one part) and the PRESIDENT OF INDIA (hereinafter called the Government of the other part.)

WHEREAS THIS agreement is supplementary to a contract (Hereinafter called the Contract) dated _____ and made between the GUARANTOR OF THE ONE PART AND the Government of the other part, whereby the contractor inter alia, undertook to render the work in the said contract recited structurally stable workmanship, finishing and use of sound materials.

AND WHEREAS THE GUARANTOR agreed to give a guarantee to the affect that the said work will remain structurally stable and guaranteed against faulty workmanship, finishing, manufacturing defects of materials and leakages, etc.

NOW THE GUARANTOR hereby guarantee that work executed by him will remain structurally stable after expiry of maintenance period prescribed in the contract for the **minimum life of ten year** to be reckoned from the date after the expiry of maintenance period prescribed in the contract.

The decision of the Engineer-in-charge with regard to nature and cause of defect shall be final.

During this period of guarantee, the guarantor shall make good all defects to the satisfaction of the Engineer-in-charge calling upon him to rectify the defects failing which the work shall be got done by the Department by some other contractor at the Guarantor's cost and risk. The decision of the Engineer-in-Charge as to the cost, payable by the Guarantor shall be final and binding.

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That if the guarantor fails to make good all the defects commits breach thereunder, then the guarantor will indemnify the principal and his successor against all loss, damage, cost expense or otherwise which may be incurred by him by reason of any default on the part of the GUARANTOR in performance and observance of this supplementary agreement. As to the amount of loss and/or damage and or cost incurred by the Government, the decision of the Engineer-in-charge will be final and binding on both the parties.

IN WITNESS WHEREOF these presents, have been executed by the obligator _____ and _____ by _____ for and on behalf of the PRESIDENT OF INDIA on the day, _____ month and year first above written.

SIGNED, sealed and delivered by OBLIGATOR in the presence of:

1. _____
2. _____

SIGNED FOR AND ON BEHALF OF THE PRESIDENT OF INDIA BY _____ in the presence of :

1. _____
2. _____

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Annexure-III

**GUARANTEE TO BE EXECUTED BY CONTRACTORS FOR REMOVAL OF
DEFECTS AFTER COMPLETION IN RESPECT OF WATER PROOFING
WORKS IN BASEMENT/TERRACE/TOILETS.**

The agreement made this _____ day of _____ Two
Thousand and _____ between _____ son of
_____ of _____ (hereinafter called the
Guarantor of the one party) and the PRESIDENT OF INDIA (hereinafter called the
Government of the other party).

WHEREAS THIS agreement is supplementary to a contract (hereinafter called the Contract), dated _____ and made between the GUARANTOR OF THE ONE part and the Government of the one other part, whereby the Contractor, inter alia, undertook to render the buildings and structures in the said contract recited completely water and leak-proof.

AND WHEREAS THE GUARANTOR agreed to give a guarantee to the effect that the said structures will remain **water and leak-proof for ten years** from the date of giving of water proofing treatment.

NOW THE GUARANTOR hereby guarantees that water proofing treatment given by him will render the structures completely leak-proof and the **minimum life of such water proofing treatment shall be ten years** to be reckoned from the date after the maintenance period prescribed in the contract.

Providing that the Guarantor will not be responsible for leakage caused by earthquake or structural defects or misuse of roof or alteration and for such purpose.

- a) Misuse of roof shall mean any operation which will damage proofing treatment, like chopping of firewood and things of the same nature which might cause damage to the roof.
- b) Alteration shall mean construction of an additional storey or a part of the roof or construction adjoining to existing roof whereby proofing treatment is removed in parts.
- c) The decision of the Engineer-in-charge with regard to cause of leakage shall be final.

During the period of guarantee, the guarantor shall make good all defects and in case of any defect being found render the building water proof to the satisfaction of the Engineer-in-charge at his cost and shall commence the work for such rectification within seven days from the date of issue of the notice from the Engineer-in-charge calling upon him to rectify the defects failing which the work shall be got done by the Department by some other contractor at the GUARANTOR'S cost and risk. The decision of the Engineer-in-charge as to the cost, payable by the Guarantor shall be final and binding.

That if the Guarantor fails to execute the water proofing or commits breach there under, then the Guarantor will indemnify the Principal and his successor against all loss,

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damage, cost, expense or otherwise which may be incurred by him by reason of any default the part of the GUARANTOR in performance and observance of this supplementary agreement. As to the amount of loss and / or damage and / or cost incurred by the Government of decision of the Engineer-in-charge will be final and binding on the parties.

IN WITNESS WHEREOF these presents have been executed by the Obligor

 and by _____ and for and on behalf of the PRESIDENT OF INDIA on the day, month and year first above written.

Signed, sealed and delivered by Obligor in the presence of-

1. _____
2. _____

_____ Singed for and on behalf of the President of India by

_____ In the presence of –

1. _____
2. _____

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Annexure-IV

**TO BE EXECUTED BY THE CONTRACTOR FOR REMOVAL OF
DEFECTS AFTER COMPLETION IN RESPECT OF ALUMINUM
WORKS**

The agreement made this _____ day of _____ Two Thousand and _____ between _____ son of _____ (hereinafter called the GUARANTOR of the one part) and the PRESIDENT OF INDIA (hereinafter called the Government of the other part.)

WHEREAS THIS agreement is supplementary to a contract (Hereinafter called the Contract) dated _____ and made between the GUARANTOR OF THE ONE PART AND the Government of the other part, whereby the contractor inter alia, undertook to render the work in the said contract recited structurally stable workmanship, finishing and use of sound materials.

AND WHEREAS THE GUARANTOR agreed to give a guarantee to the affect that the said work will remain structurally stable and guaranteed against faulty workmanship, finishing and unsound materials and other related problems.

NOW THE GUARANTOR hereby guarantee that work executed by him will remain structurally stable after the expiry of maintenance period prescribed in the contract for the **minimum life of five years** to be reckoned from the date after expiry of maintenance period prescribed in the contract.

The decision of the Engineer-in-charge with regard to nature and cause of defect shall be final.

During this period of guarantee, the guarantor shall make good all defects to the satisfaction of the Engineer-in-charge calling upon him to rectify the defects failing which the work shall be got done by the Department by some other contractor at the

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Guarantor's risk and cost. The decision of the Engineer-in-Charge as to the cost payable by the Guarantor shall be final and binding.

That if the guarantor fails to make good all the defects, commits breach thereunder, then the guarantor will indemnify the principal and his successor against all loss, damage, cost expense or otherwise which may be incurred by him by reason of any default on the part of the GUARANTOR in performance and observance of this supplementary agreement. As to the amount of loss and/or damage and/or cost incurred by the Government, the decision of the Engineer-in-charge will be final and binding on both the parties.

IIN WITNESS WHEREOF these presents, have been executed by the obligator
 _____ and
 _____ by _____ for and
 on behalf of the PRESIDENT OF INDIA on the day, month and year first above written.

SIGNED, sealed and delivered by OBLIGATOR in the presence of:

1. _____
2. _____

SIGNED FOR AND ON BEHALF OF THE PRESIDENT OF INDIA BY

_____ in the presence of :

1. _____
2. _____

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Annexure-V

**GUARANTEE TO BE EXECUTED BY THE CONTRACTOR FOR REMOVAL OF
DEFECTS AFTER COMPLETION IN RESPECT OF ALL TYPES OF ROAD
WORK**

The agreement made this _____ day of _____ Two Thousand and _____ between _____ son of _____ (hereinafter called the GUARANTOR of the one part) and the PRESIDENT OF INDIA (hereinafter called the Government of the other part.)

WHEREAS THIS agreement is supplementary to a contract (Hereinafter called the Contract) dated _____ and made between the GUARANTOR OF THE ONE PART AND the Government of the other part, whereby the contractor inter alia, undertook to render the work in the said contract recited structurally stable workmanship, finishing and use of sound materials.

AND WHEREAS THE GUARANTOR agreed to give a guarantee to the effect that the said work will remain structurally stable and guaranteed against faulty workmanship, finishing and unsound materials and other related problems.

NOW THE GUARANTOR hereby guarantee that work executed by him will remain structurally stable after the expiry of maintenance period prescribed in the contract for the **minimum life of 3 (Three) years** to be reckoned after the date of completion of the work.

The decision of the Engineer-in-charge with regard to nature and cause of defect shall be final. During this period of guarantee, the guarantor shall make good all defects to the satisfaction of the Engineer-in-charge calling upon him to rectify the defects failing which the work shall be got done by the Department by some other contractor at the Guarantor's risk and cost. The decision of the Engineer-in-Charge as to the cost payable by the Guarantor shall be final and binding.

That if the guarantor fails to make good all the defects, commits breach thereunder, then the guarantor will indemnify the principal and his successor against all loss, damage, cost expense or otherwise which may be incurred by him by reason of any default on the part of the GUARANTOR in performance and observance of this supplementary agreement. As to the amount of loss

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and/or damage and/or cost incurred by the Government, the decision of the Engineer-in-charge will be final and binding on both the parties.

IIN WITNESS WHEREOF these presents, have been executed by the obligator _____

and

_____ by _____
for and on behalf of the PRESIDENT OF INDIA on the day, month and year first above written.

SIGNED, sealed and delivered by OBLIGATOR in the presence of:

1. _____
2. _____

SIGNED FOR AND ON BEHALF OF THE PRESIDENT OF INDIA BY _____ in the presence of:

1. _____
2. _____

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Annexure-VI

On non-judicial stamp paper of minimum Rs. 100

(Guarantee offered by Bank to CPWD in connection with the execution of contracts) Form of Bank Guarantee for Earnest Money Deposit /Performance Guarantee/ Security Deposit/ Mobilization Advance/ Refund of milestone withheld amount

1. Whereas the Executive Engineer, on behalf of the President of India (here in after called "The Government") has invited bids under(NIT number)..... dated for (name of work) The Government has further agreed to accept irrevocable Bank Guarantee for Rs. (Rupees only) valid upto (date)*..... as Earnest Money Deposit from (name and address of contractor)(herein after called "the contractor") for compliance of his obligations in accordance with the terms and conditions of the said NIT.

OR**

Whereas the Executive Engineer, on behalf of the President of India (here in after called "The Government") has entered into an agreement bearing number..... with(name and address of the contractor) (hereinafter called "the Contractor") for execution of work (name of work)..... The Government has further agreed to accept an irrevocable Bank Guarantee for Rs. (Rupees only) valid upto (date).....as Performance Guarantee/Security Deposit/Mobilization Advance/Refund of mile stone withheld amount from the said Contractor for compliance of his obligations in accordance with the terms and conditions of the agreement.

2. We, (indicate the name of the bank) (herein after referred to as "the Bank"), here by undertake to pay to the Government an amount not exceeding Rs. (Rupees.....only) on demand by the Government within 10 days of the demand.

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3. We,(indicate the name of the Bank), do here by undertake to pay the amount due and payable under this guarantee without any demur, merely on a demand from the Government stating that the amount claimed is required to meet the recoveries due or likely to be due from the said Contractor. Any such demand made on the Bank shall be conclusive as regards the amount due and payable by the Bank under this Guarantee. However, our liability under this guarantee shall be restricted to an amount not exceeding Rs.(Rupeesonly)
4. We, (indicate the name of the Bank), further undertake to pay the Government any money so demanded not withstanding any dispute or disputes raised by the contractor in any suit or proceeding pending On non-judicial stamp paper of minimum Rs. 100 before any Court or Tribunal, our liability under this Bank Guarantee being absolute and unequivocal. The payment so made by us under this Bank Guarantee shall be a valid discharge of our liability for payment there under and the Contractor shall have no claim against us for making such payment.
5. We, (indicate the name of the Bank), further agree that the Government shall have the fullest liberty without our consent and without affecting in any manner our obligation here under to vary any of the terms and conditions of the said agreement or to extend time of performance by the said Contractor from time to time or to postpone for any time or from time to time any of the powers exercisable by the Government against the said contractor and to forbear or enforce any of the terms and conditions relating to the said agreement and we shall not be relieved from our liability by reason of any such variation or extension being granted to the said Contractor or for any forbearance, act of omission on the part of the Government or any indulgence by the Government to the said Contractor or by any such matter or thing what so ever which under the law relating to sureties would, but for this provision, have effect of so relieving us.
6. We, (indicate the name of the Bank), further agree that the Government at its option shall be entitled to enforce this Guarantee against the Bank as a principal debtor at the first instance without proceeding against the Contractor and not withstanding any security or other guarantee the Government may have in relation to the Contractor's liabilities.
7. This guarantee will not be discharged due to the change in the constitution of the Bank or the Contractor.

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8. We, (indicate the name of the Bank), undertake not to revoke this guarantee except with the consent of the Government in writing.
9. This Bank Guarantee shall be valid up to unless extended on demand by the Government. Not with standing anything mentioned above, our liability against this guarantee is restricted to Rs. (Rupees only) and unless a claim in writing is lodged with us within the date of expiry or extended date of expiry of this guarantee, all our liabilities under this guarantee shall stand discharged.

Date

Witnesses:

1. Signature.....
signatory
Name and address

Authorized

Name
Designation
Staff code no.

2. Signature
Name and address

Bank seal

*Date to be worked out on the basis of validity period of 90 days where only financial bids are invited and 180 days for two/three bid system from the date of submission of tender.

**In paragraph 1, strike out the portion not applicable. Bank Guarantee will be made either for earnest money or for performance guarantee/security deposit/mobilization advance/Refund of mile stone withheld amount, as the case may be.

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Annexure-VII

AFFIDAVIT

I/We have submitted a bank guarantee for the work _____

_____ (Name of work)
Agreement _____ No.

Dated _____ from

_____ (Name of the Bank with full address)

to the Executive Engineer _____ with a
view

_____ (Name of the Division)
to seek exemption from payment of performance guarantee in cash. This Bank
guarantee expires on _____. I / We
undertake to keep the validity of the bank guarantee intact by getting it extended from
time to time at my / our own initiative up to a period of _____
months after the recorded date of completion of the work or as directed by the Engineer
in charge.

I / We also indemnify the Government against any losses arising out of non-
encasement of the bank guarantee if any.

(Deponent)
Signature of Contractor

Note : The affidavit is to be given by the Executants before a first class Magistrate.

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Annexure-VIII**FORM "A"****DECLARATION ABOUT SITE INSPECTION**

To,

The Executive Engineer
CPWD, Almora

Subject:- Construction of Main Gate and Boundary wall with various miscellaneous repair works at Kendriya Vidyalaya Almora (UK).

NIT No. :- 23/EE/ACD/CPWD/Almora/2026-27

Dear Sir,

It is hereby declared that as per CPWD 6 for e-tendering, I/we the bidder inspected and examined the subject site and its surroundings and satisfy my self/our selves before submitting my/our bids as to the nature of the ground and sub-soil (so far as is practicable), the form and nature of the site, the means of access to the site, the accommodation I/we may require and in general shall myself/ourselves obtain all necessary information as to risks, contingencies and other circumstances which may influence or affect my/our bid. I/we the bidder shall have full knowledge of the site and no extra Charges consequent upon any misunderstanding or otherwise shall be claimed at later date. I/we the bidder shall be responsible for arranging and maintaining at own cost all materials, tools & plants, water, electricity, access, facilities for workers and all other services required for executing the work unless otherwise specifically provided for in the contract documents. Submission of a bid by me/us implies that I/we has read this notice and all other contract documents and has made myself/ourselves aware of the scope and specifications of the work to be done and of local conditions and other factors having a bearing on cost on the execution of the work.

Yours faithfully

(Seal & Signature)
(Bidder/Duly authorized signatory of the bidder)

Seal & Signature with date
(Assistant Engineer)

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Note:-

- (i) Before Inspection of Site, **Sh. Dhiraj Sharma (Mob. No. 9368158445)** concerned Assistant Engineer shall be contacted so that he may remain present during inspection of site by the bidder and signature of concerned Assistant Engineer shall be obtained in Form "A" as proof of site visited by the bidder and uploaded.
- (ii) Alternatively, the agency shall have liberty to take photographs of site inspection carried out by the agency (face of agency representative should be visible) along with GPS location of site displayed on the photographs and shall be uploaded along with form "A" without having signature of concerned assistant Engineer.
- (iii) If Form "A" is not having signature of concerned Assistant Engineer or Form "A" not having photographs of inspection of site along with GPS location, the bid of the bidder shall be treated as ineligible and bid of bidder shall not be opened.

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<u>SCHEDULE OF QUANTITY</u>					
Name of Work: Construction of Main Gate and Boundary wall with various miscellaneous repair works at Kendriya Vidyalaya Almora (UK).					
S.No.	Description	Qty	Unit	Rate	Amount
1	EARTH WORK				
1.1	Earth work in excavation by mechanical means (Hydraulic excavator)/manual means over areas (exceeding 30 cm in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and for all lift, as directed by Engineer-in-charge.				
1.1.1	All kinds of soil.	89.00	cum	222.80	19829.00
1.2	Earth work in excavation by mechanical means (Hydraulic excavator)/manual means over areas (exceeding 30 cm in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth lead upto 50 m and lift upto 1.5 m, as directed by Engineer-in-charge.				
1.2.1	Ordinary rock	31.00	cum	626.25	19414.00
1.3	Earth work in excavation by mechanical means (Hydraulic excavator) / manual means in foundation trenches or drains (not exceeding 1.5 m in width or 10 sqm on plan), including dressing of sides and ramming of bottoms, for all lift, including getting out the excavated soil and disposal of surplus excavated soil as directed, within a lead of 50 m.				
1.3.1	All kinds of soil.	14.00	cum	326.70	4574.00
1.4	Filling available excavated earth (excluding rock) in trenches, plinth, sides of foundations etc. in layers not exceeding 20cm in depth, consolidating each deposited layer by ramming and watering, lead up to 50 and for all lift.	81.00	cum	246.00	19926.00
2	CONCRETE WORK				
2.1	Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work up to plinth level :				
2.1.1	1:1½:3 (1 Cement: 1½ coarse sand (zone-III) derived from natural sources : 3 graded stone aggregate 20 mm nominal size	5.00	cum	10469.20	52346.00

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	derived from natural sources)				
2.1.2	1:5:10 (1 cement : 5 coarse sand (zone-III) derived from natural sources : 10 graded stone aggregate 40 mm nominal size derived from natural sources)	12.00	cum	8181.95	98183.00
2.2	Centering and shuttering including strutting, propping etc. and removal of form work for :				
2.2.1	Foundations, footings, bases for columns	21.00	sqm	492.20	10336.00
3	REINFORCED CEMENT CONCRETE				
3.1	Providing and laying in position specified grade of reinforced cement concrete, excluding the cost of centering, shuttering, finishing and reinforcement - All work up to plinth level :				
3.1.1	1:1.5:3 (1 cement : 1.5 coarse sand (zone-III) derived from natural sources : 3 graded stone aggregate 20 mm nominal size derived from natural sources)	21.00	cum	11353.95	238433.00
3.2	Reinforced cement concrete work in walls (any thickness), including attached pilasters, buttresses, plinth and string courses, fillets, columns, pillars, piers, abutments, posts and struts etc. above plinth level up to floor five level, excluding cost of centering, shuttering, finishing and reinforcement :				
3.2.1	1:1.5:3 (1 cement : 1.5 coarse sand(zone-III) derived from natural sources : 3 graded stone aggregate 20 mm nominal size derived from natural sources)	10.00	cum	13622.25	136223.00
3.3	Reinforced cement concrete work in beams, suspended floors, roofs having slope up to 15° landings, balconies, shelves, chajjas, lintels, bands, plain window sills, staircases and spiral stair cases above plinth level up to floor five level, excluding the cost of centering, shuttering, finishing and reinforcement with 1:1.5:3 (1 cement : 1.5 coarse sand(zone-III) derived from natural sources : 3 graded stone aggregate 20 mm nominal size derived from natural sources).	3.00	cum	14441.35	43324.00
3.4	Centering and shuttering including strutting, propping etc. and removal of form for				
3.4.1	Foundations, footings, bases of columns, etc. for mass concrete	31.00	sqm	492.20	15258.00

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3.4.2	Suspended floors, roofs, landings, balconies and access platform	4.00	sqm	1163.85	4655.00
3.4.3	Lintels, beams, plinth beams, girders, bressumers and cantilevers	59.00	sqm	924.30	54534.00
3.4.4	Columns, Pillars, Piers, Abutments, Posts and Struts	96.00	sqm	1206.60	115834.00
3.4.5	Small lintels not exceeding 1.5 m clear span, moulding as in cornices, window sills, string courses, bands, copings, bed plates, anchor blocks and the like	10.00	sqm	492.20	4922.00
3.5	Extra for additional height in centering, shuttering where ever required with adequate bracing, propping etc., including cost of de-shuttering and decentering at all levels, over a height of 3.5 m, for every additional height of 1 metre or part thereof (Plan area to be measured).				
3.5.1	Suspended floors, roofs, landing, beams and balconies (Plan area to be measured)	10.00	sqm	482.35	4824.00
3.6	Steel reinforcement for R.C.C. work including straightening, cutting, bending, placing in position and binding all complete upto plinth level.				
3.6.1	Thermo-Mechanically Treated bars of grade Fe-500D or more.	3400.00	Kg.	135.40	460360.00
4	MASONRY WORK				
4.1	Brick work with common burnt clay F.P.S. (non modular) bricks of class designation 7.5 in foundation and plinth in:				
4.1.1	Cement mortar 1:6 (1 cement : 6 coarse sand)	21.00	cum	8952.20	187996.00
4.2	Brick work with common burnt clay F.P.S. (non modular) bricks of class designation 7.5 in superstructure above plinth level up to floor V level in all shapes and sizes in :				
4.2.1	Cement mortar 1:6 (1 cement : 6 coarse sand)	7.00	cum	11429.55	80007.00
5	STONE WORK				
5.1	Random rubble masonry with hard stone in foundation and plinth including levelling up with cement concrete 1:6:12 (1 cement : 6 coarse sand : 12 graded stone aggregate 20 mm nominal size) upto plinth level with :				
5.1.1	Cement mortar 1:6 (1 cement : 6 coarse sand)	22.00	cum	9176.85	201891.00

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5.2	Random rubble masonry with hard stone in superstructure above plinth level and upto floor five level, including leveling up with cement concrete 1:6:12 (1 cement : 6 coarse sand : 12 graded stone aggregate 20 mm nominal size) at window sills, ceiling level and the like.				
5.2.1	Cement mortar 1:6 (1 cement : 6 coarse sand)	19.00	cum	11642.25	221203.00
6	CLADDING WORK				
6.1	Providing and fixing 18 mm thick gang saw cut, mirror polished, premoulded and prepolished, machine cut for kitchen platforms, vanity counters, window sills, facias and similar locations of required size, approved shade, colour and texture laid over 20 mm thick base cement mortar 1:4 (1 cement : 4 coarse sand), joints treated with white cement, mixed with matching pigment, epoxy touch ups, including rubbing, curing, moulding and polishing of edges to give high gloss finish etc. complete at all levels.				
6.1.1	Granite stone slab of colour black, Cherry/Ruby red				
6.1.1.1	Area of slab upto 0.50 sqm	1.00	sqm	6794.90	6795.00
6.2	Designing, fabricating, testing, installing and fixing in position Curtain Wall with Aluminium Composite Panel Cladding, with open grooves for linear as well as curvilinear portions of the building , for all heights and all levels etc. including: (a) Structural analysis & design and preparation of shop drawings for pressure equalisation or rain screen principle as required, proper drainage of water to make it watertight including checking of all the structural and functional design. (b) Providing, fabricating and supplying and fixing panels of aluminium composite panel cladding in pan shape in metallic colour of approved shades made out of 4mm thick aluminium composite panel material consisting of 3mm thick FR grade mineral core sandwiched between two Aluminium sheets (each 0.5mm thick). The aluminium composite panel cladding sheet shall be coil coated, with Kynar 500 based PVDF / Lumiflon based fluoropolymer resin coating of approved colour and shade on face # 1 and polymer (Service) coating on face # 2 as specified using stainless steel screws, nuts, bolts, washers, cleats, weather silicone sealant, backer rods etc. (c) The fastening brackets of Aluminium alloy 6005 T5 / MS with Hot Dip Galvanised with serrations and serrated washers to arrest the wind load movement, fasteners, SS 316 Pins and anchor bolts of	107.00	sqm	6294.75	673538.00

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	approved make in SS 316, Nylon separators to prevent bi-metallic contacts all complete required to perform as per specification and drawing The item includes cost of all material & labour component, the cost of all mock ups at site, cost of all samples of the individual components for testing in an approved laboratory, field tests on the assembled working curtain wall with aluminium composite panel cladding, cleaning and protection of the curtain wall with aluminium composite panel cladding till the handing over of the building for occupation. Base frame work for ACP cladding is payable under the relevant aluminium item.s The Contractor shall provide curtain wall with aluminium composite panel cladding, having all the performance characteristics all complete , as per the Architectural drawings, as per item description, as specified, as per the approved shop drawings and as directed by the Engineer-in-Charge. However, for the purpose of payment, only the actual area on the external face of the curtain wall with Aluminum Composite Panel Cladding (including width of groove) shall be measured in sqm. up to two decimal places.				
7	WOOD & PVC WORK				
7.1	Providing and fixing ISI marked flush door shutters conforming to IS : 2202 (Part I) non-decorative type, core of block board construction with frame of 1st class hard wood and well matched commercial 3 ply veneering with vertical grains or cross bands and face veneers on both faces of shutters:				
7.1.1	35 mm thick including ISI marked Stainless Steel butt hinges with necessary screws	2.00	sqm	3003.20	6006.00
7.2	Providing and fixing IS : 12817 marked stainless steel butt hinges with stainless steel screws etc. complete :				
7.2.1	100X58X1.90 mm	3.00	each	134.40	403.00
7.3	Providing and fixing aluminium sliding door bolts, ISI marked anodised (anodic coating not less than grade AC 10 as per IS : 1868), transparent or dyed to required colour or shade, with nuts and screws etc. complete :				
7.3.1	300x16 mm	2.00	each	380.60	761.00
7.4	Providing and fixing aluminium tower bolts, ISI marked, anodised (anodic coating not less than grade AC 10 as per IS : 1868) transparent or dyed to required colour or shade, with necessary screws etc. complete :				
7.4.1	250x10 mm	2.00	each	144.55	289.00

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7.5	Providing and fixing aluminium handles, ISI marked, anodised (anodic coating not less than grade AC 10 as per IS : 1868) transparent or dyed to required colour or shade, with necessary screws etc. complete :				
7.5.1	125 mm	2.00	each	83.15	166.00
7.6	Providing and fixing aluminium hanging floor door stopper, ISI marked, anodised (anodic coating not less than grade AC 10 as per IS : 1868) transparent or dyed to required colour and shade, with necessary screws etc. complete.				
7.6.1	Twin rubber stopper	2.00	each	90.80	182.00
8	STEEL WORK				
8.1	Structural steel work riveted, bolted or welded in built up sections, trusses and framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete.	700.00	Kg	167.85	117495.00
8.2	Providing and fixing 1mm thick M.S. sheet door with frame of 40x40x6 mm angle iron and 3 mm M.S. gusset plates at the junctions and corners, all necessary fittings complete, including applying a priming coat of approved steel primer.				
8.2.1	Using M.S. angels 40x40x6 mm for diagonal braces	1.00	sqm	7285.45	7285.00
8.3	Providing and fixing factory made ISI marked steel glazed doors, windows and ventilators, side /top /centre hung, with beading and all members such as F7D,F4B, K11 B and K12 B etc. complete of standard rolled steel sections, joints mitred and flash butt welded and sash bars tenoned and riveted, including providing and fixing of hinges, pivots, priming coat of approved steel primer, but excluding the cost of other fittings,glass panes complete all as per approved design, (sectional weight of only steel members shall be measured for payment).				
8.3.1	Fixing with 15x3 mm lugs 10 cm long embedded in cement concrete block 15x10x10 cm of C.C. 1:3:6 (1 Cement : 3 coarse sand : 6 graded stone aggregate 20 mm nominal size)	592.00	Kg.	164.45	97354.00

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8.4	Steel work in built up tubular (round, square or rectangular hollow tubes etc.) trusses etc., including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer, including welding and bolted with special shaped washers etc. complete.				
8.4.1	Hot finished welded type tubes	2305.00	Kg.	244.00	562420.00
8.5	Providing and fixing carbon steel galvanised (minimum coating 5 micron) dash fastener of 10 mm dia double threaded 6.8 grade (yield strength 480 N/mm ²), counter sunk head, comprising of 10 mm dia polyamide PA 6 grade sleeve, including drilling of hole in frame , concrete/ masonry, etc. as per direction of Engineer-in-charge.				
8.5.1	10 x 120 mm	16.00	Nos	214.60	3434.00
8.6	Providing & fixing fly proof wire gauze to windows, clerestory windows & doors with M.S. Flat 15x3 mm and nuts & bolts complete.				
8.6.1	Stainless steel (grade 304) wire gauze of 0.5 mm dia wire and 1.4 mm aperture on both sides	54.00	sqm	1422.80	76831.00
8.7	Providing & fixing glass panes with putty and glazing clips in steel doors, windows, clerestory windows, all complete with :				
8.7.1	4.0 mm thick glass panes	27.00	sqm	1336.30	36080.00
9	FLOORING				
9.1	Chequerred precast cement concrete tiles 22 mm thick in footpath & courtyard, jointed with neat cement slurry mixed with pigment to match the shade of tiles, including rubbing and cleaning etc. complete, on 20 mm thick bed of cement mortar 1:4 (1 cement: 4 coarse sand).				
9.1.1	Dark shade pigment using ordinary cement	6.00	sqm	1405.15	8431.00
9.2	Kota stone slab flooring over 20 mm (average) thick base laid over and jointed with grey cement slurry mixed with pigment to match the shade of the slab, including rubbing and polishing complete with base of cement mortar 1 : 4 (1 cement : 4 coarse sand) :				
9.2.1	25 mm thick	31.00	sqm	2445.40	75807.00

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9.3	Kota stone slabs 20 mm thick in risers of steps, skirting, dado and pillars laid on 12 mm (average) thick cement mortar 1:3 (1 cement: 3 coarse sand) and jointed with grey cement slurry mixed with pigment to match the shade of the slabs, including rubbing and polishing complete.	16.00	sqm	2955.50	47288.00
9.4	Providing and laying vitrified floor tiles in different sizes (thickness to be specified by the manufacturer) with water absorption less than 0.08% and conforming to IS: 15622, of approved make, in all colours and shades, laid on 20mm thick cement mortar 1:4 (1 cement : 4 coarse sand), jointing with grey cement slurry @ 3.3 kg/ sqm including grouting the joints with white cement and matching pigments etc., complete.				
9.4.1	Size of Tile 600x600 mm	4.00	sqm	1949.85	7799.00
9.5	Providing and laying Vitrified tiles in different sizes (thickness to be specified by manufacturer), with water absorption less than 0.08 % and conforming to I.S. 15622, of approved make, in all colours & shade, in skirting, riser of steps, over 12 mm thick bed of cement mortar 1:3 (1 cement: 3 coarse sand), jointing with grey cement slurry @ 3.3 kg/ sqm including grouting the joint with white cement & matching pigments etc. complete.				
9.5.1	Size of Tile 600x600 mm	1.00	sqm	2037.25	2037.00
10	ROOFING				
10.1	Providing and fixing on wall face unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion, (i) Single socketed pipes.				
10.1.1	110 mm diameter	8.00	meter	473.70	3790.00
10.2	Providing and fixing on wall face unplasticised - PVC moulded fittings/ accessories for unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion.				
10.2.1	Bend 87.5°				
10.2.1.1	110 mm	4.00	each	188.75	755.00
10.2.2	Shoe (Plain)				

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10.2.2.1	110 mm	2.00	each	165.50	331.00
10.3	Providing and fixing precoated galvanised iron profile sheets (size, shape and pitch of corrugation as approved by Engineer-in-Charge) of total coated thickness 0.50mm (base metal of minimum 0.45mm thickness with total coating thickness of 0.05mm) with zinc coating 120 grams per sqm as per IS: 277, in 240 mpa steel grade, 5-7 microns epoxy primer on both side of the sheet and polyester top coat 15-18 microns. Sheet should have protective guard film of 25 microns minimum to avoid scratches during transportation and should be supplied in single length upto 12 metre or as desired by Engineer-in-charge. The sheet shall be fixed using self drilling /self tapping screws of size (5.5x 55 mm) with EPDM seal, complete upto any pitch in horizontal/ vertical or curved surfaces, excluding the cost of purlins, rafters and trusses and including cutting to size and shape wherever required.	139.00	sqm	927.10	128867.00
10.4	Providing and fixing precoated galvanised steel sheet roofing accessories of total coated thickness 0.50mm (base metal of minimum 0.45mm thickness with total coating thickness of 0.05mm) with Zinc coating 120 grams per sqm as per IS: 277, in 240 mpa steel grade, 5-7 microns epoxy primer on both side of the sheet and polyester top coat 15-18 microns using self drilling/ self tapping screws complete :				
10.4.1	Ridges plain (500 - 600mm)	11.00	meter	668.20	7350.00
10.4.2	Gutter (600 mm over all girth)	22.00	meter	1648.90	36276.00
11	FINISHING				
11.1	12 mm cement plaster of mix :				
11.1.1	1:4 (1 cement: 4 coarse sand)	119.00	sqm	448.55	53377.00
11.2	15 mm cement plaster on rough side of single or half brick wall of mix:				
11.2.1	1:6 (1 cement: 6 coarse sand)	124.00	sqm	496.25	61535.00
11.3	Cement plaster 1:3 (1 cement: 3 coarse sand) finished with a floating coat of neat cement.				
11.3.1	12 mm cement plaster	2.00	sqm	564.25	1129.00
11.4	6 mm cement plaster of mix :				
11.4.1	1:3 (1 cement : 3 fine sand)	14.00	sqm	377.15	5280.00

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11.5	Pointing on stone work with cement mortar 1:3 (1 cement : 3 fine sand) :				
11.5.1	Raised and cut pointing	33.00	sqm	879.90	29037.00
11.6	Distempering with 1st quality acrylic distemper (ready mixed) having VOC content less than 50 gram/litre, of approved manufacturer and of required shade and colour all complete to achieve even shade and colour :				
11.6.1	New work (two or more coats) over and including water thinnable priming coat with cement primer having VOC content less than 50 gram/litre	412.00	sqm	233.05	96017.00
11.7	Finishing walls with ready mixed Premium acrylic emulsion paint (Company Depot Tinted) having VOC less than 50 gm/litre and UV resistance as per IS 15489:2004, Alkali & fungal resistance, dirt resistance exterior paint of required shade with silicon additives.				
11.7.1	New work (Two or more coats applied @ 1.43 litre/ 10 sqm over and including priming coat of exterior primer applied @ 0.90 litre/10 sqm.	266.00	sqm	227.50	60515.00
11.8	Painting with synthetic enamel paint of approved brand and manufacture to give an even shade :				
11.8.1	Two or more coats on new work	272.00	sqm	195.70	53230.00
11.9	Providing and applying white cement based putty of average thickness 1 mm, of approved brand and manufacturer, over the plastered wall surface to prepare the surface even and smooth complete.	678.00	sqm	195.90	132820.00
11.10	Removing dry or oil bound distemper, water proofing cement paint and the like by scrapping, sand papering and preparing the surface smooth including necessary repairs to scratches etc. complete.	412.00	sqm	31.55	12999.00
12	DISMANTLING AND DEMOLISHING				
12.1	Demolishing cement concrete manually/ by mechanical means including disposal of material within 50 metres lead as per direction of Engineer - in - charge.				
12.1.1	Nominal concrete 1:3:6 or richer mix (i/c equivalent design mix)	5.00	cum	3055.45	15277.00
12.2	Demolishing brick work manually/ by mechanical means including stacking of serviceable material and disposal of unserviceable material within 50 metres				

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	lead as per direction of Engineer-in-charge.				
12.2.1	In cement mortar	17.00	cum	2585.85	43959.00
12.3	Demolishing stone rubble masonry manually/ by mechanical means including stacking of serviceable material and disposal of unserviceable material within 50 metres lead as per direction of Engineer-in-charge :				
12.3.1	In cement mortar	28.00	cum	3086.40	86419.00
13	ROAD WORK				
13.1	Providing and fixing concertina coil fencing with punched tape concertina coil 600 mm dia 10 metre openable length (total length 90 m), having 50 nos rounds per 6 metre length, upto 3 m height of wall with existing angle iron 'Y' shaped placed 2.4m or 3.00 m apart and with 9 horizontal R.B.T. reinforced barbed wire, stud tied with G.I. staples and G.I. clips to retain horizontal, including necessary bolts or G.I. barbed wire tied to angle iron, all complete as per direction of Engineer-in-charge, with reinforced barbed tape(R.B.T.) / Spring core (2.5mm thick) wire of high tensile strength of 165 kg/ sq.mm with tape (0.52 mm thick) and weight 43.478 gm/ metre (cost of M.S. angle, C.C. blocks shall be paid separately)	25.00	metre	471.70	11793.00
13.2	Manufacturing, supplying and fixing retro reflective overhead signage boards made up of 2 mm thick aluminium sheet, face to be fully covered with high intensity and encapsulated lens type heat activated retro reflective sheeting conforming to type - III of ASTM-D-4956-01 as approved by Engineer-in-charge, letters, borders etc. as per IRC : 67-2001 in silver white with blue colour back ground and with high intensity grade, pasted on substrate by pressure sensitive adhesive backing which shall be activated by applying pressure conforming to class II of ASTM-D-4956-01 and fixing the same to the plate of structural frame work by means of suitable sized aluminium alloys, rivets or bolts & nuts @ 300 mm centre to centre all along the periphery as well as in two vertical rows along with theft resistant measures, including the cost of painting with two or more coats of epoxy paint in grey colour on the back side of aluminium sheet including appropriate priming coat. The rate includes				

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	the cost of rounding off the corners, lowering down the structural frame work from the gantry, fixing and erecting the same in position all complete as per drawings, specification and direction of the engineer-in-charge.(Structural frame work including M.S. plate to be provided separately. Rectangular area of the sheet only shall be measured for payment).				
13.2.1	Overhead informatory road signage	4.00	sqm	7380.30	29521.00
14	ALUMINIUM WORK				
14.1	Providing and fixing aluminium work for doors, windows, ventilators and partitions with extruded built up standard tubular sections/ appropriate Z sections and other sections of approved make conforming to IS: 733 and IS: 1285, fixing with dash fasteners of required dia and size, including necessary filling up the gaps at junctions, i.e. at top, bottom and sides with required EPDM rubber/ neoprene gasket etc. Aluminium sections shall be smooth, rust free, straight, mitred and jointed mechanically wherever required including cleat angle, Aluminium snap beading for glazing / paneling, C.P. brass / stainless steel screws, all complete as per architectural drawings and the directions of Engineer-in-charge. (Glazing, paneling and dash fasteners to be paid for separately) :				
14.1.1	For fixed portion				
14.1.1.1	Powder coated aluminium (minimum thickness of powder coating 50 micron)	19.00	Kg.	666.35	12661.00
14.2	Providing and fixing glazing in aluminium door, window, ventilator shutters and partitions etc. with EPDM rubber / neoprene gasket etc. complete as per the architectural drawings and the directions of engineer-in-charge . (Cost of aluminium snap beading shall be paid in basic item):				
14.2.1	With float glass panes of 5 mm thickness (weight not less than 12.50 kg/sqm)	3.00	sqm	1889.35	5668.00

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14.3	Providing and fixing anodised aluminium grill (anodised transparent or dyed to required shade according to IS: 1868 with minimum anodic coating of grade AC 15) of approved design/pattern, with approved standard section and fixed to the existing window frame with C.P. brass/ stainless steel screws @ 200 mm centre to centre, including cutting the grill to proper opening size for fixing and operation of handles and fixing approved anodised aluminium standard section around the opening, all complete as per requirement and direction of Engineer-in-charge. (Only weight of grill to be measured for payment).	5.00	Kg.	801.20	4006.00
15	WATER PROOFING				
15.1	Providing and applying fibre reinforced elastomeric liquid water proofing membrane with resilient acrylic polymers having Sun Reflectivity Index (SRI) of 105 on top of concrete roof in three coats @10.76 litre/ 10 sqm. One coat of self-priming of elastomeric waterproofing liquid (dilution with water in the ratio of 3:1) and two coats of undiluted elastomeric waterproofing liquid (dry film thickness of complete application/system not less than 500 microns). The operation shall be carried out after scrapping and properly cleaning the surface to remove loose particles with wire brushes, complete in all respect as per the direction of Engineer-in-Charge.	31.00	sqm	706.80	21911.00
16	RAIN WATER HARVESTING & TUBEWELLS				
16.1	Supplying, filling, spreading & leveling stone boulders of size range 5 cm to 20 cm, in recharge pit, in the required thickness, for all leads & lifts, all complete as per direction of Engineer-in-charge.	11.00	cum	1895.05	20846.00
17	STRUCTURAL GLAZING ALUMINIUM COMPOSITE PANEL				

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17.1	Providing and supplying aluminium extruded tubular and other aluminium sections as per the architectural drawings and approved shop drawings , the aluminium quality as per grade 6063 T5 or T6 as per BS 1474,including super durable powder coating of 60-80 microns conforming to AAMA 2604 of required colour and shade as approved by the Engineer-in-Charge. (The item includes cost of material such as cleats, sleeves, screws etc. necessary for fabrication of extruded aluminium frame work. Nothing extra shall be paid on this account). The weight of aluminium extruded section shall be taken for purpose of payment.	210.00	Kg.	520.00	109200.00
18	MISCELLANEOUS WORK				
18.1	Providing and Fixing Bush Male & Female of required size and strength for M.S Gate inclusive of smooth movement of door etc. complete as per direction as per direction of Engineer-in-Charge.	6.00	Nos	2580.85	15485.00
18.2	Providing CnC Cutting on M.S Plate as required as per design and shape with clean finish etc. complete as per direction of Engineer-in-Charge.	4.00	sqm	11862.20	47449.00
18.3	Desgining,Making and Fixing in position SS-304 grade Silver Matt Finished Letters in both english & hindi having height 220 to 300 mm having 50mm depth and logo with IP65 LED backlit illumination complete with acrylic diffuser,LED modules, power supply ,wiring, fixing arrangement, template including all accessories complete etc complete including installation and preparing surface as well as Stagging as approved by Engineer-in-Charge.	1.00	Job	138427.55	138428.00
18.4	Credit of old stone boulder after old rubble masonry demolished and stacked properly for all lead and lift etc. complete as per direction of Engineer-in-charge. (For measurement purpose total demolished RR Masonry is measured after deduction of 33% for mortar and from net quantity deduction 15% for voids)	16.00	cum	-1004.55	-16073.00
18.5	Providing and fixing 110mm diameter PVC Pipe (working pressure 4 kg/sqm) confirming to IS : 13592 Type A in CC retaining wall as weep hole etc all complete as per direction of Engineer-in-charge.	37.00	metre	276.25	10221.00

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18.6	Credit of old bricks after bricks demolished and stacked properly (uncleaned bricks stacks after demolishing is to be measured) complete as per direction of Engineer-in-charge.	14.00	cum	582.90	8161.00
				Total ₹	49,72,713.00

Note:- Contractor shall quote the Percentage rate according to current rate of GST applicable in works contract. Nothing extra shall be paid on this account.

**Executive Engineer-Almora,
CPWD, Almora**

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